PROOF OF FUJITA–SHOKUROV'S LOG SPECTRUM CONJECTURE

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Abstract. We prove Fujita–Shokurov's log spectrum conjecture, which predicts upper bounds for the accumulation points of all orders of the pseudo-effective thresholds of ample Cartier divisors on smooth projective varieties with reduced simple normal crossing boundaries.

Contents

1. Introduction 1
2. Preliminaries 7
3. Numerical spectra 11
4. Generalized lc thresholds 27
5. Good models and bounded families 36
6. Models of polarized log smooth pairs 41
7. Thresholds approaching an accumulation point 45
8. A discrepancy gap at a fixed accumulation point 51
9. Proof of the main theorem 54
References 58

1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Fujita [Fuj92] introduced the *Kodaira energy* of a polarized manifold (X, L), where X is a smooth projective variety and L is an ample Cartier divisor on X, and formulated the spectrum conjecture: for any positive integer d and any real number $\epsilon > 0$, there exist only finitely many Kodaira energies of polarized manifolds of dimension d that are less than $-\epsilon$. In [Fuj96], Fujita extended the Kodaira energy to polarized log varieties (X, Δ, L) by

$$\kappa\epsilon(X, \Delta, L) := -\inf\{t \in \mathbb{Q} \mid \kappa(K_X + \Delta + tL) \geq 0\},$$

and called $\kappa\epsilon(X, \Delta, L)$ the *log Kodaira energy* of (X, Δ, L). The Kodaira energy of (X, L) is $\kappa\epsilon(X, 0, L)$. In [Fuj96, (3.7)], Fujita posed the log spectrum conjecture on the k-th accumulation sets, defined below, of the set of log Kodaira energies of triples (X, Δ, L), where X is smooth, Δ is a reduced simple normal crossing divisor, and L is an ample Cartier divisor, and attributed it to Shokurov, who had proposed it in a slightly different form. In dimension two, the log Kodaira energies of such triples already have infinitely many accumulation points [Fuj96, (3.6)]. See Remark 1.3.

Following [HL21] and [Li21], we formulate the conjecture in terms of pseudo-effective thresholds. Let X be a smooth projective variety, Δ a reduced simple normal crossing divisor on X, possibly $\Delta = 0$, and L an ample Cartier divisor on X. The *pseudo-effective threshold* of L with respect to (X, Δ) is

$$\mathrm{pet}(X, \Delta; L) := \inf\{t \in \mathbb{R}_{\geq 0} \mid K_X + \Delta + tL \text{ is pseudo-effective}\},$$

Date: October 29, 2026.

2020 *Mathematics Subject Classification.* 14E30, 14C20, 14J45.

Key words and phrases. pseudo-effective threshold, log spectrum conjecture, Kodaira energy, accumulation point, generalized pair, boundedness.

(cf. [Li21, Definition 1.1]). Since L is ample, $K_X+\Delta+tL$ is ample for $t\gg 0$, so $\operatorname{pet}(X,\Delta;L)\in[0,+\infty)$. For any positive integer d, let

$$\operatorname{PET}_d:=\{\operatorname{pet}(X,\Delta;L)\mid \dim X=d\}\subset[0,+\infty),$$

where X runs over smooth projective varieties of dimension d, Δ over reduced simple normal crossing divisors on X, and L over ample Cartier divisors on X. For any subset $\Gamma\subset\mathbb{R}$, let

$$\operatorname{Acc}(\Gamma):=\left\{\gamma\in\mathbb{R}\middle|\gamma=\lim_{i\to+\infty}\gamma_i \text{ for some pairwise distinct } \gamma_i\in\Gamma\right\},$$

let $\operatorname{Acc}^0(\Gamma):=\Gamma$, and let

$$\operatorname{Acc}^k(\Gamma):=\operatorname{Acc}(\operatorname{Acc}^{k-1}(\Gamma))$$

for any positive integer k. We call $\operatorname{Acc}^k(\Gamma)$ the *k-th accumulation set* of Γ. In [HL21] and [Li21], $\operatorname{Acc}^k(\Gamma)$ is denoted by $\lim^k(\Gamma)$.

By [HL21, Section 1], $\kappa\epsilon(X,\Delta,L)=-\operatorname{pet}(X,\Delta;L)$ if $\operatorname{pet}(X,\Delta;L)>0$, since then $K_X+\Delta+tL$ is big for any $t>\operatorname{pet}(X,\Delta;L)$ (a pseudo-effective divisor plus an ample divisor is big, and the pseudo-effective cone is closed [Laz04, Theorem 2.2.26]) and is not pseudo-effective for any $t<\operatorname{pet}(X,\Delta;L)$, and $\kappa\epsilon(X,\Delta,L)\geq 0$ if $\operatorname{pet}(X,\Delta;L)=0$, that is, if $K_X+\Delta$ is pseudo-effective. Hence the negative log Kodaira energies of the triples (X,Δ,L) above are exactly the numbers $-\operatorname{pet}(X,\Delta;L)$ with $\operatorname{pet}(X,\Delta;L)>0$. Fujita's conjecture [Fuj96, (3.7)] predicts that, for any positive integer $k\leq d$, the k-th accumulation set of the set of log Kodaira energies of the triples (X,Δ,L) above with $\dim X=d$ is contained in $[k-d,+\infty)$. Since $k-d\leq 0$, $\operatorname{PET}_d\subset[0,+\infty)$, and

$$\operatorname{Acc}^k(\Gamma)\cap U=\operatorname{Acc}^k(\Gamma\cap U)\cap U$$

for any subset $\Gamma\subset\mathbb{R}$ and any open subset $U\subset\mathbb{R}$ (by induction on k, since all but finitely many terms of a sequence converging to a point of U are in U), the prediction only depends on the negative log Kodaira energies, and it is equivalent to $\operatorname{Acc}^k(\operatorname{PET}_d)\subset[0,d-k]$. We state Fujita–Shokurov's log spectrum conjecture in the form of [HL21] and [Li21].

Conjecture 1.1 ([Fuj96, (3.7)])**.** *Let d be a positive integer. Then $\operatorname{Acc}^k(\operatorname{PET}_d)\subset[0,d-k]$ for any integer $1\leq k\leq d$.*

See also [HL21, Conjecture 1.1] and [Li21, Conjecture 1.2].

Conjecture 1.1 considers the higher accumulation sets of PET_d. As described in [HL21, Section 1], the ACC for PET_d is known. Fujita's spectrum conjecture, that is, the case $\Delta=0$, is also known. [Fuj96, (3.4)] proved the spectrum conjecture [Fuj96, (3.2)] for $\mathbb{Q}$-factorial terminal varieties of dimension at most 3, in particular for smooth threefolds. In any dimension d, [DC17, Theorem 1.2] proved the spectrum conjecture, that is,

$$\{\operatorname{pet}(X,0;L)\mid\dim X=d\}\cap[\epsilon,+\infty)$$

is a finite set for any real number $\epsilon>0$, and [DC16, Theorem 1.2] proved that the log Kodaira energies of the triples (X,Δ,L) above can accumulate only from below, that is (recall that $\operatorname{PET}_d\subset[0,d+1]$, since $K_X+(d+1)L$ is nef by Lemma 6.2 and $\Delta\geq 0$), PET_d is an ACC set. We refer the reader to [HL20, Theorems 1.3 and 1.4] for more general settings. The term "log spectrum conjecture" in [DC16] and [HL20] refers to the ACC for PET_d and its generalizations, not to Conjecture 1.1 [HL21, Section 1].

Much less is known about the higher accumulation sets, and [HL21, Section 1] notes that no complete answer was known even for surfaces. To describe the higher accumulation points, [HL21] and [Li21] use a projective variety X with a boundary B and a nef divisor M on a higher model of X such that

$$K_X+B+tM_X\equiv 0$$

for some real number $t>0$, where M_X is the pushforward of M. We call (X,B,tM) a numerically trivial generalized pair in the sense of [BZ16], and we call t the *parameter* of (X,B,tM). [HL21, Theorem 1.2] showed that, for pairs with coefficients in a DCC set (see [HL21, Theorem 1.2] for

the precise hypotheses), any positive k-th accumulation point of the pseudo-effective thresholds in dimension d is the parameter of some (X, B, tM) of dimension at most $\max\{d-k, 1\}$ satisfying condition (†) of [HL21, Section 3.1]. For $k = d$, [HL21, Theorem 1.2] places the d-th accumulation points in the set in [HL21, (7)], whose positive elements are parameters of such generalized pairs on $\mathbb{P}^1$ by the remark after [HL21, (7)]. If the bound $d-k$ in Conjecture 1.1 is replaced with $d-k+1$, [HL21, Proposition 1.3] proved the weaker statement for $k = d-1$ and $k = d$, and [Li21, Theorem 1.3] proved it for any $1 \leq k \leq d$, that is,

$$\operatorname{Acc}^k(\operatorname{PET}_d) \subset [0, d-k+1].$$

[Li21, Theorem 1.3] follows from [Li21, Theorem 1.4], which proves the bound $d-k+1$ for the k-th accumulation sets of pseudo-effective thresholds of pairs whose boundaries have coefficients in DCC sets (see [Li21, Theorem 1.4] for the precise hypotheses). [Li21, Remark 3.6] says that it is unknown whether the bound in [Li21, Theorem 1.3] is sharp. The approach of [HL21] is parallel to the description of the accumulation points of lc thresholds in [HMX14, Theorem 1.11].

We now explain why the approach of [HL21] and [Li21] stops at $d-k+1$. [HL21, Proposition 3.4] first replaces accumulation points of thresholds with accumulation points of parameters of numerically trivial generalized pairs of dimension at most d. Each further accumulation lowers the dimension by one [HL21, Corollary 3.7]. For ample L, the generalized pairs so obtained satisfy condition (†) of [Li21, Section 2.3] by [Li21, Theorem 3.1], and [Li21] bounds their parameters by their dimensions plus one using [Li21, (3.1) and Propositions 3.2 and 3.4], which implies the bound $d-k+1$. The first step does not lower the dimension, and the bound loses one there. Indeed, let $d \geq 2$, let m be a positive integer, let H be a hyperplane in $\mathbb{P}^d$, and let H' be a hyperplane in $\mathbb{P}^{d-1}$. Then

$$K_{\mathbb{P}^d} + \Big(1 - \frac{1}{m}\Big)H + \Big(d + \frac{1}{m}\Big)H \equiv 0 \quad \text{and} \quad K_{\mathbb{P}^{d-1}} + dH' \equiv 0.$$

Thus the parameter of $(\mathbb{P}^d, (1-\frac{1}{m})H, (d+\frac{1}{m})H)$ is $d+\frac{1}{m}$, the numbers $d+\frac{1}{m}$ with $m \in \mathbb{N}^+$ are pairwise distinct and converge to d, and d is the parameter of $(\mathbb{P}^{d-1}, 0, dH')$. Therefore, if one only keeps track of numerically trivial generalized pairs, the first accumulation points can only be placed among parameters in dimension $d-1$, and the bound obtained is d instead of the bound $d-1$ predicted by Conjecture 1.1. The numbers $d+\frac{1}{m}$ are also pseudo-effective thresholds: defining $\operatorname{pet}(\mathbb{P}^d, (1-\frac{1}{m})H; H)$ as the infimum of the real numbers $t \geq 0$ with $K_{\mathbb{P}^d} + (1-\frac{1}{m})H + tH$ pseudo-effective, we have

$$K_{\mathbb{P}^d} + \Big(1 - \frac{1}{m}\Big)H + tH \equiv \Big(t - d - \frac{1}{m}\Big)H,$$

so the pseudo-effective threshold of H with respect to $(\mathbb{P}^d, (1-\frac{1}{m})H)$ is $d+\frac{1}{m}$ [Li21, Example 3.5]. More generally, [Li21, Example 3.5] shows that the bound $d-k+1$ in [Li21, Theorem 1.4] is optimal for boundaries with coefficients of the form $1-\frac{1}{m}$. Thus any proof of Conjecture 1.1 must use that Δ is reduced. Therefore we do not replace the triples with numerically trivial generalized pairs before taking limits. Instead, along a sequence of pseudo-effective thresholds converging to an accumulation point, we keep the triples (X_i, Δ_i, L_i) and the exact thresholds $\operatorname{pet}(X_i, \Delta_i; L_i)$.

The goal of this paper is to prove Conjecture 1.1.

The main result is formulated in terms of parameters of numerically trivial generalized pairs. We use sets of such parameters rather than numerical bounds because a numerical bound does not improve under taking accumulation points: $\operatorname{Acc}(\operatorname{PET}_d) \subset [0, d-1]$ only implies $\operatorname{Acc}^2(\operatorname{PET}_d) \subset [0, d-1]$, whereas the sets $\mathcal{N}_r$ defined below satisfy

$$\operatorname{Acc}(\mathcal{N}_r) \cap (0, +\infty) \subset \mathcal{N}_{r-1}$$

for any positive integer r (Lemma 3.7), which makes an induction on k possible.

Roughly speaking, for a positive integer d, $\mathcal{N}_d$ is the set of real numbers $t > 0$ for which there exists a projective generalized lc pair $(X, B, t\mathbf{M})$ such that $1 \leq \dim X \leq d$, $K_X + B + t\mathbf{M}_X \equiv 0$,

$B \in \Gamma_t$ for an explicit set $\Gamma_t \subset [0,1]$ depending only on t, $\mathbf{M}_X$ is $\mathbb{Q}$-Cartier, and $\mathbf{M}$ is induced by an ample Cartier divisor, that is, $\mathbf{M}$ is the nef $\boldsymbol{b}$-divisor determined by the pullback of an ample Cartier divisor under a contraction from a higher model of X (see Definitions 2.3, 2.2, and 3.1). If $\mathbf{M}_X \equiv 0$, we also ask that some coefficient of B be of the form $\frac{m-1+kt}{m}$ with $m, k \in \mathbb{N}^+$. We let $\mathcal{N}_0 := \{\frac{1}{m} \mid m \in \mathbb{N}^+\}$, and we refer to Definition 3.3 for the precise definition of $\mathcal{N}_d$. For $d \geq 1$, $\mathcal{N}_d$ is the intersection of $(0, +\infty)$ with the set defined in [Li21, (2.6)] for the coefficient set $\{0, 1\}$ (see Lemma 3.5).

Theorem 1.2. *The following hold.*

(1) Let $d \geq 2$ be an integer. Then

$$\mathrm{Acc}(\mathrm{PET}_d) \cap (0, +\infty) \subset \mathcal{N}_{d-2},$$

where $\mathcal{N}_{d-2}$ is the set in Definition 3.3.

(2) Conjecture 1.1 holds.

Remark 1.3. Theorem 1.2(1) is sharp when $d = 2$:

$$\mathrm{Acc}(\mathrm{PET}_2) \cap (0, +\infty) = \mathcal{N}_0 = \left\{ \frac{1}{m} \middle| m \in \mathbb{N}^+ \right\}.$$

Indeed, the inclusion $\subset$ is Theorem 1.2(1) for $d = 2$. For the reverse inclusion, let $e \geq 1$ be an integer, Σ_e the Hirzebruch surface, that is, the ruled surface over $\mathbb{P}^1$ with invariant e [Har77, V, Notation 2.8.1, Proposition 2.9, and Theorem 2.17], $C_0 \subset \Sigma_e$ the section with $C_0^2 = -e$, F a fiber of $\Sigma_e \to \mathbb{P}^1$, and $L_e := C_0 + (e+1)F$. Then L_e is ample and

$$\mathrm{pet}(\Sigma_e, C_0; mL_e) = \frac{e+2}{(e+1)m}$$

for any $m \in \mathbb{N}^+$ [Fuj96, (3.6)]. In fact, by [Har77, V, Corollary 2.11], $K_{\Sigma_e} \equiv -2C_0 - (e+2)F$, and by [Har77, V, Corollary 2.18], a divisor $aC_0 + bF$ with $a, b \in \mathbb{Z}$ is ample if and only if $a > 0$ and $b > ae$, so L_e is ample. By [Har77, V, Proposition 2.20(a)], the class of any irreducible curve on Σ_e is $aC_0 + bF$ with $a, b \geq 0$, so the pseudo-effective classes on Σ_e are exactly the classes in the closed cone $\{aC_0 + bF \mid a, b \geq 0\}$. Thus

$$K_{\Sigma_e} + C_0 + tL_e \equiv (t-1)C_0 + \big(t(e+1) - e - 2\big)F$$

is pseudo-effective if and only if $t \geq \frac{e+2}{e+1}$, and the formula for $\mathrm{pet}(\Sigma_e, C_0; mL_e)$ follows from $\mathrm{pet}(X, \Delta; mL) = \frac{1}{m}\mathrm{pet}(X, \Delta; L)$. Since Σ_e is a smooth projective surface, C_0 is a smooth curve, and mL_e is an ample Cartier divisor, we have $\frac{e+2}{(e+1)m} \in \mathrm{PET}_2$. For fixed m, the numbers $\frac{e+2}{(e+1)m} = \frac{1}{m} + \frac{1}{(e+1)m}$ with $e \geq 1$ are pairwise distinct and converge to $\frac{1}{m}$ from above as $e \to +\infty$. Therefore, for $d = 2$ the inclusion in Theorem 1.2(1) is an equality, and $\mathcal{N}_{d-2}$ cannot be replaced by a smaller set in general. In particular, the bound $d - k$ in Conjecture 1.1 is attained when $(d, k) = (2, 1)$.

Sketch of the proof. Theorem 1.2(2) follows from (1) by induction on k, since

$$\mathrm{Acc}(\mathcal{N}_r) \cap (0, +\infty) \subset \mathcal{N}_{r-1} \quad \text{and} \quad \mathcal{N}_r \subset (0, r+1]$$

(Lemmas 3.7 and 3.8). We prove (1) by contradiction. Suppose that $c \in \mathrm{Acc}(\mathrm{PET}_d)$, $c > 0$, and $c \notin \mathcal{N}_{d-2}$.

(i) Theorem 7.1, proved from Proposition 5.1 by MMPs, yields thresholds $u_i = \mathrm{pet}(X_i, \Delta_i; L_i)$ strictly decreasing to c and klt varieties Y_i birational to X_i with $\rho(Y_i) = 1$, on which the birational transform of Δ_i is a prime divisor S_i, $K_{Y_i} + (1-\alpha_i)S_i + u_i\overline{L_i}_{Y_i} \equiv 0$, and $K_{Y_i} + S_i + c\overline{L_i}_{Y_i} \equiv 0$, with $\alpha_i \to 0$ (Set-up 6.1), where $\overline{L_i}$ is the nef $\boldsymbol{b}$-divisor determined by L_i and $\overline{L_i}_{Y_i}$ is the trace of $\overline{L_i}$ on Y_i (Definition 2.2).

(ii) By Theorem 8.2, for $i \gg 0$, S_i is the unique generalized lc place of $(Y_i, S_i, c\overline{L_i})$, and the other generalized log discrepancies are at least some $\epsilon = \epsilon(d, c) > 0$. Thus the Y_i are ϵ-lc Fano varieties.

(iii) By Theorem 2.12, [Bir21a, Theorem 1.2], and [Siu93, Theorem (0.1)], the triples (X_i, Δ_i, L_i) are bounded (Section 9), so the u_i take finitely many values (Lemma 5.2), a contradiction.

The key step is (ii), explained in (c). The proof excludes one c at a time and does not identify the accumulation points. In Theorem 1.2(2), only the first accumulation step is new, and of the nine cases excluded in Theorem 7.1, only one needs a new result, Proposition 4.4. The others follow from known results (Section 7). The proof also uses the ACC for PET_d [HL20, Theorem 1.4] in Theorem 7.1, the bounds for pseudo-effective thresholds [Li21, Propositions 3.2 and 3.4] in Lemma 3.8, and the deformation invariance of volumes [HMX13, Theorem 1.8(3)] in Lemma 5.2.

(a) Models at the exact threshold. Since [BZ16, Lemma 4.4(2)] needs a generalized klt pair, [HL21, Proof of Proposition 3.4, Step 2] replaces Δ_i with a klt boundary $\mathbb{R}$-linearly equivalent to Δ_i plus a small positive multiple of an ample divisor, which lowers the threshold. Proposition 5.1 keeps Δ_i and u_i: it replaces u_iL_i with $\frac{u_i}{m}H$ for a general $H \in |mL_i|$ and uses [BCHM10, Theorem 1.2(1) and Lemma 3.9.3] for a klt boundary $\mathbb{R}$-linearly equivalent to $\Delta_i + \frac{u_i}{m}H$.

Since $(X_i, \Delta_i + H)$ is lc, the prime divisors on the variety Y of Proposition 5.1 exceptional over X_i are lc places of (X_i, Δ_i), so B_Y, their sum plus the birational transform of Δ_i, is reduced. On a common resolution, the pullback of $K_{X_i} + \Delta_i + u_iL_i$ is the pullback of $K_Y + B_Y + u_i\overline{L_i}_Y$ plus an $\mathbb{R}$-divisor $G \geq 0$ exceptional over Y (Proposition 5.1(2)). After the MMPs in Theorem 7.1, the equality of Proposition 5.1(2) becomes the comparison in Set-up 6.1(9) used in (c).

(b) Rounding up the boundary. If $(Y_i, S_i, u_i\overline{L_i})$ is not generalized lc, the generalized lc thresholds $\lambda_i \in [c, u_i)$ of $\overline{L_i}$ with respect to (Y_i, S_i) may all equal c. Proposition 4.2 places λ_i in $\mathcal{N}_{d-1}$, and since Proposition 3.6 needs pairwise distinct parameters, we only obtain $c \in \mathcal{N}_{d-1}$. Proposition 4.4 shows $\lambda_i \in \mathcal{N}_{d-2}$ directly: since $(Y_i, (1-\alpha_i)S_i, u_i\overline{L_i})$ is generalized klt and $\lambda_i < u_i$, a prime divisor E over Y_i computing λ_i has center in S_i. After extracting E, the strict transform of S_i is ample over Y_i, so on a general fiber F of E over its center, adjunction to E is followed by adjunction to the strict transform of S_i.

For $\lambda_i \leq 1$, which cannot be avoided (Remark 1.3), the nef part may become numerically trivial. On F, the restriction H_F of the trace of $\overline{L_i}$ on the extraction is ample, and is the sum of the trace on F of the restricted nef part and an $\mathbb{R}$-divisor $R_F \geq 0$ that enters the boundary with coefficient λ_i (4.5). If the restricted nef part becomes numerically trivial after the second adjunction, then R_F does not, since H_F is ample, so some boundary coefficient has the form $\frac{m-1+f+k\lambda_i}{m}$ with $k \geq 1$ (Definition 3.3).

(c) A discrepancy gap at a fixed accumulation point. The global ACC [BZ16, Theorem 1.6] (Lemma 8.1) bounds from below only the positive generalized log discrepancies of $(Y_i, S_i, c\overline{L_i})$, and the components of Δ_i contracted on Y_i might be generalized lc places. Set-up 6.1(9) compares $K_{X_i} + \Delta_i + u_iL_i$ with $K_{Y_i} + (1-\alpha_i)S_i + u_i\overline{L_i}_{Y_i}$ up to the pullback of an $\mathbb{R}$-divisor $N_i \geq \alpha_i\Delta_i$, and passing to $(Y_i, S_i, c\overline{L_i})$ adds $(u_i - c)e(Q, Y_i, \overline{L_i}) - \alpha_i \operatorname{mult}_Q S_i$ to the generalized log discrepancy of a prime divisor Q (Definition-Proposition 2.4), which the data on X_i do not control. The nef divisor $K_{X_i} + (d+1)L_i$ (Lemma 6.2) turns the comparison into the identity of Proposition 6.5, whose correction terms have fixed signs, so the center on X_i of a generalized lc place is a stratum of Δ_i, and the components of Δ_i through it have coefficient α_i in N_i. Since L_i is ample, a contracted component P' has $e(P', Y_i, \overline{L_i}) > 0$ (Lemma 6.4), and Corollary 4.3 implies $\operatorname{mult}_{P'} N_i > \frac{\epsilon}{2} > \alpha_i$. Hence only S_i remains. The constants depend on c, since the global ACC needs a DCC set of coefficients containing c.

(d) Boundedness of the triples. Theorem 2.12 bounds the Y_i, but u_i depends on (X_i, Δ_i, L_i), and all components of Δ_i but one are contracted on Y_i. [Bir21a, Theorem 1.2] descends nef parts to bounded log smooth models whose boundaries contain the divisors exceptional over Y_i, but it needs a big adjoint divisor of bounded volume, while $K_{Y_i} + S_i + c\overline{L_i}_{Y_i} \equiv 0$. We add $\overline{A_i}$ to the nef part, where A_i is a very ample divisor of bounded degree on Y_i (Theorem 2.12): the generalized

log discrepancies do not change, and the adjoint divisor becomes numerically equivalent to A_i. Then $\overline{L_i}$ descends to a bounded log smooth couple (X_i', Σ_i), Lemma 2.9 provides a morphism $X_i' \to X_i$, and Σ_i contains the birational transform of Δ_i. Thus L_i^d, $K_{X_i} \cdot L_i^{d-1}$, and $\Delta_i \cdot L_i^{d-1}$ are bounded (Lemma 2.14), and by [Siu93, Theorem (0.1)], a fixed multiple of L_i is very ample.

Reducedness of Δ_i is used three times. First, the log discrepancies of (X_i, Δ_i) are integers and its lc places have strata of Δ_i as centers (Lemma 2.7), which locates the smallest generalized log discrepancy less than 1 in the proof of Theorem 7.1 on a component of Δ_i, and the generalized lc places in Proposition 6.5 over strata. Second, $a(P', X_i, \Delta_i) = 0$ for any component P' of Δ_i, so the generalized log discrepancy of P' with respect to $(Y_i, (1-\alpha_i)S_i, u_i\overline{L_i})$ is $\mathrm{mult}_{P'} N_i$ (Theorem 8.2). Third, (iii) needs it: for the example $(\mathbb{P}^d, (1-\frac{1}{m})H)$ above, $(\mathbb{P}^d, (1-\frac{1}{m})H, (d+\frac{1}{m})\overline{H})$ has the properties of $(Y_i, (1-\alpha_i)S_i, u_i\overline{L_i})$ in (i) and (ii) with $c = d \notin \mathcal{N}_{d-2}$, while the thresholds $d+\frac{1}{m}$ are pairwise distinct on one polarized variety, so Lemma 5.2 needs fixed coefficients.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, for a nef $\boldsymbol{b}$-divisor $\mathbf{M}$ over a normal projective variety X and a prime divisor E over X, we introduce the multiplicity along E of the difference between the pullback of the trace of $\mathbf{M}$ on X and the trace of $\mathbf{M}$ on a higher model (Definition-Proposition 2.4), and we establish uniform constants for a bounded family of log smooth couples (Lemma 2.14).

In Section 3 we study nef $\boldsymbol{b}$-divisors induced by ample Cartier divisors and the sets $\mathcal{N}_d$: we prove that

$$\mathrm{Acc}(\mathcal{N}_d) \cap (0, +\infty) \subset \mathcal{N}_{d-1}$$

and $\mathcal{N}_d \subset (0, d+1]$ (Lemmas 3.7 and 3.8), and two lemmas that realize a parameter by generalized pairs of smaller dimension: Lemma 3.9 for restrictions to a general fiber of a contraction either of the underlying variety or of the normalization of a component of the boundary with coefficient one, and Lemma 3.10 for generalized pairs that are not generalized klt.

In Section 4 we prove that a positive and finite generalized lc threshold of a nef $\boldsymbol{b}$-divisor induced by an ample Cartier divisor, with respect to a $\mathbb{Q}$-factorial klt projective variety of dimension d with a reduced boundary, belongs to $\mathcal{N}_{d-1}$ (Proposition 4.2), and that after the fractional coefficients of a boundary are rounded up, the generalized lc threshold belongs to $\mathcal{N}_{d-2}$ whenever it is positive and smaller than a parameter at which the original generalized pair is generalized klt (Proposition 4.4), and we deduce that, for a real number $c > 0$ with $c \notin \mathcal{N}_{d-2}$, there exists a real number $\eta > 0$ depending only on d and c such that, in the setting of Proposition 4.2, generalized lc at a parameter in $(c, c+\eta)$ implies generalized lc at $c + 2\eta$ (Corollary 4.3), which is used in Section 8.

In Section 5, for $c = \mathrm{pet}(X, \Delta; L) > 0$ with $\dim X = d$, we construct a $\mathbb{Q}$-factorial klt birational model Y of X with a reduced boundary on which the generalized pair with nef part c times the nef $\boldsymbol{b}$-divisor determined by the pullback of L is generalized lc, such that on a common resolution the pullback of $K_X + \Delta + cL$ exceeds the pullback of the corresponding $\mathbb{R}$-divisor on Y by an $\mathbb{R}$-divisor $N \geq 0$ exceptional over Y, and $c \in \mathcal{N}_{d-1}$ unless the generalized pair on Y is numerically trivial (Proposition 5.1). We also prove that the triples (X, Δ, L) in a bounded family have only finitely many values $\mathrm{pet}(X, \Delta; L)$ (Lemma 5.2).

In Section 6, for a birational map from a smooth projective variety X of dimension d with an ample Cartier divisor L to a normal projective variety Y such that no prime divisor on Y is exceptional over X and both K_Y and the birational transform of L on Y are $\mathbb{Q}$-Cartier, we bound the log discrepancy of a prime divisor with respect to Y minus the log discrepancy of the prime divisor with respect to X by $d+1$ times the multiplicity of Definition-Proposition 2.4 for the nef $\boldsymbol{b}$-divisor over Y determined by L (Definition-Proposition 6.3), and, for the data in Set-up 6.1, we characterize the generalized lc places of the numerically trivial generalized pair on Y with reduced boundary (Proposition 6.5).

In Section 7 we prove that a positive accumulation point $c \notin \mathcal{N}_{d-2}$ of PET_d is the limit of a strictly decreasing sequence of pseudo-effective thresholds u_i, each realized on a $\mathbb{Q}$-factorial

klt projective variety Y_i of Picard number one by a numerically trivial generalized pair with boundary $(1-\alpha_i)S_i$, where $\lim_{i\to+\infty}\alpha_i=0$ (Theorem 7.1).

In Section 8 we prove that, at a fixed accumulation point c and for i large, the generalized pairs on Y_i obtained by replacing the coefficient $1-\alpha_i$ of S_i with 1 and the coefficient u_i of the nef part with c have S_i as the unique generalized lc place, and any other prime divisor over Y_i has generalized log discrepancy at least a positive real number depending only on d and c (Theorem 8.2).

In Section 9 we prove Theorem 1.2.

Acknowledgements. The work was partially supported by the National Key R&D Program of China #2024YFA1014400. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun, Bin Wu, and Leheng Chen for AI assistance. Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and generalized pairs.

Definition 2.1. A *pair* (X,B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B\geq 0$ on X such that K_X+B is $\mathbb{R}$-Cartier. If we do not require that $B\geq 0$, then we say that (X,B) is a *sub-pair*. Let $h:X'\to X$ be any log resolution of (X,B) and E a prime divisor on X', and write $K_{X'}+B':=h^*(K_X+B)$. Then we define

$$a(E,X,B):=1-\operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X,B).

For any (sub-)pair (X,B), we say that (X,B) is *(sub-)lc* if $a(E,X,B)\geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E,X,B)>0$ for any prime divisor E over X. An *lc place* of (X,B) is a prime divisor E over X such that $a(E,X,B)=0$. An *lc center* of (X,B) is the center of an lc place of (X,B) on X.

For an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor G on a normal variety X and a prime divisor E over X, we denote by $\operatorname{mult}_E G$ the multiplicity of E in the pullback of G to a normal variety on which E is a divisor.

Definition 2.2. Let X be a normal projective variety. A *nef $\boldsymbol{b}$-divisor* $\mathbf{M}$ over X is determined by a projective birational morphism $f:X'\to X$ from a normal variety and a nef $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $\mathbf{M}_{X'}$ on X', where two such choices $(f,\mathbf{M}_{X'})$ and $(f_1,\mathbf{M}_{X'_1})$, with $f_1:X'_1\to X$, determine one nef $\boldsymbol{b}$-divisor if the pullbacks of $\mathbf{M}_{X'}$ and $\mathbf{M}_{X'_1}$ to a common resolution of X' and X'_1 coincide. We also regard $\mathbf{M}$ as a nef $\boldsymbol{b}$-divisor over any normal projective variety Y birational to X, determined by the pullback of $\mathbf{M}_{X'}$ to a common resolution of X' and Y.

Let Y be a normal variety birational to X. We say that $\mathbf{M}$ *descends* to Y if there exists an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $\mathbf{M}_Y$ on Y such that the pullbacks of $\mathbf{M}_{X'}$ and $\mathbf{M}_Y$ to a common resolution of X' and Y coincide. For a projective birational morphism $h:X''\to Y$ from a normal variety X'' to which $\mathbf{M}$ descends, we call $\mathbf{M}_Y:=h_*\mathbf{M}_{X''}$ the *trace* of $\mathbf{M}$ on Y. For a nef $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D on a normal projective variety Y birational to X, we denote by $\overline{D}$ the nef $\boldsymbol{b}$-divisor determined by the pullback of D to a common resolution of X and Y. For nef $\boldsymbol{b}$-divisors $\mathbf{M}$ and $\mathbf{N}$ over X and a real number $s\geq 0$, we denote by $\mathbf{M}+s\mathbf{N}$ the nef $\boldsymbol{b}$-divisor determined by a projective birational morphism $f_2:X'_2\to X$ from a normal variety to which both $\mathbf{M}$ and $\mathbf{N}$ descend and the $\mathbb{R}$-divisor $\mathbf{M}_{X'_2}+s\mathbf{N}_{X'_2}$ on X'_2, and by $\mathbf{0}$ the nef $\boldsymbol{b}$-divisor $\overline{0}$, where 0 is the zero divisor on X.

See [BZ16, Definition 1.4] and [Bir21a, 2.3, 2.6(3), and 3.1(1)].

Definition 2.3. A *generalized pair* $(X, B, \mathbf{M})$ consists of a normal projective variety X, an $\mathbb{R}$-divisor $B \geq 0$ on X, and a nef $\boldsymbol{b}$-divisor $\mathbf{M}$ over X, such that $K_X + B + \mathbf{M}_X$ is $\mathbb{R}$-Cartier. Let E be a prime divisor over X, and let $f : X' \to X$ be a projective birational morphism from a normal variety such that E is a divisor on X' and $\mathbf{M}$ descends to X'. Write

$$K_{X'} + B' + \mathbf{M}_{X'} := f^*(K_X + B + \mathbf{M}_X).$$

We define $a(E, X, B, \mathbf{M}) := 1 - \mathrm{mult}_E B'$ as the *generalized log discrepancy* of E with respect to $(X, B, \mathbf{M})$. Let $\epsilon \geq 0$ be a real number. We say that $(X, B, \mathbf{M})$ is *generalized ϵ-lc* (resp. *generalized klt*) if $a(E, X, B, \mathbf{M}) \geq \epsilon$ (resp. $a(E, X, B, \mathbf{M}) > 0$) for any prime divisor E over X, and *generalized lc* if it is generalized 0-lc. A *generalized lc place* of $(X, B, \mathbf{M})$ is a prime divisor E over X such that $a(E, X, B, \mathbf{M}) = 0$. When $\mathbf{M} = \mathbf{0}$, we write (X, B) for $(X, B, \mathbf{0})$ and drop the word generalized. We say that a pair (X, B) is *ϵ-lc* if $a(E, X, B) \geq \epsilon$ for any prime divisor E over X. We say that a normal quasi-projective variety X is *ϵ-lc* (resp. *klt*) if $(X, 0)$ is a pair that is ϵ-lc (resp. klt).

Let $(X, B, \mathbf{M})$ be a generalized pair, $D \geq 0$ an $\mathbb{R}$-divisor on X, and $\mathbf{N}$ a nef $\boldsymbol{b}$-divisor over X such that $D + \mathbf{N}_X$ is $\mathbb{R}$-Cartier. The *generalized lc threshold* of $D + \mathbf{N}$ with respect to $(X, B, \mathbf{M})$ is

$$\sup\{s \geq 0 \mid (X, B + sD, \mathbf{M} + s\mathbf{N}) \text{ is generalized lc}\}.$$

When $D = 0$ (resp. $\mathbf{N} = \mathbf{0}$), we call it the *generalized lc threshold* of $\mathbf{N}$ (resp. D) with respect to $(X, B, \mathbf{M})$, and we drop the word generalized only when $\mathbf{M} = \mathbf{N} = \mathbf{0}$.

See [BZ16, Definitions 4.1 and 4.3] and [Bir21a, 2.6 and 2.9].

Definition-Proposition 2.4. Let X be a normal projective variety, E a prime divisor over X, and $\mathbf{M}$ a nef $\boldsymbol{b}$-divisor over X such that $\mathbf{M}_X$ is $\mathbb{R}$-Cartier. Let $f : X' \to X$ be a projective birational morphism from a normal variety such that E is a divisor on X' and $\mathbf{M}$ descends to X'. We define

$$e(E, X, \mathbf{M}) := \mathrm{mult}_E(f^*\mathbf{M}_X - \mathbf{M}_{X'}).$$

Then $e(E, X, \mathbf{M})$ does not depend on the choice of f, $e(E, X, \mathbf{M}) \geq 0$, and $e(E, X, \mathbf{M}) = 0$ if E is a divisor on X. Moreover, if K_X is $\mathbb{R}$-Cartier and $B \geq 0$ is an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X, then

$$a(E, X, B, t\mathbf{M}) = a(E, X, 0) - \mathrm{mult}_E B - te(E, X, \mathbf{M})$$

for any real number $t \geq 0$. In particular, if X is $\mathbb{Q}$-factorial, $B \geq 0$ is an $\mathbb{R}$-divisor on X, and $(X, B, s\mathbf{M})$ is generalized lc for some real number $s \geq 0$, then $(X, B, s'\mathbf{M})$ is generalized lc for any real number $0 \leq s' \leq s$, and (X, B) is lc.

Proof. Let $f_1 : X'_1 \to X$ be another projective birational morphism from a normal variety such that E is a divisor on X'_1 and $\mathbf{M}$ descends to X'_1, and let $g : X'_2 \to X'$ and $g_1 : X'_2 \to X'_1$ be projective birational morphisms from a normal variety X'_2 such that $f \circ g = f_1 \circ g_1$. Since $\mathbf{M}$ descends to X' and X'_1, $g^*\mathbf{M}_{X'} = g_1^*\mathbf{M}_{X'_1}$, so

$$g^*(f^*\mathbf{M}_X - \mathbf{M}_{X'}) = g_1^*(f_1^*\mathbf{M}_X - \mathbf{M}_{X'_1}).$$

Thus $e(E, X, \mathbf{M})$ does not depend on the choice of f.

Let $F := f^*\mathbf{M}_X - \mathbf{M}_{X'}$. Since $\mathbf{M}_X = f_*\mathbf{M}_{X'}$, F is f-exceptional. Since $f^*\mathbf{M}_X = \mathbf{M}_{X'} + F$ and $\mathbf{M}_{X'}$ is nef, $F \geq 0$ by the negativity lemma [BCHM10, Lemma 3.6.2]. Thus $e(E, X, \mathbf{M}) = \mathrm{mult}_E F \geq 0$, and $e(E, X, \mathbf{M}) = 0$ if E is a divisor on X, because then E is not f-exceptional.

Now assume that K_X and B are $\mathbb{R}$-Cartier, and let $t \geq 0$ be a real number. Then $(X, B, t\mathbf{M})$ is a generalized pair, and $t\mathbf{M}$ descends to X' with trace $t\mathbf{M}_{X'}$. Write

$$K_{X'} + B_0 := f^*K_X, \quad K_{X'} + B_t + t\mathbf{M}_{X'} := f^*(K_X + B + t\mathbf{M}_X).$$

Subtracting the first equality from the second one, we obtain

$$B_t = B_0 + f^*B + t(f^*\mathbf{M}_X - \mathbf{M}_{X'}).$$

Since $a(E,X,0)=1-\operatorname{mult}_E B_0$ and $a(E,X,B,t\mathbf{M})=1-\operatorname{mult}_E B_t$ by Definition 2.3, taking the multiplicities along E implies the formula.

If X is $\mathbb{Q}$-factorial, then K_X, B, and $\mathbf{M}_X$ are $\mathbb{R}$-Cartier. Thus, for any prime divisor E over X and any real numbers $0\leq s'\leq s$, the formula and $e(E,X,\mathbf{M})\geq 0$ imply that

$$a(E,X,B,s'\mathbf{M})=a(E,X,B,s\mathbf{M})+(s-s')e(E,X,\mathbf{M})\geq a(E,X,B,s\mathbf{M}).$$

Taking $s'=0$, we obtain that (X,B) is lc if $(X,B,s\mathbf{M})$ is generalized lc. □

Definition-Proposition 2.5 ([BZ16, Definition 4.7 and Remark 4.8])**.** Let $(X,B,\mathbf{M})$ be a generalized pair, S a component of $\lfloor B\rfloor$ with $\operatorname{mult}_S B=1$, and S^ν the normalization of S. Let $f:X'\to X$ be a log resolution of $(X,\operatorname{Supp}B)$ such that $\mathbf{M}$ descends to X', let $S':=f_*^{-1}S$, and write $K_{X'}+B'+\mathbf{M}_{X'}:=f^*(K_X+B+\mathbf{M}_X)$. Let B_{S^ν} be the pushforward of $(B'-S')|_{S'}$ to S^ν, and let $\mathbf{M}|_{S^\nu}$ be the nef $\boldsymbol{b}$-divisor over S^ν that descends to S' with $(\mathbf{M}|_{S^\nu})_{S'}=\mathbf{M}_{X'}|_{S'}$. Then

$$K_{S^\nu}+B_{S^\nu}+(\mathbf{M}|_{S^\nu})_{S^\nu}=(K_X+B+\mathbf{M}_X)|_{S^\nu}.$$

If $(X,B,\mathbf{M})$ is generalized lc, then $B_{S^\nu}\geq 0$ and $(S^\nu,B_{S^\nu},\mathbf{M}|_{S^\nu})$ is a generalized lc pair. In this case, we call $(S^\nu,B_{S^\nu},\mathbf{M}|_{S^\nu})$ the *generalized adjunction* of $(X,B,\mathbf{M})$ to S.

Proof. See [BZ16, Remark 4.8]. □

Definition 2.6. Let X be a smooth variety and Δ a reduced simple normal crossing divisor on X. A *stratum* of Δ is an irreducible component of $P_1\cap\cdots\cap P_r$ for some positive integer r and distinct components $P_1,\dots,P_r$ of Δ.

Lemma 2.7 (cf. [KM98, Lemma 2.29 and Corollary 2.31(3)])**.** *Let X be a smooth variety, Δ a reduced simple normal crossing divisor on X, and E a prime divisor over X. Then $a(E,X,\Delta)$ is a non-negative integer. If $a(E,X,\Delta)=0$, then $\operatorname{center}_X E$ is a stratum of Δ.*

Lemma 2.8 ([BZ16, Lemma 4.6], [HL21, Proposition 2.5])**.** *Let $(X,B,\mathbf{M})$ be a generalized lc pair such that X is $\mathbb{Q}$-factorial and klt, and let E be a prime divisor over X such that E is exceptional over X and $a(E,X,B,\mathbf{M})<1$. Then there exists an extremal divisorial contraction $h:Y\to X$ from a $\mathbb{Q}$-factorial klt projective variety Y such that E is the unique h-exceptional prime divisor and*

$$K_Y+h_*^{-1}B+\big(1-a(E,X,B,\mathbf{M})\big)E+\mathbf{M}_Y=h^*(K_X+B+\mathbf{M}_X).$$

In particular, $-E$ is ample over X.

Proof. We only need to show that $-E$ is ample over X. Since Y is $\mathbb{Q}$-factorial, E is $\mathbb{Q}$-Cartier. If E were h-nef, then [KM98, Lemma 3.39] applied to $-E$ would imply $-E\geq 0$ because $h_*(-E)=0$, which is impossible since $E\neq 0$ and $E\geq 0$. Thus there exists a curve C contracted by h such that $(E\cdot C)<0$. Since h is projective, an h-ample divisor is positive on $\overline{\mathrm{NE}}(Y/X)\backslash\{0\}$ by [KM98, Theorem 1.44], so $\overline{\mathrm{NE}}(Y/X)$ contains no line. Since h is extremal, $\rho(Y/X)=1$, and hence $\overline{\mathrm{NE}}(Y/X)$ is the ray spanned by the class of C. Thus $-E$ is positive on $\overline{\mathrm{NE}}(Y/X)\backslash\{0\}$. Let l be a positive integer such that lE is Cartier. By [KM98, Theorem 1.44], $-lE$ is h-ample, so $-E$ is ample over X. □

Lemma 2.8 is the case of a single divisor in [HL21, Proposition 2.5]. That Y is $\mathbb{Q}$-factorial and klt follows from [HL21, Proposition 2.4] and its item (3).

Lemma 2.9. *Let $h:X'\to X$ be a projective birational morphism between normal projective varieties, $g:X'\to Y$ a morphism to a projective variety Y, H an ample Cartier divisor on Y, and D a Cartier divisor on X such that $h^*D\sim g^*H$. Then there exists a morphism $\bar{g}:X\to Y$ such that $g=\bar{g}\circ h$. In particular, $D\sim\bar{g}^*H$.*

Proof. Let $\Gamma\subset X\times Y$ be the image of $(h,g):X'\to X\times Y$ with the reduced structure, and let $p:\Gamma\to X$ and $q:\Gamma\to Y$ be the projections. Since X' is an irreducible projective variety, Γ is an integral closed subvariety of $X\times Y$ and p is projective. Since h is birational and $h=p\circ(h,g)$, p is birational.

Let $x \in X$ be a closed point and Z an irreducible component of $h^{-1}(x)$. Suppose that $g(Z)$ is not a point. Then $g(Z)$ contains a curve C', and cutting an irreducible component of $Z \cap g^{-1}(C')$ dominating C' with general hyperplane sections, we obtain a curve $C \subset Z$ such that $g(C) = C'$. Since $h(C) = \{x\}$ and $h^*D \sim g^*H$, the projection formula [Ful98, Proposition 2.5(c)] implies

$$0 = (h^*D \cdot C) = (g^*H \cdot C) = \deg(C \to C')(H \cdot C') > 0,$$

a contradiction. Thus $g(Z)$ is a point for any irreducible component Z of $h^{-1}(x)$. Since $h^{-1}(x)$ is connected by Zariski's main theorem [Har77, III, Corollary 11.4], $g(h^{-1}(x))$ is a point. Since (h, g) maps X' onto Γ, $p^{-1}(x) = \{x\} \times g(h^{-1}(x))$ is a point. Hence p is quasi-finite and projective, so p is finite [Har77, III, Exercise 11.2]. Since X is normal, the finite birational morphism p is an isomorphism. Let $\bar{g} := q \circ p^{-1}$. Since $p \circ (h, g) = h$, we have $(h, g) = p^{-1} \circ h$, and hence $g = q \circ (h, g) = \bar{g} \circ h$.

Since $h^*(D - \bar{g}^*H) = h^*D - g^*H \sim 0$ and h is birational, $D - \bar{g}^*H = h_*h^*(D - \bar{g}^*H) \sim 0$, because the pushforward of a principal divisor under h is principal and $h_*h^*G = G$ for any Cartier divisor G on X. □

2.2. **Boundedness and ACC.**

Definition 2.10. A set $\Lambda \subset \mathbb{R}$ is a *DCC set* (resp. an *ACC set*) if any non-increasing (resp. non-decreasing) sequence in Λ is eventually constant.

Definition 2.11. A *couple* (X, Σ) consists of a normal projective variety X and a reduced divisor Σ on X. It is *log smooth* if X is smooth and Σ is a simple normal crossing divisor. A set $\mathcal{P}$ of couples is *bounded* if there exist positive integers d and r such that for any $(X, \Sigma) \in \mathcal{P}$, $\dim X \leq d$ and there exists a very ample Cartier divisor A on X with $A^{\dim X} \leq r$ and $\Sigma \cdot A^{\dim X - 1} \leq r$. A set of normal projective varieties X is *bounded* if the set of couples $(X, 0)$ is bounded.

By [Bir21a, 2.21], a set of couples is bounded if and only if there exist finitely many projective morphisms between varieties and a reduced divisor on the total space of each of them, such that any member of the set is isomorphic to a fiber of one of the morphisms, and under the isomorphism the restriction of the corresponding reduced divisor to the fiber is not smaller than the reduced divisor of the member. This is the notion of boundedness in [Bir21b, Theorem 1.1].

Theorem 2.12 ([Bir21b, Theorem 1.1])**.** *Let d be a positive integer and ϵ a positive real number. Then the set of projective varieties X of dimension d such that there exists an ϵ-lc pair (X, B) with $-(K_X + B)$ nef and big is bounded.*

Theorem 2.13 ([BZ16, Theorem 1.6])**.** *Let d be a positive integer and Λ a DCC set of non-negative real numbers. Then there exists a finite subset $\Lambda_0 \subset \Lambda$ depending only on d and Λ satisfying the following. Let $(X, B, \mu\mathbf{M})$ be a generalized lc pair of dimension d, where $\mu \in \Lambda$ and $\mathbf{M}$ descends to a nef Cartier divisor $\mathbf{M}_{X'}$ on a higher model X' of X. Assume that $\mu = 0$ if $\mathbf{M}_{X'} \equiv 0$, $B \in \Lambda$, and $K_X + B + \mu\mathbf{M}_X \equiv 0$. Then $B \in \Lambda_0$ and $\mu \in \Lambda_0$.*

Theorem 2.13 is the case of a single nef part in [BZ16, Theorem 1.6].

Lemma 2.14. *Let $\mathcal{P}$ be a bounded set of log smooth couples. Then there exists a positive integer b depending only on $\mathcal{P}$ satisfying the following. For any $(X, \Sigma) \in \mathcal{P}$, there exist a very ample Cartier divisor A on X and an $\mathbb{R}$-divisor $C \geq \Sigma$ on X such that $A^{\dim X} \leq b$, and $bA - C$, $bA - K_X$, and $bA + K_X$ are nef.*

Proof. By [Bir21a, 2.21], there exist finitely many projective morphisms $V^j \to T^j$ of varieties and reduced divisors C^j on V^j satisfying the following: for any $(X, \Sigma) \in \mathcal{P}$, there exist an index j, a closed point $t \in T^j$, and an isomorphism $\varphi : V^j_t \to X$ such that (V^j_t, C^j_t) is a couple and $\varphi_* C^j_t \geq \Sigma$.

Fix j. We show that there exist finitely many smooth irreducible locally closed subsets $T' \subset T^j$ satisfying the following, where $V' := V^j \times_{T^j} T'$:

(1) $V' \to T'$ is smooth, and any irreducible component of $C^j \cap V'$ is flat over T', and
(2) any closed point $t \in T^j$ such that V_t^j is smooth belongs to one of the subsets T'.

By Noetherian induction, it suffices to show that for any irreducible closed subset $T_0 \subset T^j$, there exist a non-empty open subset $T_1 \subset T_0$ and an open subset $T' \subset T_1$ satisfying (1) such that T' contains any closed point $t \in T_1$ for which V_t^j is smooth. By generic flatness [Eis95, Theorem 14.4], applied to $V^j \times_{T^j} T_0$ and to the irreducible components of C^j over T_0, there exists a smooth non-empty open subset $T_1 \subset T_0$ over which V^j and the irreducible components of C^j over T_0 are flat. Let $T' \subset T_1$ be the complement of the image of the closed subset of $V^j \times_{T^j} T_1$ at which $V^j \times_{T^j} T_1 \to T_1$ is not smooth. By [Har77, III, Theorem 10.2], T' contains any closed point $t \in T_1$ for which V_t^j is smooth, and $V' \to T'$ is smooth.

Fix one of the subsets T' satisfying (1) and (2), and let $V' = V^j \times_{T^j} T'$. Since $V' \to T'$ is smooth and T' is smooth, V' is smooth. Let C_V be the sum of the irreducible components of $C^j \cap V'$ of codimension one in V', $K_{V'/T'}$ a relative canonical divisor of $V' \to T'$, and A_V a Cartier divisor on V' that is very ample over T'. For a closed point $t \in T'$, we write V_t' for the fiber of $V' \to T'$ over t, and $A_t := A_V|_{V_t'}$.

Let $t \in T'$ be a closed point such that V_t' is a variety of dimension n and C_t^j is a reduced divisor on V_t'. By (1) and [Har77, III, Corollary 9.6], for any irreducible component W of $C^j \cap V'$, the fiber of W over t is of pure dimension $\dim W - \dim T'$. Since $\dim V' = n + \dim T'$ and C_t^j is of pure dimension $n-1$, any component of C_t^j is a component of the fiber of a component of C_V, and no component of C_V contains V_t'. Thus $C_V|_{V_t'} \geq 0$ is a Cartier divisor whose support contains C_t^j, so $C_V|_{V_t'} \geq C_t^j$.

For any closed point $s \in T'$, there exists a positive integer b_s such that $b_s A_s - C_V|_{V_s'}$, $b_s A_s - K_{V'/T'}|_{V_s'}$, and $b_s A_s + K_{V'/T'}|_{V_s'}$ are ample. By [Laz04, Theorem 1.2.17], there exists an open neighborhood U_s of s in T' such that for any closed point $t \in U_s$, the restrictions of $b_s A_V - C_V$, $b_s A_V - K_{V'/T'}$, and $b_s A_V + K_{V'/T'}$ to V_t' are ample. Since T' is quasi-compact, finitely many of the subsets U_s cover T'. Moreover, embedding V' into $\mathbb{P}^N_{T'}$ by A_V, since T' is integral and $V' \to T'$ is flat, the Hilbert polynomial of V_t' does not depend on t [Har77, III, Theorem 9.9]. If V_t' is a variety, then the leading coefficient of the Hilbert polynomial of V_t' is $A_t^{\dim V_t'}/(\dim V_t')!$. Thus $A_t^{\dim V_t'}$ takes only one value for the closed points $t \in T'$ such that V_t' is a variety.

Let b be a positive integer that is not smaller than the integers b_s for the finitely many subsets U_s and the finitely many values of $A_t^{\dim V_t'}$ above, for all j and all subsets T'. Let $(X, \Sigma) \in \mathcal{P}$, and let j, t, and φ be as in the first paragraph. Since X is smooth, t belongs to one of the subsets T' by (2), and $t \in U_s$ for one of the finitely many points s. Let

$$A := \varphi_* A_t, \quad C := \varphi_*(C_V|_{V_t'}).$$

Then A is a very ample Cartier divisor, $A^{\dim X} = A_t^{\dim V_t'} \leq b$, and $C \geq \varphi_* C_t^j \geq \Sigma$. Since A_t is ample and $b \geq b_s$, the divisors $bA_t - C_V|_{V_t'}$ and $bA_t \pm K_{V'/T'}|_{V_t'}$ are ample. Since $K_{V'/T'}|_{V_t'}$ is a canonical divisor of V_t', $bA - C$, $bA - K_X$, and $bA + K_X$ are ample, hence nef. □

3. Numerical spectra

In this section, we define the numerical spectra $\mathcal{N}_d$ and prove the properties of $\mathcal{N}_d$ used in the rest of the paper: positive accumulation points of $\mathcal{N}_d$ belong to $\mathcal{N}_{d-1}$, $\mathcal{N}_d$ is closed in $(0, +\infty)$ and bounded from above, and two lemmas realize a parameter by generalized pairs of smaller dimension.

Definition 3.1. Let X be a normal projective variety. A nef $\boldsymbol{b}$-divisor $\mathbf{M}$ over X is *induced by an ample Cartier divisor* if there exist a projective birational morphism $f : X' \to X$ from a smooth variety, a contraction $g : X' \to Z$ to a normal projective variety, and an ample Cartier divisor H on Z, such that $\mathbf{M} = \overline{g^* H}$.

We ask $\mathbf{M}$ to be the pullback of an ample Cartier divisor on the base of a contraction, and not only to descend to a nef Cartier divisor on a higher model, in order to bound the numerical spectra from above (Lemma 3.8): if one only knows that the trace of $\mathbf{M}$ on a variety of Picard number one is an ample Weil divisor, which need not be Cartier, then no upper bound for the parameter is known [HL21, Remark 3.9]. The condition in Definition 3.1 corresponds to conditions (3) and (4) of [Li21, Section 2.3] (see Lemma 3.5).

Lemma 3.2. *Let X and V be normal projective varieties, $f : V \to X$ a morphism, $g : X \to Z$ a contraction to a normal projective variety, and H an ample Cartier divisor on Z. Then there exist a contraction $g_V : V \to Z_V$ to a normal projective variety and a finite morphism $h : Z_V \to Z$ such that $g \circ f = h \circ g_V$. In particular, h^*H is an ample Cartier divisor on Z_V.*

Proof. Let $Z_V := \mathbf{Spec}(g \circ f)_*\mathcal{O}_V$ [Har77, II, Exercise 5.17], and let $h : Z_V \to Z$ be the natural morphism. Then $g \circ f$ factors as $h \circ g_V$ for a morphism $g_V : V \to Z_V$ with $(g_V)_*\mathcal{O}_V = \mathcal{O}_{Z_V}$, and since $(g \circ f)_*\mathcal{O}_V$ is coherent [Har77, III, Theorem 8.8(b)], h is finite. This is the Stein factorization [Har77, III, Corollary 11.5]. Since V is normal, so is Z_V. Since h is finite and Z is projective, Z_V is projective and h^*H is an ample Cartier divisor [Laz04, Proposition 1.2.13]. □

By Lemma 3.2 (cf. [Li21, Section 3]), a nef $\boldsymbol{b}$-divisor induced by an ample Cartier divisor is still induced by an ample Cartier divisor when it is regarded as a nef $\boldsymbol{b}$-divisor over a higher model, when it is restricted to a general fiber of a contraction, and when it is restricted to the normalization of a component of the boundary (via the strict transform on a smooth higher model and a resolution of the strict transform).

Definition 3.3. Let $\mathcal{S} := \{1 - \frac{1}{m} \mid m \in \mathbb{N}^+\} \cup \{1\}$, and for a real number $t > 0$, let

$$\mathcal{S}_t := \left\{ \frac{m - 1 + f + kt}{m} \middle| m, k \in \mathbb{N}^+,\ f \in \{0, 1\},\ \frac{m - 1 + f + kt}{m} \leq 1 \right\}.$$

Let $t > 0$ be a real number. A generalized pair $(X, B, t\mathbf{M})$ satisfies *condition* (†) if

(1) $(X, B, t\mathbf{M})$ is generalized lc and $K_X + B + t\mathbf{M}_X \equiv 0$,
(2) $B \in \mathcal{S} \cup \mathcal{S}_t$,
(3) $\mathbf{M}$ is induced by an ample Cartier divisor and $\mathbf{M}_X$ is $\mathbb{Q}$-Cartier, and
(4) if $\mathbf{M}_X \equiv 0$, then at least one coefficient of B belongs to $\mathcal{S}_t$.

For a positive integer d, let $\mathcal{N}_d$ be the set of real numbers $t > 0$ for which there exists a generalized pair $(X, B, t\mathbf{M})$ satisfying condition (†) with $1 \leq \dim X \leq d$. Let $\mathcal{N}_0 := \{\frac{1}{m} \mid m \in \mathbb{N}^+\}$.

The set $\mathcal{S}_t$ consists of the coefficients $\frac{m-1+f+kt}{m}$ of the generalized adjunction formula [HL21, Definition 2.2 and (2)] in which the nef part $t\mathbf{M}$ contributes a term kt with $k \geq 1$, and $\mathcal{S} \cup \mathcal{S}_t$ is closed under generalized adjunction by Lemma 3.4 (cf. [HL21, (4)]). Condition (4) excludes realizations in which t does not appear: without it, for two distinct points $p, q \in \mathbb{P}^1$, the generalized pair $(\mathbb{P}^1, p + q, t\mathbf{0})$ would satisfy conditions (1)–(3) for any $t > 0$.

The requirement in condition (3) that $\mathbf{M}_X$ is $\mathbb{Q}$-Cartier is the second half of condition (†)(3) of [Li21, Section 2.3] (Lemma 3.5(1)). It is used in Case 2 of the proof of Lemma 3.8, where X has Picard number one and need not be $\mathbb{Q}$-factorial, to deduce from $\mathbf{M}_X \not\equiv 0$ that $\mathbf{M}_X$ is ample. It also gives the hypothesis $\mathbf{M}_X \equiv 0$ of condition (4) its usual meaning.

The set $\mathcal{N}_0$ is not defined by generalized pairs of dimension zero: by Lemma 3.7 for $d = 1$ and by the generalized pairs $(\mathbb{P}^1, (1 - \frac{1}{n})p, \frac{n+1}{nm}\overline{mz})$ with $p \neq z$, which satisfy condition (†) and whose parameters converge to $\frac{1}{m}$ as $n \to +\infty$, it is the set of positive accumulation points of $\mathcal{N}_1$, that is, the set of $t > 0$ with $1 \in \mathcal{S}_t$.

Lemma 3.4. *Let $t > 0$ be a real number, m a positive integer, $k, n_1, \ldots, n_l$ non-negative integers, and $b_1, \ldots, b_l$ elements of $(\mathcal{S} \cup \mathcal{S}_t) \setminus \{0\}$ such that*

$$b := \frac{m - 1 + \sum_{j=1}^{l} n_j b_j + kt}{m} \leq 1.$$

Then $b \in \mathcal{S} \cup \mathcal{S}_t$. Moreover, $b \in \mathcal{S}_t$ if $k \geq 1$, and $b \in \mathcal{S}$ if $k = 0$ and $b_j \in \mathcal{S}$ for any j with $n_j \geq 1$.

Proof. Since $t > 0$, any element of $\mathcal{S}_t$ is of the form $\frac{m'-1+k't}{m'}$ with $m', k' \in \mathbb{N}^+$. Since $b \leq 1$, we have $\sum_j n_j b_j + kt \leq 1$. Any element of $(\mathcal{S} \cup \mathcal{S}_t)\backslash\{0\}$ is 1, or $k't$ for some $k' \in \mathbb{N}^+$, or $\frac{m'-1+k't}{m'}$ for some integers $m' \geq 2$ and $k' \geq 0$. An element of the third form is at least $\frac{1}{2}$, with equality if and only if $m' = 2$ and $k' = 0$.

If $b_j = 1$ and $n_j \geq 1$ for some j, then $n_j = 1$, $k = 0$, and $n_{j'} = 0$ for any $j' \neq j$, so $b = 1 \in \mathcal{S}$. If $\sum n_j \geq 2$, where the sum runs over all j such that b_j is of the third form, then two of the terms equal $\frac{1}{2}$, all other terms vanish, and $k = 0$, so $b = 1 \in \mathcal{S}$. Otherwise, suppose first that some b_j with $n_j \neq 0$ is of the third form. Then there exists exactly one such index j_0, $n_{j_0} = 1$, and $\sum_{j \neq j_0} n_j b_j = Kt$ for a non-negative integer K. Write $b_{j_0} = \frac{m'-1+k't}{m'}$. A direct calculation shows

$$b = \frac{mm' - 1 + (k' + m'(K + k))t}{mm'}.$$

If no b_j with $n_j \neq 0$ is of the third form, then $\sum_j n_j b_j = Kt$ for a non-negative integer K and

$$b = \frac{m - 1 + (K + k)t}{m}.$$

In both cases, $b \in \mathcal{S}$ if the coefficient of t in the numerator vanishes, and $b \in \mathcal{S}_t$ otherwise. If $k \geq 1$, then only the last two formulas are possible, $k' + m'(K + k) \geq 1$, and $K + k \geq 1$, so $b \in \mathcal{S}_t$. If $k = 0$ and $b_j \in \mathcal{S}$ for any j with $n_j \geq 1$, then writing each such b_j as 1 or as $\frac{m'-1}{m'}$ with $m' \geq 2$, the coefficient of t in the numerator vanishes, so $b \in \mathcal{S}$. □

Lemma 3.5. *Let $I := \{0, 1\}$. Then the following hold.*

(1) $\mathcal{S}$ and $\mathcal{S}_t$ are the sets $D(I)$ and $D_t(I)$ of [Li21, (2.4)]*, and condition (†) is equivalent to condition (†) of* [Li21, Section 2.3] *for I.*
(2) $\mathcal{N}_d = N_d(I) \cap (0, +\infty)$ for any positive integer d, and $\mathcal{N}_0 = N_0(I) \cap (0, +\infty)$, where $N_d(I)$ is the set in [Li21, (2.6)] *and $N_0(I)$ is the set in* [HL21, (7)]*.*
(3) $\mathcal{N}_0 \subset \mathcal{N}_1$, and $\mathcal{N}_d \subset \mathcal{N}_{d+1}$ for any positive integer d.

Proof. (1) Let I_+ be the set of real numbers $\sum_j n_j a_j \leq 1$ with $a_j \in I$ and non-negative integers n_j [Li21, (2.3)]. Since $I_+ = \{0, 1\}$, the sets $D(I)$ and $D_t(I)$ of [Li21, (2.4)] are $\mathcal{S}$ and $\mathcal{S}_t$. Clauses (3) and (4) of condition (†) in [Li21, Section 2.3] ask that $\mathbf{M}$ descends to a semi-ample Cartier divisor $\mathbf{M}_{X''}$ on a normal higher model X'' of X, that $\mathbf{M}_X$ is $\mathbb{Q}$-Cartier, and that $\mathbf{M}_{X''}$ is the pullback of an ample Cartier divisor by the fibration defined by a sufficiently divisible multiple of $\mathbf{M}_{X''}$. The other clauses are conditions (1), (2), and (4) of Definition 3.3, since $\mathbf{M}_X$ is $\mathbb{Q}$-Cartier and thus $\mathbf{M}_{X''} \equiv 0$ if and only if $\mathbf{M}_X \equiv 0$ by the negativity lemma [KM98, Lemma 3.39] (cf. the remark after [Li21, (2.5)]). If $\mathbf{M}_{X''} = \varphi^* H$ for a contraction $\varphi : X'' \to Z$ and an ample Cartier divisor H on Z, then the pullback of $\mathbf{M}_{X''}$ to a resolution $\mu : X' \to X''$ is $(\varphi \circ \mu)^* H$, and $\varphi \circ \mu$ is a contraction. Thus $\mathbf{M}$ is induced by an ample Cartier divisor. Conversely, if $\mathbf{M} = \overline{g^* H}$ as in Definition 3.1, then $g^* H$ is a semi-ample Cartier divisor, and since $g_* \mathcal{O}_{X'} = \mathcal{O}_Z$ and H is ample, the fibration defined by a sufficiently divisible multiple of $g^* H$ is g. Thus condition (†) is equivalent to condition (†) of [Li21, Section 2.3] for I.

(2) By [HL21, (7)], $N_0(I) = \{0\} \cup \{\frac{1}{m} \mid m \in \mathbb{N}^+\}$. Thus (2) follows from (1) and the definitions of $N_d(I)$ and $N_0(I)$.

(3) For any positive integer d, $\mathcal{N}_d \subset \mathcal{N}_{d+1}$ by definition. Let $l \in \mathbb{N}^+$, and let p, q, and z be points on $\mathbb{P}^1$ such that $p \neq q$. The generalized pair $(\mathbb{P}^1, \frac{1}{2}p + \frac{1}{2}q, \frac{1}{l}\overline{lz})$ satisfies condition (†): $\frac{1}{2} \in \mathcal{S}$, $\overline{lz}$ is induced by the ample Cartier divisor lz on $\mathbb{P}^1$, $(\mathbb{P}^1, \frac{1}{2}p + \frac{1}{2}q)$ is lc, $\overline{lz}$ descends to $\mathbb{P}^1$, and

$$\deg\left(K_{\mathbb{P}^1} + \frac{1}{2}p + \frac{1}{2}q + \frac{1}{l} \cdot lz\right) = -2 + \frac{1}{2} + \frac{1}{2} + 1 = 0.$$

Thus $\frac{1}{l} \in \mathcal{N}_1$, and $\mathcal{N}_0 \subset \mathcal{N}_1$ (cf. [HL21, (7)]). □

Proposition 3.6 (cf. [HL21, Proposition 3.5]). *Let d be a positive integer, and let t_i be a strictly decreasing sequence of real numbers with limit $t > 0$. For each i, let $(X_i, D_i + B_i, t_i\mathbf{M}_i)$ be a generalized pair of dimension d such that*

(1) $(X_i, D_i + B_i, t_i\mathbf{M}_i)$ is generalized lc and $K_{X_i} + D_i + B_i + t_i(\mathbf{M}_i)_{X_i} \equiv 0$,
(2) $D_i \geq 0$ and B_i have no common components, $B_i \in \mathcal{S} \cup \mathcal{S}_{t_i}$, and for any real number $\delta > 0$, $D_i \in (1-\delta, 1]$ for any i sufficiently large,
(3) $\mathbf{M}_i$ is induced by an ample Cartier divisor and $(\mathbf{M}_i)_{X_i}$ is $\mathbb{Q}$-Cartier, and
(4) if $(\mathbf{M}_i)_{X_i} \equiv 0$, then at least one coefficient of B_i belongs to $\mathcal{S}_{t_i}$.

Then $t \in \mathcal{N}_{d-1}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove the proposition when $d = 1$.

Let $\theta := \min\{\frac{1}{2}, t\}$. Any non-zero element of $\mathcal{S}$ is at least $\frac{1}{2}$. Since $t_i > 0$, any element of $\mathcal{S}_{t_i}$ is of the form $\frac{m-1+kt_i}{m}$ with $m, k \in \mathbb{N}^+$ and $kt_i \leq 1$, so it is at least θ, and $k < \frac{1}{t}$. Possibly passing to a subsequence, we may assume that $D_i \in (\frac{1}{2}, 1]$ for any i (condition (2)).

For any i, X_i is a smooth projective curve, so the projective birational morphism in Definition 3.1 is an isomorphism, and $(\mathbf{M}_i)_{X_i}$ is a nef Cartier divisor. Let $l_i := \deg(\mathbf{M}_i)_{X_i}$. Then l_i is a non-negative integer, and $l_i = 0$ if and only if $(\mathbf{M}_i)_{X_i} \equiv 0$. By condition (4), $l_i > 0$ or $B_i \neq 0$, so $\deg(D_i + B_i + t_i(\mathbf{M}_i)_{X_i}) > 0$, and condition (1) implies $\deg K_{X_i} < 0$. Thus $X_i \cong \mathbb{P}^1$ and

$$\deg D_i + \deg B_i + l_i t_i = 2.$$

In particular, D_i has at most 3 components, B_i has at most $\frac{2}{\theta}$ components, and $l_i \leq \frac{2}{t}$.

Possibly passing to a subsequence, we may assume that the number p of components of D_i, the number q of components of B_i, and $l := l_i$ do not depend on i, and that there exist a subset $J \subset \{1, \dots, q\}$ and positive integers k_j for $j \in J$, such that for any i, $B_i = \sum_{j=1}^{q} b_{ij} B_{ij}$, where the B_{ij} are distinct prime divisors, $b_{ij} = \frac{m_{ij}-1+k_j t_i}{m_{ij}}$ for some $m_{ij} \in \mathbb{N}^+$ if $j \in J$, and $b_{ij} \in \mathcal{S} \backslash (\mathcal{S}_{t_i} \cup \{0\})$ if $j \notin J$. Let $k_j := 0$ for $j \notin J$. If $j \notin J$, then $b_{ij} = 1$, or $b_{ij} = \frac{m_{ij}-1}{m_{ij}}$ for an integer $m_{ij} \geq 2$. Possibly passing to a subsequence again, we may assume that for any j, either $\lim_{i \to +\infty} b_{ij} = 1$, or there exists $m_j \in \mathbb{N}^+$ such that $b_{ij} = \frac{m_j - 1 + k_j t_i}{m_j}$ for any i. Let U be the set of j of the first kind. For $j \notin U$, let

$$\beta_j := \lim_{i \to +\infty} b_{ij} = \frac{m_j - 1 + k_j t}{m_j},$$

where $m_j \geq 2$ if $j \notin J$. Since $k_j t < k_j t_i \leq 1$ for $j \in J$, we have $0 < \beta_j < 1$ for $j \notin U$. Let $p' := p + |U|$. Since $\lim_{i \to +\infty} \deg D_i = p$ by condition (2), we obtain

$$p' + \sum_{j \notin U} \beta_j + lt = 2.$$

We show that $p' \geq 1$. If $p' = 0$, then $D_i = 0$, $U = \emptyset$, and for any i,

$$\Big(\sum_{j \in J} \frac{k_j}{m_j} + l\Big) t_i = 2 - \sum_{j=1}^{q} \frac{m_j - 1}{m_j}.$$

By condition (4), $l \geq 1$ or $J \neq \emptyset$, so t_i does not depend on i, a contradiction.

We show that $p' \neq 2$. If $p' = 2$, then $U = \{1, \dots, q\}$ and $l = 0$, since $\beta_j > 0$ for $j \notin U$ and $t > 0$. By condition (4), $J \neq \emptyset$. Since $p + q = 2$ and $D_i + B_i \in [0, 1]$, we obtain that all the coefficients of $D_i + B_i$ are equal to 1. Thus $k_j t_i = 1$ for $j \in J$ and any i, a contradiction.

Therefore, $p' = 1$ and

$$\sum_{j \notin U} \beta_j + lt = 1.$$

Let J_1 (resp. J_2) be the set of $j \notin U$ with $m_j \geq 2$ (resp. $m_j = 1$). Then $J_2 \subset J$, $\beta_j = k_j t$ for $j \in J_2$, and $\beta_j \geq \frac{1}{2}$ for $j \in J_1$. Let $K := \sum_{j \in J_2} k_j$.

Suppose that $|J_1| \geq 2$. Then $|J_1| = 2$, $\beta_j = \frac{1}{2}$ for $j \in J_1$, $J_2 = \emptyset$, and $l = 0$. Thus $m_j = 2$ and $k_j = 0$ for $j \in J_1$, so $J \subset U$. By condition (4), $J \neq \emptyset$, so $p' = 1$ implies $p = 0$ and $U = J = \{j_0\}$ for some j_0. Since $b_{ij} = \frac{1}{2}$ for $j \in J_1$, we obtain $b_{ij_0} = 1$, so $k_{j_0} t_i = 1$ for any i, a contradiction.

If $J_1 = \{j_1\}$ for some j_1, then

$$\frac{m_{j_1} - 1 + k_{j_1} t}{m_{j_1}} + (K + l)t = 1,$$

so $\big(k_{j_1} + m_{j_1}(K + l)\big)t = 1$. If $J_1 = \emptyset$, then $(K + l)t = 1$. Since $t > 0$, in both cases $t = \frac{1}{n}$ for a positive integer n, so $t \in \mathcal{N}_0$.

Step 2. In this step, we fix notation for $d \geq 2$ and show five facts (a)–(e), which are used in the rest of the proof.

From now on, we assume that $d \geq 2$ and that the proposition holds for generalized pairs of dimension r for any integer $1 \leq r \leq d-1$. Since $t_i > t$ and the t_i are pairwise distinct, any of the finitely many positive integers smaller than $\frac{1}{t}$ is equal to $\frac{1}{t_i}$ for at most one i. Possibly passing to a subsequence, we may assume that $\frac{1}{t_i} \notin \mathbb{Z}$ for any i.

For any i, write $B_i = \sum_j b_{ij} B_{ij}$, where the B_{ij} are distinct prime divisors, let J_i be the set of j with $b_{ij} \in \mathcal{S}_{t_i}$, and for $j \in J_i$ write $b_{ij} = \frac{m_{ij} - 1 + k_{ij} t_i}{m_{ij}}$ with $m_{ij}, k_{ij} \in \mathbb{N}^+$. Let

$$R_i := \sum_{j \in J_i} \frac{k_{ij}}{m_{ij}} B_{ij} \quad \text{and} \quad B_i(s) := B_i - (t_i - s)R_i$$

for any real number $0 \leq s \leq t_i$. Then $B_i(t_i) = B_i$, $B_i(0) \in \mathcal{S}$, and for $0 < s \leq t_i$, $B_i(s) \geq 0$, $B_i(s) \in \mathcal{S} \cup \mathcal{S}_s$, and the coefficient $\frac{m_{ij} - 1 + k_{ij} s}{m_{ij}}$ in $B_i(s)$ of any component B_{ij} of R_i belongs to $\mathcal{S}_s$. When we replace X_i with a birational model below, we replace D_i and B_i with their pushforwards or strict transforms, and each B_{ij} with the corresponding prime divisor, keeping m_{ij} and k_{ij}. Thus R_i and $B_i(s)$ are replaced with their pushforwards or strict transforms. When X_i is $\mathbb{Q}$-factorial, let $G_i := R_i + (\mathbf{M}_i)_{X_i}$.

(a) Let $s > 0$ be a real number, m a positive integer, $k, n_1, \dots, n_l$ non-negative integers, and $b_1, \dots, b_l$ elements of $(\mathcal{S} \cup \mathcal{S}_s) \backslash \{0\}$ such that

$$b := \frac{m - 1 + \sum_{j=1}^{l} n_j b_j + ks}{m} \leq 1.$$

Assume that $k \geq 1$, or that $b_j \in \mathcal{S}_s$ for some j with $n_j \geq 1$. Then $b \in \mathcal{S}_s$.

To show (a), we may assume that $n_j \geq 1$ for any j. For any j, write $b_j = \frac{m_j - 1 + k_j s}{m_j}$ with $m_j \in \mathbb{N}^+$ and $k_j \in \mathbb{Z}_{\geq 0}$, where $k_j \geq 1$ if $b_j \in \mathcal{S}_s$, and $k_j = 0$ and $m_j \geq 2$ if $b_j \in \mathcal{S} \backslash \mathcal{S}_s$ and $b_j \neq 1$. If $b_{j_0} = 1$ and $b_{j_0} \notin \mathcal{S}_s$ for some j_0, then $b \leq 1$ implies $n_{j_0} = 1$, $l = 1$, and $k = 0$, which is not possible by assumption. Since $b \leq 1$, we have $\sum_j n_j b_j \leq 1$, and $b_j \geq \frac{1}{2}$ if $m_j \geq 2$. If the sum of the n_j with $m_j \geq 2$ is at least 2, then exactly two of the terms b_j, counted with multiplicity n_j, are equal to $\frac{1}{2}$, all the other terms vanish, and $k = 0$. Since $b_j = \frac{1}{2}$ and $m_j \geq 2$ imply $m_j = 2$ and $k_j = 0$, this contradicts the assumption. Thus there exists at most one index j_1 with $m_{j_1} \geq 2$, and $n_{j_1} = 1$ if j_1 exists. Let $K := k + \sum_{j \neq j_1} n_j k_j$. Then

$$b = \frac{m m_{j_1} - 1 + (k_{j_1} + m_{j_1} K)s}{m m_{j_1}}$$

if j_1 exists, and $b = \frac{m - 1 + Ks}{m}$ otherwise. By assumption, $k_{j_1} + m_{j_1} K \geq 1$ (resp. $K \geq 1$). Since $b \leq 1$, $b \in \mathcal{S}_s$.

(b) Let Y and Y'' be birational normal projective varieties of dimension $n \geq 1$, and $\mathbf{N}$ a nef $\boldsymbol{b}$-divisor over Y induced by an ample Cartier divisor, with $f : Y' \to Y$, $g : Y' \to Z$, and H as in Definition 3.1. Then $\mathbf{N} = \mathbf{0}$ if and only if Z is a point. If Z is not a point, then there exists a

non-zero $\mathbb{Q}$-divisor $N'' \geq 0$ on Y'' such that $N'' \sim_{\mathbb{Q}} \mathbf{N}_{Y''}$, and $(\mathbf{N}_{Y''} \cdot A^{n-1}) > 0$ for any ample Cartier divisor A on Y'', where the intersection number is computed as cycles. In particular, $\mathbf{N}_{Y''} \not\equiv 0$ if $\mathbf{N}_{Y''}$ is $\mathbb{Q}$-Cartier. Moreover, if Y'' is $\mathbb{Q}$-factorial and $R'' \geq 0$ is an $\mathbb{R}$-divisor on Y'' such that $R'' \neq 0$ or $\mathbf{N} \neq \mathbf{0}$, then $-(R'' + \mathbf{N}_{Y''})$ is not pseudo-effective, and $R'' + \mathbf{N}_{Y''}$ is ample if $\rho(Y'') = 1$.

For (b), if Z is a point, then $H = 0$ since any Cartier divisor on a point is zero, so $\mathbf{N} = \mathbf{0}$. Assume that Z is not a point. Let $p : Y''' \to Y'$ be a resolution such that there exists a morphism $q : Y''' \to Y''$. Then $\mathbf{N}_{Y''} = q_* p^* g^* H$. Let $l \in \mathbb{N}^+$ such that $|lH|$ is base point free. Let $H_l \in |lH|$ be a general member. Then H_l contains the image in Z of no q-exceptional prime divisor, so no component of $p^* g^* H_l$ is q-exceptional. Since $g \circ p$ is surjective and $H_l \neq 0$, $p^* g^* H_l \neq 0$. Thus $N'' := \frac{1}{l} q_* p^* g^* H_l$ is a non-zero $\mathbb{Q}$-divisor, $N'' \geq 0$, and $N'' \sim_{\mathbb{Q}} \mathbf{N}_{Y''}$ [Ful98, Theorem 1.4]. A non-zero $\mathbb{R}$-divisor $N \geq 0$ on Y'' satisfies $(N \cdot A^{n-1}) > 0$, and intersection numbers with A^{n-1} depend only on rational equivalence classes [Ful98, Corollary 2.4.1]. Thus $(\mathbf{N}_{Y''} \cdot A^{n-1}) > 0$, so $\mathbf{N} \neq \mathbf{0}$, and under the assumption of the last statement, $((R'' + \mathbf{N}_{Y''}) \cdot A^{n-1}) > 0$, which implies that $-(R'' + \mathbf{N}_{Y''})$ is not pseudo-effective, and that, if $\rho(Y'') = 1$, then $R'' + \mathbf{N}_{Y''} \equiv cA$ for a real number $c > 0$, so $R'' + \mathbf{N}_{Y''}$ is ample [Laz04, Proposition 1.3.13].

By condition (4), for any i, $R_i \neq 0$ or $\mathbf{M}_i \neq \mathbf{0}$.

(c) If there exist infinitely many i and real numbers $s_i \in \mathcal{N}_{d-1}$ such that $t \leq s_i \leq t_i$, then $t \in \mathcal{N}_{d-1}$.

For (c), if $s_i = t$ for some i, then $t \in \mathcal{N}_{d-1}$. Otherwise, $s_i > t$ and $\lim_{i \to +\infty} s_i = t$, so possibly passing to a subsequence, we may assume that s_i is strictly decreasing. By Definition 3.3, for any i there exists a generalized pair $(X_i', B_i', s_i \mathbf{M}_i')$ satisfying condition (†) with $1 \leq \dim X_i' \leq d-1$. Possibly passing to a subsequence, we may assume that $r := \dim X_i'$ does not depend on i. Then the generalized pairs $(X_i', 0 + B_i', s_i \mathbf{M}_i')$ satisfy conditions (1)–(4) with r, s_i, 0, and B_i' in place of d, t_i, D_i, and B_i. By the induction hypothesis and Lemma 3.5(3), $t \in \mathcal{N}_{r-1} \subset \mathcal{N}_{d-1}$.

(d) Let X be a $\mathbb{Q}$-factorial normal projective variety, and $\phi : X \dashrightarrow X'$ a divisorial contraction or a flip of an extremal ray Γ, where X' is $\mathbb{Q}$-factorial. Let $(X, \Delta, \mathbf{N})$ be a generalized lc pair such that $K_X + \Delta + \mathbf{N}_X \equiv 0$. Then $K_{X'} + \phi_* \Delta + \mathbf{N}_{X'} \equiv 0$, and for any prime divisor E over X,

$$a(E, X, \Delta, \mathbf{N}) = a(E, X', \phi_* \Delta, \mathbf{N}).$$

Moreover, if $R \geq 0$ is a non-zero $\mathbb{R}$-divisor on X such that $(R \cdot \Gamma) > 0$, then $\phi_* R \neq 0$.

For (d), let $p : W \to X$ and $q : W \to X'$ be a common resolution, and

$$\Xi := p^*(K_X + \Delta + \mathbf{N}_X) - q^*(K_{X'} + \phi_* \Delta + \mathbf{N}_{X'}).$$

Since ϕ does not extract any divisor, $q_* \Xi = 0$. Since $\Xi \equiv 0$ over X', the negativity lemma [BCHM10, Lemma 3.6.2(2)], applied to Ξ and to $-\Xi$, implies $\Xi = 0$. This implies the first two statements. If $\phi_* R = 0$, then ϕ is a divisorial contraction whose exceptional prime divisor P satisfies $R = cP$ for some real number $c > 0$. By [KM98, Lemma 3.39], P is not nef over X', so $(P \cdot \Gamma) < 0$ and $(R \cdot \Gamma) < 0$, a contradiction.

(e) Let X be a $\mathbb{Q}$-factorial normal projective variety, and $(X, \Delta, \mathbf{N})$ a generalized lc pair such that $(X, 0)$ is klt. Then any variety of a $(K_X + \Delta + \mathbf{N}_X)$-MMP with scaling of a general ample divisor [BZ16, Lemma 4.4] is $\mathbb{Q}$-factorial and klt.

For (e), by condition (ii) of [BZ16, Section 4], property $(*)$ there holds for $(X, \Delta, \mathbf{N})$, it is preserved by the steps of the MMP, and for any step $X_j \dashrightarrow X_{j+1}$ of the MMP and the extremal ray Γ_j contracted by it, there exists an $\mathbb{R}$-divisor $\Psi_j \geq 0$ on X_j such that (X_j, Ψ_j) is klt and $((K_{X_j} + \Psi_j) \cdot \Gamma_j) < 0$ [BZ16, Section 4]. By induction on j, X_j is $\mathbb{Q}$-factorial [BCHM10, Lemma 3.10.2(3) and Remark 3.10.6]. Moreover, since $(*)$ holds on X_j, there exists an $\mathbb{R}$-divisor $\Phi_j \geq 0$ on X_j such that (X_j, Φ_j) is klt, so $a(E, X_j, 0) \geq a(E, X_j, \Phi_j) > 0$ for any prime divisor E over X_j, that is, X_j is klt.

Step 3. In this step, we show the following. Let $s > 0$ be a real number, X a $\mathbb{Q}$-factorial normal projective variety of dimension $r \geq 2$, $\mathbf{M}$ a nef $\boldsymbol{b}$-divisor over X induced by an ample Cartier divisor, $k_1, \ldots, k_l$ positive integers, and $\Delta_0 \geq 0$ an $\mathbb{R}$-divisor on X such that $\Delta_0 \in \mathcal{S}$. Let

$P_1, \dots, P_l$ be distinct prime divisors on X such that $\operatorname{mult}_{P_j} \Delta_0 = \frac{m_j - 1}{m_j}$ for some $m_j \in \mathbb{N}^+$, and let

$$R := \sum_{j=1}^{l} \frac{k_j}{m_j} P_j \quad \text{and} \quad \Delta_u := \Delta_0 + uR$$

for any real number $u \geq 0$. Assume that $(X, \Delta_s, s\mathbf{M})$ is generalized lc. Let F be one of the following:

(1) a general fiber of a contraction $X \to Z$, or
(2) a general fiber of a contraction $S^\nu \to Z$, where S^ν is the normalization of a component S of $\lfloor \Delta_s \rfloor$ such that $S \not\subset \operatorname{Supp} R$, and Z may be a point.

Assume that $\dim F \geq 1$, $(K_X + \Delta_s + s\mathbf{M}_X)|_F \equiv 0$, and $(R + \mathbf{M}_X)|_F \not\equiv 0$. Then $s \in \mathcal{N}_{\dim F}$.

Since $(X, \Delta_s, s\mathbf{M})$ is generalized lc, $\Delta_s \in [0, 1]$, so $\Delta_s \in \mathcal{S} \cup \mathcal{S}_s$, and the coefficient $\frac{m_j - 1 + k_j s}{m_j}$ of P_j in Δ_s belongs to $\mathcal{S}_s$. If $\mathbf{M}_X|_F \not\equiv 0$, then $s \in \mathcal{N}_{\dim F}$ by Lemma 3.9, whose proof does not use Proposition 3.6. Thus we may assume that $\mathbf{M}_X|_F \equiv 0$, so $R|_F \not\equiv 0$.

Case (1). By [BZ16, Definition 4.1 and Remark 4.2(6)], the restriction $(F, \Delta_s|_F, s\mathbf{M}|_F)$ of $(X, \Delta_s, s\mathbf{M})$ to F is generalized lc, $(\mathbf{M}|_F)_F = \mathbf{M}_X|_F$, and

$$K_F + \Delta_s|_F + s(\mathbf{M}|_F)_F = (K_X + \Delta_s + s\mathbf{M}_X)|_F \equiv 0.$$

By Lemma 3.2, $\mathbf{M}|_F$ is induced by an ample Cartier divisor. Since $(\mathbf{M}|_F)_F = \mathbf{M}_X|_F$ is $\mathbb{Q}$-Cartier and $(\mathbf{M}|_F)_F \equiv 0$, **Step 2**(b) implies $\mathbf{M}|_F = \mathbf{0}$. Since F is general, any coefficient of $\Delta_s|_F$ is a coefficient of Δ_s, so $\Delta_s|_F \in \mathcal{S} \cup \mathcal{S}_s$. Since $R|_F \neq 0$, some P_j dominates Z, and the components of $P_j|_F$ have coefficient $\frac{m_j - 1 + k_j s}{m_j} \in \mathcal{S}_s$ in $\Delta_s|_F$. Thus $(F, \Delta_s|_F, s\mathbf{0})$ satisfies condition (†), and $s \in \mathcal{N}_{\dim F}$.

Case (2). By Lemma 3.2, there exist a log resolution $f : X'' \to X$ of $(X, \operatorname{Supp} \Delta_s)$, a contraction $g : X'' \to T$ to a normal projective variety, and an ample Cartier divisor H on T such that $\mathbf{M} = \overline{g^* H}$. Let $E_{\mathbf{M}} := f^* \mathbf{M}_X - \mathbf{M}_{X''}$, which is ≥ 0 by Definition-Proposition 2.4. Let $S'' := f_*^{-1} S$, $f_S : S'' \to S^\nu$ the induced morphism, and $Q := (f_S)_*(E_{\mathbf{M}}|_{S''}) \geq 0$. For any real number $0 \leq u \leq s$, write

$$K_{X''} + \Psi_u + u\mathbf{M}_{X''} := f^*(K_X + \Delta_u + u\mathbf{M}_X),$$

and let Θ_u be the pushforward of $(\Psi_u - S'')|_{S''}$ to S^ν. By Definition-Proposition 2.5, $(S^\nu, \Theta_s, s\mathbf{M}|_{S^\nu})$ is the generalized adjunction of $(X, \Delta_s, s\mathbf{M})$ to S. By Definition-Proposition 2.4, (X, Δ_s) is lc, so (X, Δ_0) is lc, and Θ_0 is the boundary of the adjunction of (X, Δ_0) to S. Since $\Psi_u = \Psi_0 + u(f^* R + E_{\mathbf{M}})$ and $S \not\subset \operatorname{Supp} R$, we have

$$\Theta_u = \Theta_0 + u(R|_{S^\nu} + Q).$$

Let F'' be the fiber of the induced morphism $S'' \to Z$ corresponding to F, and $\pi : F'' \to F$ the induced morphism. Since F is general, F'' is smooth [Har77, III, Corollary 10.7] and π is birational. By [BZ16, Definition 4.1 and Remark 4.2(6)], the restriction $(F, \Theta_s|_F, s\mathbf{M}|_F)$ of $(S^\nu, \Theta_s, s\mathbf{M}|_{S^\nu})$ to F is generalized lc, $\mathbf{M}|_F$ descends to F'' with $(\mathbf{M}|_F)_{F''} = \mathbf{M}_{X''}|_{F''}$, and

$$K_F + \Theta_s|_F + s(\mathbf{M}|_F)_F = (K_X + \Delta_s + s\mathbf{M}_X)|_F \equiv 0.$$

Since F is general, restricting $f^* \mathbf{M}_X = \mathbf{M}_{X''} + E_{\mathbf{M}}$ to F'' and pushing it forward to F, we obtain

$$\mathbf{M}_X|_F = (\mathbf{M}|_F)_F + Q|_F.$$

Let $n := \dim F$ and A an ample Cartier divisor on F. By Lemma 3.2, $\mathbf{M}|_F$ is induced by an ample Cartier divisor, so by **Step 2**(b), $((\mathbf{M}|_F)_F \cdot A^{n-1}) \geq 0$, with equality only if $\mathbf{M}|_F = \mathbf{0}$, where the intersection number is computed as cycles. Since $Q|_F \geq 0$ and $(\mathbf{M}_X|_F \cdot A^{n-1}) = 0$, we obtain $\mathbf{M}|_F = \mathbf{0}$ and $Q|_F = 0$. Thus $(F, \Theta_s|_F)$ is lc and $K_F + \Theta_s|_F \equiv 0$.

Since F is general, any component of $\Theta_s|_F$ is a component of $V|_F$ for a prime divisor V on S^ν dominating Z, with coefficient $\operatorname{mult}_V \Theta_s$. Let V be a prime divisor on S^ν dominating Z. Since $Q|_F = 0$, $\operatorname{mult}_V Q = 0$, so

$$\operatorname{mult}_V \Theta_s = \operatorname{mult}_V \Theta_0 + sc_V, \quad \text{where } c_V := \operatorname{mult}_V(R|_{S^\nu}) \geq 0.$$

If $\operatorname{mult}_V \Theta_0 = 1$, then $c_V = 0$ since $\operatorname{mult}_V \Theta_s \leq 1$, so $\operatorname{mult}_V \Theta_s = 1 \in \mathcal{S}$. Assume that $\operatorname{mult}_V \Theta_0 < 1$. Let $P_{l+1}, \dots, P_{l'}$ be the components of $\Delta_0 - S$ other than $P_1, \dots, P_l$, and for $1 \leq j \leq l'$, let $\delta_j := \operatorname{mult}_{P_j} \Delta_0$ and $\delta_j(s) := \operatorname{mult}_{P_j} \Delta_s$. By [HL21, Definition 2.2 and (2)], applied to the lc pair (X, Δ_0), and by [Kol+92, (16.5) and (16.6.3)],

$$\operatorname{mult}_V \Theta_0 = \frac{m - 1 + \sum_{j=1}^{l'} n_j \delta_j}{m},$$

where m is the Cartier index along V, that is, the order of the local class group of X at the generic point of the image of V in X [Kol+92, (16.6.3)], and $n_j := m \operatorname{mult}_V(P_j|_{S^\nu})$ is a non-negative integer for any j. Since $c_V = \sum_{j=1}^{l} \frac{k_j}{m_j} \cdot \frac{n_j}{m}$ and $\delta_j(s) = \delta_j + \frac{k_j s}{m_j}$ for $j \leq l$, we obtain

$$\operatorname{mult}_V \Theta_s = \frac{m - 1 + \sum_{j=1}^{l'} n_j \delta_j(s)}{m}.$$

Since $\delta_j(s) \in \mathcal{S} \cup \mathcal{S}_s$ for any j, Lemma 3.4 implies $\operatorname{mult}_V \Theta_s \in \mathcal{S} \cup \mathcal{S}_s$. If $c_V > 0$, then $n_j \geq 1$ for some $j \leq l$, and $\delta_j(s) \in \mathcal{S}_s$, so $\operatorname{mult}_V \Theta_s \in \mathcal{S}_s$ by **Step 2**(a).

Thus $\Theta_s|_F \in \mathcal{S} \cup \mathcal{S}_s$. Since $\mathbf{M}_X|_F \equiv 0$ and $(R + \mathbf{M}_X)|_F \not\equiv 0$, we have $R|_F \neq 0$, so some component V of $R|_{S^\nu}$ dominates Z. Then $c_V > 0$, so $\operatorname{mult}_V \Theta_s \in \mathcal{S}_s$, and the components of $V|_F$ have coefficient $\operatorname{mult}_V \Theta_s$ in $\Theta_s|_F$. Therefore, $(F, \Theta_s|_F, s\mathbf{0})$ satisfies condition (†), and $s \in \mathcal{N}_{\dim F}$.

Step 4. In this step, we show the following. Let X be a normal projective variety of dimension $r \geq 2$, and $\mathbf{M}$, k_j, Δ_0, P_j, m_j, R, and Δ_u as in **Step 3**, except that X need not be $\mathbb{Q}$-factorial. Let $s > 0$ be a real number such that $\frac{1}{s} \notin \mathbb{Z}$. Assume that $(X, \Delta_s, s\mathbf{M})$ is generalized lc but not generalized klt, $K_X + \Delta_s + s\mathbf{M}_X \equiv 0$, and $R \neq 0$ or $\mathbf{M} \neq \mathbf{0}$. Then $s \in \mathcal{N}_{r-1}$.

Let E be a generalized lc place of $(X, \Delta_s, s\mathbf{M})$. By [HL21, Proposition 2.4], applied to the list $\{E\}$ if E is exceptional over X and to the empty list otherwise, there exists a projective birational morphism $h : X_1 \to X$ such that X_1 is $\mathbb{Q}$-factorial, $(X_1, 0)$ is klt, E is a divisor on X_1, and any h-exceptional prime divisor is E or a generalized lc place of $(X, \Delta_s, s\mathbf{M})$. Let E_h be the sum of the h-exceptional prime divisors, $R_1 := h_*^{-1} R$, and $\Delta_{1,u} := h_*^{-1}\Delta_0 + E_h + uR_1$ for $u \geq 0$. Then

$$K_{X_1} + \Delta_{1,s} + s\mathbf{M}_{X_1} = h^*(K_X + \Delta_s + s\mathbf{M}_X) \equiv 0,$$

so $(X_1, \Delta_{1,s}, s\mathbf{M})$ is generalized lc, and E is a component of $\lfloor \Delta_{1,s} \rfloor$. Let $G_1 := R_1 + \mathbf{M}_{X_1}$, and fix a real number $0 < s_0 < s$. Since X_1 is $\mathbb{Q}$-factorial and $R_1 \geq 0$, the formula in Definition-Proposition 2.4 implies that $(X_1, \Delta_{1,s_0}, s_0\mathbf{M})$ is generalized lc. By **Step 2**(b), $-G_1$ is not pseudo-effective, so

$$K_{X_1} + \Delta_{1,s_0} + s_0\mathbf{M}_{X_1} \equiv -(s - s_0)G_1$$

is not pseudo-effective. Since $(X_1, 0)$ is klt, by [BZ16, Lemma 4.4(1)], under condition (ii) there, we may run a $(K_{X_1} + \Delta_{1,s_0} + s_0\mathbf{M}_{X_1})$-MMP with scaling of a general ample divisor, which terminates with a Mori fiber space $\pi : Y \to Z$. For any variety X_j of the MMP, let $\Delta_{j,u}$, R_j, and $G_j := R_j + \mathbf{M}_{X_j}$ be the pushforwards. Any step $X_j \dashrightarrow X_{j+1}$ of the MMP contracts an extremal ray Γ_j with $(G_j \cdot \Gamma_j) > 0$. By **Step 2**(e), X_j is $\mathbb{Q}$-factorial, and by **Step 2**(d), $(X_j, \Delta_{j,s}, s\mathbf{M})$ is generalized lc, $K_{X_j} + \Delta_{j,s} + s\mathbf{M}_{X_j} \equiv 0$, and $R_j \neq 0$ or $\mathbf{M} \neq \mathbf{0}$. Moreover, $\Delta_{j,s} = \Delta_{j,0} + sR_j$ with $\Delta_{j,0} \in \mathcal{S}$, and for any component S_j of $\lfloor \Delta_{j,s} \rfloor$, $S_j \not\subset \operatorname{Supp} R_j$: otherwise $\operatorname{mult}_{S_j} \Delta_{j,s} = \frac{m-1+ks}{m} = 1$ for some $m, k \in \mathbb{N}^+$, so $\frac{1}{s} = k \in \mathbb{Z}$. We distinguish three cases.

Case 1. $\dim Z > 0$. Let F be a general fiber of π. Then $\dim F \geq 1$ and $(K_Y + \Delta_{Y,s} + s\mathbf{M}_Y)|_F \equiv 0$. Since $(s - s_0)G_Y \equiv -(K_Y + \Delta_{Y,s_0} + s_0\mathbf{M}_Y)$ is ample over Z, $G_Y|_F$ is ample, so $(R_Y + \mathbf{M}_Y)|_F \not\equiv 0$. By **Step 3**(1), applied to Y, $\Delta_{Y,0}$, the components of R_Y, $\mathbf{M}$, and F, we obtain $s \in \mathcal{N}_{\dim F}$.

Case 2. $\dim Z = 0$, and some step $X_j \to X_{j+1}$ of the MMP is a divisorial contraction whose exceptional prime divisor S_j is a component of $\lfloor \Delta_{j,s} \rfloor$. Let S_j^ν be the normalization of S_j, $S_j^\nu \to Z_j$ the Stein factorization of the induced morphism from S_j^ν to the image of S_j in X_{j+1}, and F a general fiber of $S_j^\nu \to Z_j$. Then $\dim F \geq 1$ and $(K_{X_j} + \Delta_{j,s} + s\mathbf{M}_{X_j})|_F \equiv 0$. Any curve in F maps to a curve on X_j whose class belongs to Γ_j, so $G_j|_F \not\equiv 0$. By **Step 3**(2), applied to X_j, $\Delta_{j,0}$, the components of R_j, $\mathbf{M}$, S_j, and F, we obtain $s \in \mathcal{N}_{\dim F}$.

Case 3. $\dim Z = 0$, and the MMP contracts no component of $\lfloor \Delta_{1,s} \rfloor$. Then $\rho(Y) = 1$, and the strict transform S_Y of E on Y is a component of $\lfloor \Delta_{Y,s} \rfloor$. Since $R_Y \neq 0$ or $\mathbf{M} \neq \mathbf{0}$, G_Y is ample by **Step 2**(b), so $G_Y|_{S_Y^\nu} \not\equiv 0$, where S_Y^ν is the normalization of S_Y and $\dim S_Y^\nu = r - 1 \geq 1$. By **Step 3**(2), applied to Y, $\Delta_{Y,0}$, the components of R_Y, $\mathbf{M}$, S_Y, and Z a point, we obtain $s \in \mathcal{N}_{r-1}$.

In all cases, $s \in \mathcal{N}_{\dim F}$ with $1 \leq \dim F \leq r - 1$, or $s \in \mathcal{N}_{r-1}$. By Definition 3.3, $s \in \mathcal{N}_{r-1}$.

Step 5. In this step, we reduce to the case where, for any i, X_i is $\mathbb{Q}$-factorial and klt, $\rho(X_i) = 1$, and G_i is ample.

By [HL21, Proposition 2.4], applied to $(X_i, D_i + B_i, t_i\mathbf{M}_i)$ and the empty list, there exists a projective birational morphism $h_i : X_i' \to X_i$ such that X_i' is $\mathbb{Q}$-factorial, $(X_i', 0)$ is klt, and any h_i-exceptional prime divisor is a generalized lc place of $(X_i, D_i + B_i, t_i\mathbf{M}_i)$. Let E_i be the sum of the h_i-exceptional prime divisors and $D_i' := h_{i*}^{-1}D_i + E_i$. Then

$$K_{X_i'} + D_i' + h_{i*}^{-1}B_i + t_i(\mathbf{M}_i)_{X_i'} = h_i^*(K_{X_i} + D_i + B_i + t_i(\mathbf{M}_i)_{X_i}).$$

Thus $(X_i', D_i' + h_{i*}^{-1}B_i, t_i\mathbf{M}_i)$ satisfies conditions (1) and (2), since $1 \in (1 - \delta, 1]$ for any real number $\delta > 0$. By Lemma 3.2, applied to a resolution of X_i' dominating the smooth higher model in Definition 3.1, $\mathbf{M}_i$ is induced by an ample Cartier divisor over X_i', and $(\mathbf{M}_i)_{X_i'}$ is $\mathbb{Q}$-Cartier since X_i' is $\mathbb{Q}$-factorial, so condition (3) holds. Since $h_{i*}^{-1}R_i \neq 0$ or $\mathbf{M}_i \neq \mathbf{0}$, condition (4) holds by **Step 2**(b). Possibly replacing $(X_i, D_i + B_i, t_i\mathbf{M}_i)$ with $(X_i', D_i' + h_{i*}^{-1}B_i, t_i\mathbf{M}_i)$, we may assume that X_i is $\mathbb{Q}$-factorial and $(X_i, 0)$ is klt.

Since X_i is $\mathbb{Q}$-factorial and $R_i \geq 0$, the formula in Definition-Proposition 2.4 implies that $(X_i, D_i + B_i(t), t\mathbf{M}_i)$ is generalized lc. Since $K_{X_i} + D_i + B_i + t_i(\mathbf{M}_i)_{X_i} \equiv 0$, we have

$$K_{X_i} + D_i + B_i(t) + t(\mathbf{M}_i)_{X_i} \equiv -(t_i - t)G_i,$$

which is not pseudo-effective by **Step 2**(b). By [BZ16, Lemma 4.4(1)], under condition (ii) there, we may run a $(K_{X_i} + D_i + B_i(t) + t(\mathbf{M}_i)_{X_i})$-MMP with scaling of a general ample divisor, which terminates with a Mori fiber space $\pi_i : Y_i \to Z_i$. By **Step 2**(e), the varieties of the MMP are $\mathbb{Q}$-factorial and klt. Any step of the MMP contracts an extremal ray Γ with $(G \cdot \Gamma) > 0$, where G is the pushforward of G_i. Let D_{Y_i}, B_{Y_i}, R_{Y_i}, and G_{Y_i} be the pushforwards of D_i, B_i, R_i, and G_i. By **Step 2**(d), $(Y_i, D_{Y_i} + B_{Y_i}, t_i\mathbf{M}_i)$ is generalized lc, $K_{Y_i} + D_{Y_i} + B_{Y_i} + t_i(\mathbf{M}_i)_{Y_i} \equiv 0$, and $R_{Y_i} \neq 0$ or $\mathbf{M}_i \neq \mathbf{0}$.

Suppose that $\dim Z_i > 0$ for infinitely many i. Possibly passing to a subsequence, we may assume that $\dim Z_i > 0$ for any i and that $r := \dim Y_i - \dim Z_i$ does not depend on i. Then $1 \leq r \leq d - 1$. Let F_i be a general fiber of π_i. By [BZ16, Definition 4.1 and Remark 4.2(6)], the restriction $(F_i, D_{Y_i}|_{F_i} + B_{Y_i}|_{F_i}, t_i\mathbf{M}_i|_{F_i})$ of $(Y_i, D_{Y_i} + B_{Y_i}, t_i\mathbf{M}_i)$ to F_i is generalized lc, $(\mathbf{M}_i|_{F_i})_{F_i} = (\mathbf{M}_i)_{Y_i}|_{F_i}$, and

$$K_{F_i} + D_{Y_i}|_{F_i} + B_{Y_i}|_{F_i} + t_i(\mathbf{M}_i|_{F_i})_{F_i} \equiv 0.$$

Since F_i is general, $D_{Y_i}|_{F_i}$ and $B_{Y_i}|_{F_i}$ have no common components, and their coefficients are coefficients of D_{Y_i} and B_{Y_i} respectively. By Lemma 3.2, $\mathbf{M}_i|_{F_i}$ is induced by an ample Cartier divisor, and $(\mathbf{M}_i|_{F_i})_{F_i}$ is $\mathbb{Q}$-Cartier. Since

$$-(K_{Y_i} + D_{Y_i} + B_{Y_i}(t) + t(\mathbf{M}_i)_{Y_i}) \equiv (t_i - t)G_{Y_i}$$

is ample over Z_i, $G_{Y_i}|_{F_i} = (R_{Y_i} + (\mathbf{M}_i)_{Y_i})|_{F_i}$ is ample. If $(\mathbf{M}_i|_{F_i})_{F_i} \equiv 0$, then $R_{Y_i}|_{F_i} \neq 0$, so some component P of R_{Y_i} dominates Z_i, and the components of $P|_{F_i}$ have coefficients in $\mathcal{S}_{t_i}$ in $B_{Y_i}|_{F_i}$.

Thus the generalized pairs $(F_i, D_{Y_i}|_{F_i} + B_{Y_i}|_{F_i}, t_i\mathbf{M}_i|_{F_i})$ satisfy conditions (1)–(4) with r in place of d. By the induction hypothesis and Lemma 3.5(3), $t \in \mathcal{N}_{r-1} \subset \mathcal{N}_{d-1}$.

Thus, possibly passing to a subsequence, we may assume that Z_i is a point for any i, so $\rho(Y_i) = 1$. Since $R_{Y_i} \neq 0$ or $\mathbf{M}_i \neq \mathbf{0}$, G_{Y_i} is ample by **Step 2**(b). By Lemma 3.2 and **Step 2**(b), $(Y_i, D_{Y_i} + B_{Y_i}, t_i\mathbf{M}_i)$ satisfies conditions (1)–(4). Possibly replacing $(X_i, D_i + B_i, t_i\mathbf{M}_i)$ with $(Y_i, D_{Y_i} + B_{Y_i}, t_i\mathbf{M}_i)$, we may assume that X_i is $\mathbb{Q}$-factorial and klt, $\rho(X_i) = 1$, and G_i is ample. Possibly passing to a subsequence, we may also assume that either $D_i = 0$ for any i, or $D_i \neq 0$ for any i.

Step 6. In this step, we assume that $D_i = 0$ for any i, and we show that $t \in \mathcal{N}_{d-1}$, or that, after replacing the generalized pairs, the properties in **Step 5** hold and $D_i \neq 0$ for any i.

Suppose that $(X_i, B_i, t_i\mathbf{M}_i)$ is not generalized klt for infinitely many i. By **Step 4**, applied to t_i, X_i, $B_i(0)$, the components of R_i, and $\mathbf{M}_i$, we obtain $t_i \in \mathcal{N}_{d-1}$ for infinitely many i, so $t \in \mathcal{N}_{d-1}$ by **Step 2**(c). Thus, possibly passing to a subsequence, we may assume that $(X_i, B_i, t_i\mathbf{M}_i)$ is generalized klt for any i.

Suppose that there exist a real number $\epsilon > 0$ and infinitely many i such that $(X_i, B_i, t_i\mathbf{M}_i)$ is generalized ϵ-lc. Possibly passing to a subsequence, we may assume that $(X_i, B_i, t_i\mathbf{M}_i)$ is generalized ϵ-lc for any i. Then $B_i \in [0, 1-\epsilon]$, and X_i is ϵ-lc by Definition-Proposition 2.4. Since $-K_{X_i} \equiv B_i(0) + t_iG_i$, $B_i(0) \geq 0$, $\rho(X_i) = 1$, and G_i is ample, $-K_{X_i}$ is ample. By Theorem 2.12 and Definition 2.11, there exists a positive integer v such that for any i, there exists a very ample Cartier divisor A_i on X_i with $A_i^d \leq v$.

Let C_i be the intersection of $d-1$ general members of $|A_i|$. Since X_i is normal, C_i is a smooth projective curve contained in the smooth locus of X_i [Har77, III, Corollary 10.9 and Remark 10.9.2], and C_i has at most $(A_i \cdot C_i) = A_i^d$ connected components. By adjunction [Har77, II, Proposition 8.20], $K_{C_i} = (K_{X_i} + (d-1)A_i)|_{C_i}$, and the degree of the canonical divisor of a smooth connected projective curve is at least -2 [Har77, IV, Example 1.3.3]. Thus $(K_{X_i} \cdot A_i^{d-1}) + (d-1)A_i^d \geq -2A_i^d$, so

$$0 < -(K_{X_i} \cdot A_i^{d-1}) \leq (d+1)v.$$

Any Weil divisor on X_i is Cartier along C_i. Thus $(K_{X_i} \cdot A_i^{d-1})$, $\beta_{ij} := (B_{ij} \cdot A_i^{d-1})$, and $\mu_i := ((\mathbf{M}_i)_{X_i} \cdot A_i^{d-1})$ are integers, where we use that $(\mathbf{M}_i)_{X_i}$ has integer coefficients. Moreover, $\beta_{ij} \geq 1$, and by **Step 2**(b), $\mu_i \geq 0$, with $\mu_i > 0$ if $\mathbf{M}_i \neq \mathbf{0}$. Since $t_i > 0$, any non-zero element of $\mathcal{S} \cup \mathcal{S}_{t_i}$ is at least $\theta := \min\{\frac{1}{2}, t\}$. Since

$$\sum_j b_{ij}\beta_{ij} + t_i\mu_i = -(K_{X_i} \cdot A_i^{d-1}) \leq (d+1)v,$$

the number of components of B_i, the integers β_{ij}, and the integers μ_i are bounded from above independently of i. Since $b_{ij} \leq 1-\epsilon$, we have $m_{ij} \leq \frac{1}{\epsilon}$ and $k_{ij} < \frac{1}{t}$ for $j \in J_i$, and $b_{ij} = \frac{m_{ij}-1}{m_{ij}}$ for an integer $2 \leq m_{ij} \leq \frac{1}{\epsilon}$ for $j \notin J_i$. Let $k_{ij} := 0$ for $j \notin J_i$. Possibly passing to a subsequence and renumbering the components, we may assume that the number of components of B_i, the set $J := J_i$, and the integers $m_j := m_{ij}$, $k_j := k_{ij}$, $\beta_j := \beta_{ij}$, $\mu := \mu_i$, and $(K_{X_i} \cdot A_i^{d-1})$ do not depend on i. Then

$$\Big(\sum_{j\in J} \frac{k_j\beta_j}{m_j} + \mu\Big)t_i = -(K_{X_i} \cdot A_i^{d-1}) - \sum_j \frac{(m_j-1)\beta_j}{m_j}$$

does not depend on i. Since $R_i \neq 0$ or $\mathbf{M}_i \neq \mathbf{0}$, we have $J \neq \emptyset$ or $\mu > 0$. Thus t_i does not depend on i, a contradiction.

Therefore, possibly passing to a subsequence, there exist prime divisors P_i over X_i such that

$$0 < a_i := a(P_i, X_i, B_i, t_i\mathbf{M}_i) < \frac{1}{i}$$

for any i. Let $\Lambda := \mathcal{S} \cup \{1 - a_i \mid i \in \mathbb{N}^+\}$, which is a DCC set since $\mathcal{S}$ is a DCC set and $\lim_{i\to+\infty} a_i = 0$. We show the following statement ($\diamondsuit$): for any positive integer r and any infinite set I of positive integers, there exist no lc pairs $(V_i, (1-a_i)P_i' + \Xi_i)$, $i \in I$, such that $\dim V_i = r$, $P_i' \neq 0$ is a reduced divisor without common components with $\Xi_i \in \mathcal{S}$, and $K_{V_i} + (1-a_i)P_i' + \Xi_i \equiv 0$. Otherwise, by Theorem 2.13, applied to r and Λ, the coefficient $1 - a_i$ belongs to a finite subset of Λ for any $i \in I$, which contradicts $0 < a_i < \frac{1}{i}$.

Suppose that P_i is a divisor on X_i for infinitely many i. Possibly passing to a subsequence, we may assume that P_i is a divisor on X_i for any i. Then P_i is a component of B_i with coefficient $1 - a_i$. Let $D_i' := (1-a_i)P_i$, $B_i' := B_i - D_i'$, and $R_i' := R_i - (\mathrm{mult}_{P_i} R_i)P_i$. If $\mathbf{M}_i = \mathbf{0}$ and $R_i' = 0$ for infinitely many i, then for these i, $B_i' \in \mathcal{S}$, and the lc pairs $(X_i, (1-a_i)P_i + B_i')$ with $K_{X_i} + (1-a_i)P_i + B_i' \equiv 0$ contradict ($\diamondsuit$). Thus, possibly passing to a subsequence, $R_i' \neq 0$ or $\mathbf{M}_i \neq \mathbf{0}$ for any i. Then $(X_i, D_i' + B_i', t_i\mathbf{M}_i)$ satisfies conditions (1)–(4) by **Step 2**(b), R_i' is the $\mathbb{Q}$-divisor R_i of **Step 2** for B_i', and $R_i' + (\mathbf{M}_i)_{X_i}$ is ample by **Step 2**(b). Possibly replacing D_i and B_i with D_i' and B_i', we obtain the properties in **Step 5** and $D_i \neq 0$ for any i.

Thus we may assume that P_i is exceptional over X_i for any i. Since X_i is $\mathbb{Q}$-factorial and klt, by Lemma 2.8, there exists an extremal divisorial contraction $h_i : X_i' \to X_i$ from a $\mathbb{Q}$-factorial klt projective variety X_i' such that P_i is the unique h_i-exceptional prime divisor and

$$K_{X_i'} + (1-a_i)P_i + h_{i*}^{-1}B_i + t_i(\mathbf{M}_i)_{X_i'} = h_i^*(K_{X_i} + B_i + t_i(\mathbf{M}_i)_{X_i}) \equiv 0.$$

The generalized pair $(X_i', h_{i*}^{-1}B_i, t_i\mathbf{M}_i)$ is generalized klt, and

$$K_{X_i'} + h_{i*}^{-1}B_i + t_i(\mathbf{M}_i)_{X_i'} \equiv -(1-a_i)P_i$$

is not pseudo-effective. By [BZ16, Lemma 4.4(1)], under condition (i) there, we may run a $(K_{X_i'} + h_{i*}^{-1}B_i + t_i(\mathbf{M}_i)_{X_i'})$-MMP with scaling of a general ample divisor, which terminates with a Mori fiber space $\pi_i' : Y_i' \to Z_i'$. Any step of the MMP contracts an extremal ray Γ with $(P \cdot \Gamma) > 0$, where P is the pushforward of P_i, so P_i is not contracted by **Step 2**(d), and the strict transform $P_{Y_i'}$ of P_i is ample over Z_i'. Let $B_{Y_i'}$ and $R_{Y_i'}$ be the pushforwards of $h_{i*}^{-1}B_i$ and $h_{i*}^{-1}R_i$. By **Step 2**(d) and (e), Y_i' is $\mathbb{Q}$-factorial and klt, $(Y_i', (1-a_i)P_{Y_i'} + B_{Y_i'}, t_i\mathbf{M}_i)$ is generalized klt, and $K_{Y_i'} + (1-a_i)P_{Y_i'} + B_{Y_i'} + t_i(\mathbf{M}_i)_{Y_i'} \equiv 0$.

Suppose that $\dim Z_i' > 0$ for infinitely many i. Possibly passing to a subsequence, we may assume that $\dim Z_i' > 0$ for any i and that $r := \dim Y_i' - \dim Z_i'$ does not depend on i. Then $1 \leq r \leq d-1$. Let F_i be a general fiber of π_i'. By [BZ16, Definition 4.1 and Remark 4.2(6)] and Lemma 3.2, the restrictions

$$(F_i, (1-a_i)P_{Y_i'}|_{F_i} + B_{Y_i'}|_{F_i}, t_i\mathbf{M}_i|_{F_i})$$

satisfy conditions (1)–(3) with r in place of d, and $(\mathbf{M}_i|_{F_i})_{F_i} = (\mathbf{M}_i)_{Y_i'}|_{F_i}$. If they satisfy condition (4) for infinitely many i, then $t \in \mathcal{N}_{r-1} \subset \mathcal{N}_{d-1}$ by the induction hypothesis and Lemma 3.5(3). Otherwise, possibly passing to a subsequence, $(\mathbf{M}_i)_{Y_i'}|_{F_i} \equiv 0$ and no coefficient of $B_{Y_i'}|_{F_i}$ belongs to $\mathcal{S}_{t_i}$ for any i. Then $\mathbf{M}_i|_{F_i} = \mathbf{0}$ by **Step 2**(b), and $R_{Y_i'}|_{F_i} = 0$, so $B_{Y_i'}|_{F_i} \in \mathcal{S}$. Since $P_{Y_i'}$ is ample over Z_i', $P_{Y_i'}|_{F_i} \neq 0$. Thus the lc pairs $(F_i, (1-a_i)P_{Y_i'}|_{F_i} + B_{Y_i'}|_{F_i})$ contradict ($\diamondsuit$).

Thus, possibly passing to a subsequence, we may assume that Z_i' is a point for any i, so $\rho(Y_i') = 1$. If $\mathbf{M}_i = \mathbf{0}$ and $R_{Y_i'} = 0$ for infinitely many i, then $B_{Y_i'} \in \mathcal{S}$ and the lc pairs $(Y_i', (1-a_i)P_{Y_i'} + B_{Y_i'})$ contradict ($\diamondsuit$). Thus, possibly passing to a subsequence, $R_{Y_i'} \neq 0$ or $\mathbf{M}_i \neq \mathbf{0}$ for any i, so $R_{Y_i'} + (\mathbf{M}_i)_{Y_i'}$ is ample by **Step 2**(b). By Lemma 3.2 and **Step 2**(b), $(Y_i', (1-a_i)P_{Y_i'} + B_{Y_i'}, t_i\mathbf{M}_i)$ satisfies conditions (1)–(4). Possibly replacing $(X_i, D_i + B_i, t_i\mathbf{M}_i)$ with $(Y_i', (1-a_i)P_{Y_i'} + B_{Y_i'}, t_i\mathbf{M}_i)$, we obtain the properties in **Step 5** and $D_i \neq 0$ for any i.

Step 7. In this step, we assume that $D_i \neq 0$ for any i, and we show that $t \in \mathcal{N}_{d-1}$. By **Steps 5 and 6**, this finishes the proof.

For any i, let δ_i be the minimum of the coefficients of D_i. By condition (2), $\lim_{i\to+\infty} \delta_i = 1$. We first show that, possibly passing to a subsequence, $(X_i, \lceil D_i \rceil + B_i(t), t\mathbf{M}_i)$ is generalized

lc for any i. Since X_i is $\mathbb{Q}$-factorial and $R_i \geq 0$, the formula in Definition-Proposition 2.4 implies that $(X_i, D_i + B_i(t), t\mathbf{M}_i)$ is generalized lc, and that the generalized log discrepancies of $(X_i, c\lceil D_i\rceil + B_i(t), t\mathbf{M}_i)$ are affine in the real number $c \geq 0$. Let τ_i be the generalized lc threshold of $\lceil D_i\rceil$ with respect to $(X_i, B_i(t), t\mathbf{M}_i)$. Then $\tau_i \geq \delta_i$, and $(X_i, c\lceil D_i\rceil + B_i(t), t\mathbf{M}_i)$ is generalized lc for any $0 \leq c \leq \tau_i$. Let $\Lambda_t := \mathcal{S} \cup \mathcal{S}_t \cup \{t\}$. For any real number $\eta > 0$, only finitely many elements of $\mathcal{S} \cup \mathcal{S}_t$ are smaller than $1 - \eta$, so Λ_t is a DCC set. Since $B_i(t) \in \Lambda_t$, $\lceil D_i\rceil \in \Lambda_t$, and $t\mathbf{M}_i$ descends to t times a nef Cartier divisor on a higher model of X_i, [BZ16, Theorem 1.5] implies that τ_i belongs to an ACC set depending only on d and Λ_t. Since $\tau_i \geq \delta_i$ and $\lim_{i\to+\infty} \delta_i = 1$, $\tau_i < 1$ holds for only finitely many i, as otherwise these τ_i would contain a strictly increasing sequence. Thus, possibly passing to a subsequence, $\tau_i \geq 1$, and $(X_i, \lceil D_i\rceil + B_i(t), t\mathbf{M}_i)$ is generalized lc for any i.

For any i and any real number $0 \leq s \leq t_i$, by Definition-Proposition 2.4, the generalized log discrepancies of $(X_i, \lceil D_i\rceil + B_i(s), s\mathbf{M}_i)$ are affine and non-increasing in s. Thus there exists a real number $\lambda_i \in [t, t_i]$ such that, for $0 \leq s \leq t_i$, $(X_i, \lceil D_i\rceil + B_i(s), s\mathbf{M}_i)$ is generalized lc if and only if $s \leq \lambda_i$.

Suppose that $\lambda_i < t_i$ for infinitely many i. We show that $\lambda_i \in \mathcal{N}_{d-1}$ for any such i, which implies $t \in \mathcal{N}_{d-1}$ by **Step 2**(c). Fix such an i, and let $\lambda := \lambda_i$. Let $f : X'' \to X_i$ be a log resolution of $(X_i, \mathrm{Supp}(D_i + B_i))$ such that $\mathbf{M}_i$ descends to X''. For any real number $0 \leq s \leq t_i$, write

$$K_{X''} + \Psi_s + s(\mathbf{M}_i)_{X''} := f^*(K_{X_i} + \lceil D_i\rceil + B_i(s) + s(\mathbf{M}_i)_{X_i}).$$

Then

$$\Psi_s = \Psi_0 + s\big(f^*R_i + f^*(\mathbf{M}_i)_{X_i} - (\mathbf{M}_i)_{X''}\big),$$

and $\mathrm{Supp}\,\Psi_s \subset \mathrm{Exc}(f) \cup \mathrm{Supp}\, f_*^{-1}(D_i + B_i)$, which is a simple normal crossing divisor. Since $\mathbf{M}_i$ descends to X'', for any prime divisor E over X_i, $a(E, X_i, \lceil D_i\rceil + B_i(s), s\mathbf{M}_i)$ is the log discrepancy of E with respect to the sub-pair (X'', Ψ_s). By [KM98, Corollary 2.31(3)] and [Kol13, Corollary 2.11 and (2.20)], $(X_i, \lceil D_i\rceil + B_i(s), s\mathbf{M}_i)$ is generalized lc if and only if all the coefficients of Ψ_s are at most 1. By Definition-Proposition 2.4, $f^*(\mathbf{M}_i)_{X_i} - (\mathbf{M}_i)_{X''} \geq 0$. Since $\lambda < t_i$ and Ψ_s has finitely many components, there exists a prime divisor E on X'' such that $\mathrm{mult}_E \Psi_\lambda = 1$ and

$$c_E := \mathrm{mult}_E\left(f^*R_i + f^*(\mathbf{M}_i)_{X_i} - (\mathbf{M}_i)_{X''}\right) > 0.$$

If E is a divisor on X_i, then $\mathrm{mult}_E(f^*(\mathbf{M}_i)_{X_i} - (\mathbf{M}_i)_{X''}) = 0$ by Definition-Proposition 2.4, so $\mathrm{mult}_E R_i = c_E > 0$. Thus $E = B_{ij}$ for some $j \in J_i$, and $\mathrm{mult}_E \Psi_\lambda = \frac{m_{ij}-1+k_{ij}\lambda}{m_{ij}} = 1$, so $\lambda = \frac{1}{k_{ij}}$. Thus $\lambda \in \mathcal{N}_0 \subset \mathcal{N}_{d-1}$ by Lemma 3.5(3).

Otherwise, E is exceptional over X_i, and $a(E, X_i, \lceil D_i\rceil + B_i(\lambda), \lambda\mathbf{M}_i) = 0$. By Lemma 2.8, there exists an extremal divisorial contraction $h : X' \to X_i$ from a $\mathbb{Q}$-factorial klt projective variety X' such that E is the unique h-exceptional prime divisor,

$$K_{X'} + h_*^{-1}(\lceil D_i\rceil + B_i(\lambda)) + E + \lambda(\mathbf{M}_i)_{X'} = h^*(K_{X_i} + \lceil D_i\rceil + B_i(\lambda) + \lambda(\mathbf{M}_i)_{X_i}),$$

and $-E$ is ample over X_i. Since $h_*(\mathbf{M}_i)_{X'} = (\mathbf{M}_i)_{X_i}$, $h^*(\mathbf{M}_i)_{X_i} - (\mathbf{M}_i)_{X'}$ is h-exceptional, and since $e(E, X', \mathbf{M}_i) = 0$, Definition-Proposition 2.4 implies that the coefficient of $h^*(\mathbf{M}_i)_{X_i} - (\mathbf{M}_i)_{X'}$ along E is $e(E, X_i, \mathbf{M}_i) = \mathrm{mult}_E(f^*(\mathbf{M}_i)_{X_i} - (\mathbf{M}_i)_{X''})$. Thus

$$h^*G_i = h_*^{-1}R_i + (\mathbf{M}_i)_{X'} + c_E E,$$

so $h_*^{-1}R_i + (\mathbf{M}_i)_{X'}$ is ample over X_i. Let E^ν be the normalization of E, $E^\nu \to Z$ the Stein factorization of the induced morphism from E^ν to $h(E)$, and F a general fiber of $E^\nu \to Z$. Since E is exceptional over X_i, $\dim F \geq 1$. Since h maps the image of F in X' to a point,

$$\big(K_{X'} + h_*^{-1}(\lceil D_i\rceil + B_i(\lambda)) + E + \lambda(\mathbf{M}_i)_{X'}\big)|_F \equiv 0,$$

and $(h_*^{-1}R_i + (\mathbf{M}_i)_{X'})|_F$ is ample [Laz04, Theorem 1.7.8 and Proposition 1.2.13]. By **Step 3**(2), applied to λ, X', $h_*^{-1}(\lceil D_i\rceil + B_i(0)) + E$, the components of $h_*^{-1}R_i$, $\mathbf{M}_i$, E, and F, we obtain $\lambda \in \mathcal{N}_{\dim F} \subset \mathcal{N}_{d-1}$.

Thus, possibly passing to a subsequence, we may assume that $\lambda_i = t_i$, that is, $(X_i, \lceil D_i \rceil + B_i(s), s\mathbf{M}_i)$ is generalized lc for any $0 \leq s \leq t_i$ and any i. Since $\rho(X_i) = 1$, G_i is ample, and $\lceil D_i \rceil - D_i \geq 0$, there exists a real number $\kappa_i \geq 0$ such that $\lceil D_i \rceil - D_i \equiv \kappa_i G_i$. Let $\sigma_i := t_i - \kappa_i$. Since $K_{X_i} + D_i + B_i + t_i(\mathbf{M}_i)_{X_i} \equiv 0$ and $B_i - B_i(s) + (t_i - s)(\mathbf{M}_i)_{X_i} = (t_i - s)G_i$, we have

$$K_{X_i} + \lceil D_i \rceil + B_i(s) + s(\mathbf{M}_i)_{X_i} \equiv (s - \sigma_i)G_i$$

for any $0 \leq s \leq t_i$.

Suppose that $\sigma_i < t$ for infinitely many i. Possibly passing to a subsequence, we may assume that $\sigma_i < t$ for any i. Then $\kappa_i > t_i - t > 0$, so $\lceil D_i \rceil \neq D_i$. Let $r_i := \frac{t_i - t}{\kappa_i} \in (0, 1)$. Since $K_{X_i} + D_i + B_i(t) + t(\mathbf{M}_i)_{X_i} \equiv -(t_i - t)G_i$,

$$K_{X_i} + r_i\lceil D_i \rceil + (1 - r_i)D_i + B_i(t) + t(\mathbf{M}_i)_{X_i} \equiv \big(r_i\kappa_i - (t_i - t)\big)G_i = 0.$$

Since $r_i\lceil D_i \rceil + (1 - r_i)D_i \leq \lceil D_i \rceil$, $(X_i, r_i\lceil D_i \rceil + (1 - r_i)D_i + B_i(t), t\mathbf{M}_i)$ is generalized lc. Let Λ' be the union of Λ_t and the set of the coefficients of $r_i\lceil D_i \rceil + (1 - r_i)D_i$ for all i. For any i, the coefficients of $r_i\lceil D_i \rceil + (1 - r_i)D_i$ are at least δ_i, and $\lim_{i \to +\infty} \delta_i = 1$, so Λ' is a DCC set. By Theorem 2.13, applied to d, Λ', and $\mu = t$ if $\mathbf{M}_i \neq \mathbf{0}$ and $\mu = 0$ if $\mathbf{M}_i = \mathbf{0}$, the coefficients of $r_i\lceil D_i \rceil + (1 - r_i)D_i$ belong to a finite subset of Λ' for any i. However, any component of D_i with coefficient $d_i < 1$ has coefficient $r_i + (1 - r_i)d_i \in [\delta_i, 1)$ in $r_i\lceil D_i \rceil + (1 - r_i)D_i$, and $\lim_{i \to +\infty} \delta_i = 1$, a contradiction.

Thus, possibly passing to a subsequence, we may assume that $t \leq \sigma_i \leq t_i$ for any i. Then $(X_i, \lceil D_i \rceil + B_i(\sigma_i), \sigma_i\mathbf{M}_i)$ is generalized lc and

$$K_{X_i} + \lceil D_i \rceil + B_i(\sigma_i) + \sigma_i(\mathbf{M}_i)_{X_i} \equiv 0.$$

Let S_i be a component of $\lceil D_i \rceil$ and S_i^ν the normalization of S_i. Then $S_i \not\subset \operatorname{Supp} R_i$ by condition (2), $\dim S_i^\nu = d - 1 \geq 1$, and $G_i|_{S_i^\nu}$ is ample. By **Step 3**(2), applied to σ_i, X_i, $\lceil D_i \rceil + B_i(0)$, the components of R_i, $\mathbf{M}_i$, S_i, and Z a point, we obtain $\sigma_i \in \mathcal{N}_{d-1}$. By **Step 2**(c), $t \in \mathcal{N}_{d-1}$. □

Lemma 3.7. *Let d be a positive integer. Then*

$$\operatorname{Acc}(\mathcal{N}_d) \cap (0, +\infty) \subset \mathcal{N}_{d-1}.$$

Moreover, for any non-negative integer d, if $t_i \in \mathcal{N}_d$ and $\lim_{i \to +\infty} t_i = t > 0$, then $t \in \mathcal{N}_d$.

Proof. A generalized pair realizing an element of $\mathcal{N}_d$ satisfies condition (†) of [HL21, Section 3.1] for $I = \{0, 1\}$: conditions (1) and (2) of Definition 3.3 coincide with conditions (1) and (2) of [HL21, Section 3.1], and conditions (3) and (4) of Definition 3.3 imply conditions (3) and (4) of [HL21, Section 3.1], where for a higher model X'' of X on which $\mathbf{M}$ descends, since $\mathbf{M}_X$ is $\mathbb{Q}$-Cartier, $\mathbf{M}_{X''} \equiv 0$ if and only if $\mathbf{M}_X \equiv 0$ by the negativity lemma [KM98, Lemma 3.39] (cf. the remark after [Li21, (2.5)]). Thus $\mathcal{N}_d$ is an ACC set by [HL21, Lemma 3.6].

Let $t_i \in \mathcal{N}_d$ be pairwise distinct real numbers such that $\lim_{i \to +\infty} t_i = t > 0$. Possibly passing to a subsequence, we may assume that t_i is monotonic. Since $\mathcal{N}_d$ is an ACC set and the t_i are pairwise distinct, t_i is strictly decreasing. For each i, let $(X_i, B_i, t_i\mathbf{M}_i)$ be a generalized pair satisfying condition (†) with $1 \leq \dim X_i \leq d$. Possibly passing to a subsequence, we may assume that $\dim X_i = r$ for any i. By Proposition 3.6 with $D_i = 0$, $t \in \mathcal{N}_{r-1}$, so $t \in \mathcal{N}_{d-1}$ by Lemma 3.5(3).

For the second part, let $t_i \in \mathcal{N}_d$ such that $\lim_{i \to +\infty} t_i = t > 0$. If $\{t_i\}_i$ is a finite set, then $t = t_i$ for some i, and $t \in \mathcal{N}_d$. Otherwise, $\{t_i\}_i$ contains a sequence of pairwise distinct elements converging to t. If $d = 0$, this is impossible since $\mathcal{N}_0 \cap (\frac{t}{2}, +\infty)$ is finite. If $d \geq 1$, then $t \in \mathcal{N}_{d-1} \subset \mathcal{N}_d$ by the first part and Lemma 3.5(3). □

Lemma 3.8. *Let d be a non-negative integer. Then $\mathcal{N}_d \subset (0, d + 1]$.*

Proof. Since $\mathcal{N}_0 = \{\frac{1}{m} \mid m \in \mathbb{N}^+\} \subset (0, 1]$, we may assume that $d \geq 1$. Let $t \in \mathcal{N}_d$ and $I := \{0, 1\}$. The set I is finite, $I = I_+$, and $1 \in I$, so $N_d(I) = K_d(I)$ by [Li21, (3.1)], where $K_d(I)$ is the set of real numbers c such that there exists a generalized pair of dimension at most d that satisfies condition (†) of [Li21, Section 2.3] for I and c, is generalized klt, and has Picard number one

[Li21, (2.5) and (2.6)]. By Lemma 3.5(1)(2) and the definition of $K_d(I)$, there exists a generalized pair $(X, B, t\mathbf{M})$ satisfying condition (†) such that $(X, B, t\mathbf{M})$ is generalized klt, $\rho(X) = 1$, and $1 \leq \dim X \leq d$.

Replacing the smooth higher model in Definition 3.1 with a log resolution of $(X, \operatorname{Supp} B)$ dominating it (Lemma 3.2), we may assume that there exist a log resolution $f : X' \to X$ of $(X, \operatorname{Supp} B)$, a contraction $g : X' \to Z$ to a normal projective variety, and an ample Cartier divisor H on Z such that $\mathbf{M} = \overline{g^*H}$. If Z is a point, then $\mathbf{M} = \mathbf{0}$. There are two cases.

Case 1. $\mathbf{M}_X \equiv 0$. By condition (†)(4), some coefficient b of B is of the form $\frac{m-1+f+kt}{m}$ with $m, k \in \mathbb{N}^+$ and $f \in \{0, 1\}$. Since $(X, B, t\mathbf{M})$ is generalized klt, $b < 1$, so $f + kt < 1$. Thus $t < 1 \leq d + 1$.

Case 2. $\mathbf{M}_X \not\equiv 0$. Then Z is not a point. Let $l \in \mathbb{N}^+$ such that $|lH|$ is base point free, and let $H_l \in |lH|$ be a general member. Then $\mathbf{M}_X = f_*g^*H \sim_{\mathbb{Q}} \frac{1}{l} f_*g^*H_l \geq 0$, and $f_*g^*H_l \neq 0$ since $\mathbf{M}_X \not\equiv 0$. Since $\rho(X) = 1$ and $\mathbf{M}_X$ is $\mathbb{Q}$-Cartier, $\mathbf{M}_X$ is ample. In particular, for any real number $0 \leq s < t$,

$$K_X + B + s\mathbf{M}_X \equiv -(t - s)\mathbf{M}_X$$

is not pseudo-effective, while $K_X + B + t\mathbf{M}_X \equiv 0$ is pseudo-effective.

If $\dim Z = \dim X$, then g is birational and g^*H is a nef and big Cartier divisor. By [Li21, Proposition 3.2] applied to the generalized pair $(X, B, \mathbf{M})$, where $\mathbf{M}_X$ is $\mathbb{Q}$-Cartier and $\mathbf{M} = \overline{g^*H}$ with g^*H a nef and big Cartier divisor on X', the infimum of all $s \geq 0$ such that $K_X + B + s\mathbf{M}_X$ is pseudo-effective is at most $\dim X + 1$. By the previous paragraph, the infimum equals t, so $t \leq \dim X + 1 \leq d + 1$.

Suppose that $0 < \dim Z < \dim X$. Write

$$K_{X'} + B' + t\mathbf{M}_{X'} := f^*(K_X + B + t\mathbf{M}_X),$$

let $\tilde{B}$ be the sum of the components of B' with positive coefficients, counted with their coefficients, and let $G := \tilde{B} - B'$. Then $G \geq 0$ is f-exceptional, G and $\tilde{B}$ have no common components, and

$$K_{X'} + \tilde{B} + t\mathbf{M}_{X'} = f^*(K_X + B + t\mathbf{M}_X) + G \equiv G.$$

Since $(X, B, t\mathbf{M})$ is generalized klt and f is a log resolution of $(X, \operatorname{Supp} B)$, $\tilde{B}$ is an SNC divisor with coefficients in $[0, 1)$, so $(X', \tilde{B})$ is klt. For any real number $0 \leq s < t$, $f_*(K_{X'} + \tilde{B} + s\mathbf{M}_{X'}) = K_X + B + s\mathbf{M}_X$ is not pseudo-effective, and f_* maps pseudo-effective divisors to pseudo-effective divisors, so $K_{X'} + \tilde{B} + s\mathbf{M}_{X'}$ is not pseudo-effective. Since $K_{X'} + \tilde{B} + t\mathbf{M}_{X'} \equiv G \geq 0$, the infimum of all $s \geq 0$ such that $K_{X'} + \tilde{B} + s\mathbf{M}_{X'}$ is pseudo-effective is $t > 0$. For a very general point $z \in Z$, let X'_z be the fiber of g over z. Then $\mathbf{M}_{X'}|_{X'_z} = (g^*H)|_{X'_z} \equiv 0$, so

$$(K_{X'} + \tilde{B})|_{X'_z} \equiv G|_{X'_z} \geq 0$$

is pseudo-effective. By [Li21, Proposition 3.4] applied to $g : X' \to Z$, the klt pair $(X', \tilde{B})$, and the big divisor H, for a resolution $\mu : \tilde{Z} \to Z$, the infimum of all $s \geq 0$ such that $K_{\tilde{Z}} + s\mu^*H$ is pseudo-effective is at least t (the threshold of [Li21, Definition 1.1] coincides here with the pseudo-effective one, since $K_{X'} + \tilde{B} + s\mathbf{M}_{X'} \equiv G + (s - t)g^*H$ is effective for $s \geq t$ and μ^*H is big). Since μ^*H is a nef and big Cartier divisor on the smooth variety $\tilde{Z}$, by [Li21, Proposition 3.2] applied to $(\tilde{Z}, 0)$ and μ^*H, the infimum of all $s \geq 0$ such that $K_{\tilde{Z}} + s\mu^*H$ is pseudo-effective is at most $\dim Z + 1$. Thus $t \leq \dim Z + 1 \leq \dim X \leq d$. □

The next two lemmas place a parameter t in $\mathcal{N}_r$ for r smaller than the dimension of a given generalized pair. Lemma 3.9 restricts to a general fiber of a contraction of the variety, or, after generalized adjunction, of the normalization of a component of $\lfloor B \rfloor$, and Lemma 3.10 reduces generalized pairs that are not generalized klt to it. In case (2) of Lemma 3.9, although $\mathbf{M}_X|_F \not\equiv 0$, the trace of the restricted nef part on a $\mathbb{Q}$-factorial modification of F may be numerically trivial, and then the parameter survives only in the boundary coefficients (the last step of the proof).

Lemma 3.9. *Let $t > 0$ be a real number and $(X, B, t\mathbf{M})$ a generalized lc pair such that X is $\mathbb{Q}$-factorial, $B \in \mathcal{S} \cup \mathcal{S}_t$, and $\mathbf{M}$ is induced by an ample Cartier divisor. Let F be one of the following:*

(1) a general fiber of a contraction $X \to Z$, or
(2) a general fiber of a contraction $S^\nu \to Z$, where S^ν is the normalization of a component S of $\lfloor B \rfloor$, and Z may be a point.

Assume that $\dim F \geq 1$, $(K_X + B + t\mathbf{M}_X)|_F \equiv 0$, and $\mathbf{M}_X|_F \not\equiv 0$. Then $t \in \mathcal{N}_{\dim F}$.

Proof. The proof is long, so we divide it into several steps. Let $r := \dim F$.

Step 1. In this step, we construct a generalized pair $(F, B_F, t\mathbf{M}|_F)$ and show that it is generalized lc, $K_F + B_F + t(\mathbf{M}|_F)_F \equiv 0$, and $\mathbf{M}|_F$ is induced by an ample Cartier divisor.

Replacing the smooth higher model in Definition 3.1 with a log resolution of $(X, \operatorname{Supp} B)$ dominating it (Lemma 3.2), we may assume that there exist a log resolution $f : X' \to X$ of $(X, \operatorname{Supp} B)$, a contraction $g : X' \to T$ to a normal projective variety, and an ample Cartier divisor H on T such that $\mathbf{M} = \overline{g^*H}$. Write

$$f^*\mathbf{M}_X = \mathbf{M}_{X'} + E_{\mathbf{M}},$$

where $\mathbf{M}_{X'} = g^*H$. By Definition-Proposition 2.4, $E_{\mathbf{M}} \geq 0$ is f-exceptional. Write

$$K_{X'} + B' + t\mathbf{M}_{X'} := f^*(K_X + B + t\mathbf{M}_X).$$

In case (1), let F' be the fiber of the induced morphism $X' \to Z$ corresponding to F, and let $(F, B_F, t\mathbf{M}|_F)$ be the restriction of $(X, B, t\mathbf{M})$ to F [BZ16, Remark 4.2(6)]. In case (2), let $S' := f_*^{-1}S$, let $f_S : S' \to S^\nu$ be the induced morphism, and let $(S^\nu, B_{S^\nu}, t\mathbf{M}|_{S^\nu})$ be the generalized adjunction of $(X, B, t\mathbf{M})$ to S (Definition-Proposition 2.5). Let F' be the fiber of the induced morphism $S' \to Z$ corresponding to F, and let $(F, B_F, t\mathbf{M}|_F)$ be the restriction of $(S^\nu, B_{S^\nu}, t\mathbf{M}|_{S^\nu})$ to F [BZ16, Remark 4.2(6)].

In both cases, since F is general, F' is smooth [Har77, III, Corollary 10.7], the induced morphism $\pi : F' \to F$ is a projective birational morphism, and $\operatorname{Supp} B_{F'}$ is a simple normal crossing divisor, where $B_{F'} := B'|_{F'}$ in case (1) and $B_{F'} := (B' - S')|_{F'}$ in case (2). Moreover, $\mathbf{M}|_F$ descends to F' with $(\mathbf{M}|_F)_{F'} = \mathbf{M}_{X'}|_{F'}$, $(F, B_F, t\mathbf{M}|_F)$ is generalized lc, and

$$K_F + B_F + t(\mathbf{M}|_F)_F = (K_X + B + t\mathbf{M}_X)|_F \equiv 0.$$

In case (1), we also have $(\mathbf{M}|_F)_F = \mathbf{M}_X|_F$ [BZ16, Remark 4.2(6)]. By Lemma 3.2 applied to the morphism $F' \to X'$ and to g, there exist a contraction $g_{F'} : F' \to T_{F'}$ and a finite morphism $h : T_{F'} \to T$ such that $\mathbf{M}_{X'}|_{F'} = g_{F'}^*h^*H$ and h^*H is an ample Cartier divisor. Thus $\mathbf{M}|_F = \overline{g_{F'}^*h^*H}$ is induced by an ample Cartier divisor. By Lemma 3.2 applied to a resolution of any higher model of F dominating F', $\mathbf{M}|_F$ is also induced by an ample Cartier divisor when it is regarded as a nef $\boldsymbol{b}$-divisor over any higher model of F.

Step 2. In this step, we show that $B_F \in \mathcal{S} \cup \mathcal{S}_t$. In case (2), we also show that the coefficient of a prime divisor V on S^ν in B_{S^ν} belongs to $\mathcal{S}_t$ if V is a component of $(f_S)_*(E_{\mathbf{M}}|_{S'})$.

Since F is general, any coefficient of B_F is a coefficient of B in case (1), and a coefficient of B_{S^ν} in case (2). Thus we may assume that we are in case (2). Since S' is not f-exceptional, $E_{\mathbf{M}}|_{S'}$ is defined and $E_{\mathbf{M}}|_{S'} \geq 0$. Let V be a prime divisor on S^ν, let b_V be the coefficient of V in B_{S^ν}, and let e_V be the coefficient of V in $(f_S)_*(E_{\mathbf{M}}|_{S'})$. If $b_V < 1$, then by [HL21, Definition 2.2 and (2)],

$$b_V = \frac{m - 1 + \sum_j n_j b_j + kt}{m},$$

where $m \in \mathbb{N}^+$ is the Cartier index along V, n_j and k are non-negative integers, b_j are coefficients of B, and $e_V = \frac{k}{m}$. Here the Cartier index along V is the order of the local class group of X at the generic point of the image of V in X, which is a cyclic group [Kol+92, (16.6.3)]. We show that b_V has such an expression also if $b_V = 1$ and $e_V > 0$. Since $f^*\mathbf{M}_X = \mathbf{M}_{X'} + E_{\mathbf{M}}$, we have $f^*(K_X + B) = K_{X'} + B' - tE_{\mathbf{M}}$, and (X, B) is lc by Definition-Proposition 2.4. The coefficient

of V in $(f_S)_*((B' - tE_{\mathbf{M}} - S')|_{S'})$ is $b_V - te_V < 1$, so [HL21, Definition 2.2 and (2)] applied to (X, B) implies that

$$b_V - te_V = \frac{m - 1 + \sum_j n_j b_j}{m},$$

where m is the Cartier index along V. By the choice of m, m times any Weil divisor on X is Cartier at the generic point of the image of V in X. Since $m\mathbf{M}_X$ and $\mathbf{M}_{X'}$ are Cartier over the generic point of V, $mE_{\mathbf{M}}$ is Cartier over the generic point of V, so me_V is a non-negative integer k.

By Lemma 3.4, $b_V \in \mathcal{S} \cup \mathcal{S}_t$, and $b_V \in \mathcal{S}_t$ if $e_V > 0$. Since $1 \in \mathcal{S}$, $B_{S^\nu} \in \mathcal{S} \cup \mathcal{S}_t$, and **Step 2** follows.

Step 3. In this step, we conclude the proof in case (1), and in case (2) we construct a generalized pair of dimension r satisfying conditions (1)–(3) of Definition 3.3.

In case (1), $(\mathbf{M}|_F)_F = \mathbf{M}_X|_F$ is $\mathbb{Q}$-Cartier since X is $\mathbb{Q}$-factorial, and $\mathbf{M}_X|_F \not\equiv 0$. By **Steps 1 and 2**, $(F, B_F, t\mathbf{M}|_F)$ satisfies condition (†). Since $\dim F = r \geq 1$, $t \in \mathcal{N}_r$.

In case (2), by [BZ16, Lemma 4.5] (cf. [HL21, Proposition 2.4(2)]), applied with an empty list of divisors to extract, there exists a $\mathbb{Q}$-factorial modification $\rho : \tilde{F} \to F$, that is, a projective birational morphism such that $\tilde{F}$ is $\mathbb{Q}$-factorial and, writing

$$K_{\tilde{F}} + B_{\tilde{F}} + t(\mathbf{M}|_F)_{\tilde{F}} := \rho^*(K_F + B_F + t(\mathbf{M}|_F)_F),$$

any ρ-exceptional prime divisor is a component of $\lfloor B_{\tilde{F}} \rfloor$, that is, a generalized lc place of $(F, B_F, t\mathbf{M}|_F)$. Then $B_{\tilde{F}}$ is the sum of the strict transform of B_F and the reduced ρ-exceptional divisor. By **Steps 1 and 2**, $(\tilde{F}, B_{\tilde{F}}, t\mathbf{M}|_F)$ is generalized lc, $K_{\tilde{F}} + B_{\tilde{F}} + t(\mathbf{M}|_F)_{\tilde{F}} \equiv 0$, $B_{\tilde{F}} \in \mathcal{S} \cup \mathcal{S}_t$, $\mathbf{M}|_F$ is induced by an ample Cartier divisor, and $(\mathbf{M}|_F)_{\tilde{F}}$ is $\mathbb{Q}$-Cartier. If $(\mathbf{M}|_F)_{\tilde{F}} \not\equiv 0$, then $(\tilde{F}, B_{\tilde{F}}, t\mathbf{M}|_F)$ satisfies condition (†), so $t \in \mathcal{N}_r$.

Step 4. In this step, we show that in case (2), if $(\mathbf{M}|_F)_{\tilde{F}} \equiv 0$, then some coefficient of $B_{\tilde{F}}$ belongs to $\mathcal{S}_t$. This finishes the proof.

Let A be an ample Cartier divisor on F. Since $(\mathbf{M}|_F)_F = \rho_*(\mathbf{M}|_F)_{\tilde{F}}$, by the projection formula [Ful98, Proposition 2.5(c)],

$$((\mathbf{M}|_F)_F \cdot A^{r-1}) = ((\mathbf{M}|_F)_{\tilde{F}} \cdot \rho^* A^{r-1}) = 0,$$

where the intersection number of the Weil divisor $(\mathbf{M}|_F)_F$ with A^{r-1} is computed as cycles. We have $(f^*\mathbf{M}_X)|_{F'} = \pi^*(\mathbf{M}_X|_F)$ and $\pi_*(\mathbf{M}_{X'}|_{F'}) = (\mathbf{M}|_F)_F$. Restricting $f^*\mathbf{M}_X = \mathbf{M}_{X'} + E_{\mathbf{M}}$ to F' and intersecting with $\pi^* A^{r-1}$, the projection formula implies

$$(E_{\mathbf{M}}|_{F'} \cdot \pi^* A^{r-1}) = (\mathbf{M}_X|_F \cdot A^{r-1}) - ((\mathbf{M}|_F)_F \cdot A^{r-1}) = (\mathbf{M}_X|_F \cdot A^{r-1}).$$

We show that $(\mathbf{M}_X|_F \cdot A^{r-1}) > 0$. Let $l \in \mathbb{N}^+$ such that $|lH|$ is base point free, let $H_l \in |lH|$ be a general member, and let $D := \frac{1}{l} f_* g^* H_l$. Then $D \geq 0$ and $D \sim_{\mathbb{Q}} \mathbf{M}_X$. Since $g^* H_l$ is a general member of a base point free linear system on X', $S' \not\subset \operatorname{Supp} g^* H_l$, so $S \not\subset \operatorname{Supp} D$. Since X is $\mathbb{Q}$-factorial and F is general, $D|_F \geq 0$. Since $\mathbf{M}_X|_F \not\equiv 0$, $D|_F \neq 0$. Thus

$$(\mathbf{M}_X|_F \cdot A^{r-1}) = (D|_F \cdot A^{r-1}) > 0.$$

Since $E_{\mathbf{M}}|_{S'} \geq 0$ and F is general, $E_{\mathbf{M}}|_{F'} \geq 0$. By the projection formula, $(P \cdot \pi^* A^{r-1}) = 0$ for any π-exceptional prime divisor P on F'. Since $(E_{\mathbf{M}}|_{F'} \cdot \pi^* A^{r-1}) > 0$, some component P' of $E_{\mathbf{M}}|_{F'}$ is not π-exceptional, so $V_F := \pi(P')$ is a prime divisor on F with positive coefficient in $\pi_*(E_{\mathbf{M}}|_{F'})$. Since F is general, $\pi_*(E_{\mathbf{M}}|_{F'}) = ((f_S)_*(E_{\mathbf{M}}|_{S'}))|_F$, so V_F is a component of $V|_F$ for some component V of $(f_S)_*(E_{\mathbf{M}}|_{S'})$. By **Step 2**, the coefficient b_V of V in B_{S^ν} belongs to $\mathcal{S}_t$. Since F is general, b_V is the coefficient of V_F in B_F, and of the strict transform of V_F in $B_{\tilde{F}}$. Thus $(\tilde{F}, B_{\tilde{F}}, t\mathbf{M}|_F)$ satisfies condition (†), and $t \in \mathcal{N}_r$. □

Lemma 3.10. *Let $d \geq 2$ be an integer, $t > 0$ a real number, and $(X, B, t\mathbf{M})$ a generalized pair of dimension d satisfying condition (†) such that $\mathbf{M}_X \not\equiv 0$. Assume that $(X, B, t\mathbf{M})$ is not generalized klt. Then $t \in \mathcal{N}_{d-1}$.*

Proof. Since $(X, B, t\mathbf{M})$ is generalized lc but not generalized klt, it has a generalized lc place E. By [HL21, Proposition 2.4], applied to the list $\{E\}$ if E is exceptional over X, and to the empty list otherwise, there exists a $\mathbb{Q}$-factorial modification $h : X_1 \to X$ such that $(X_1, 0)$ is klt and E is a divisor on X_1. Write

$$K_{X_1} + B_1 + t\mathbf{M}_{X_1} := h^*(K_X + B + t\mathbf{M}_X).$$

Any h-exceptional prime divisor other than E is a component of $\lfloor B_1 \rfloor$ [HL21, Proposition 2.4(2)], and E is a generalized lc place, so any h-exceptional prime divisor is a generalized lc place of $(X, B, t\mathbf{M})$. Thus B_1 is the sum of the strict transform of B and the reduced h-exceptional divisor, so $B_1 \in \mathcal{S} \cup \mathcal{S}_t$ and E is a component of $\lfloor B_1 \rfloor$. Moreover, $(X_1, B_1, t\mathbf{M})$ is generalized lc and $K_{X_1} + B_1 + t\mathbf{M}_{X_1} \equiv 0$. Since X_1 is $\mathbb{Q}$-factorial, (X_1, B_1) is lc by Definition-Proposition 2.4.

Let $f : X' \to X_1$, $g : X' \to T$, and H be as in Definition 3.1 for $\mathbf{M}$ over X_1. Let $l \in \mathbb{N}^+$ such that $|lH|$ is base point free, and let $H_l \in |lH|$ be a general member. Then $\mathbf{M}_{X_1} \sim_{\mathbb{Q}} D_1 := \frac{1}{l} f_* g^* H_l \geq 0$. Since $h_* D_1 \sim_{\mathbb{Q}} \mathbf{M}_X \not\equiv 0$, $D_1 \neq 0$. Thus $-\mathbf{M}_{X_1}$ is not pseudo-effective, so

$$K_{X_1} + B_1 \equiv -t\mathbf{M}_{X_1}$$

is not pseudo-effective. By [BZ16, Lemma 4.4(1)], applied to the lc pair (X_1, B_1) with $(X_1, 0)$ klt, we may run a $(K_{X_1} + B_1)$-MMP with scaling of an ample divisor, which terminates with a Mori fiber space $\pi : Y \to Z$. Let $\phi : X_1 \dashrightarrow Y$ be the induced birational map, and let $B_Y := \phi_* B_1$.

Since $K_{X_1} + B_1 + t\mathbf{M}_{X_1} \equiv 0$, each step of the MMP is $(K_{X_1} + B_1 + t\mathbf{M}_{X_1})$-trivial. Thus, for any variety X_j appearing in the MMP and the pushforward B_j of B_1 to X_j, X_j is $\mathbb{Q}$-factorial, $(X_j, B_j, t\mathbf{M})$ is generalized lc, $K_{X_j} + B_j + t\mathbf{M}_{X_j} \equiv 0$, $B_j \in \mathcal{S} \cup \mathcal{S}_t$, and $\mathbf{M}$ is induced by an ample Cartier divisor. Since $-(K_{X_j} + B_j) \equiv t\mathbf{M}_{X_j}$, $(\mathbf{M}_{X_j} \cdot C) > 0$ for any curve C contracted by the contraction of the extremal ray of the step $X_j \dashrightarrow X_{j+1}$, and $\mathbf{M}_Y$ is ample over Z. There are three cases.

Case 1. $\dim Z > 0$. Let F be a general fiber of π. Then $1 \leq \dim F \leq d-1$, $(K_Y + B_Y + t\mathbf{M}_Y)|_F \equiv 0$, and $\mathbf{M}_Y|_F$ is ample, so $\mathbf{M}_Y|_F \not\equiv 0$. By Lemma 3.9(1), $t \in \mathcal{N}_{\dim F}$.

Case 2. $\dim Z = 0$ and ϕ contracts a component of $\lfloor B_1 \rfloor$. Then there exists a step $X_j \to X_{j+1}$ of the MMP which is a divisorial contraction whose exceptional divisor S_j is a component of $\lfloor B_j \rfloor$. Let S_j^ν be the normalization of S_j, let $S_j^\nu \to Z_j$ be the Stein factorization of the induced morphism from S_j^ν to the image of S_j in X_{j+1}, and let F be a general fiber of $S_j^\nu \to Z_j$. Since S_j is contracted, $1 \leq \dim F \leq d-1$. Any curve $C \subset F$ maps to a curve contracted by $X_j \to X_{j+1}$, so $(\mathbf{M}_{X_j}|_F \cdot C) > 0$ and $\mathbf{M}_{X_j}|_F \not\equiv 0$. Moreover, $(K_{X_j} + B_j + t\mathbf{M}_{X_j})|_F \equiv 0$. By Lemma 3.9(2), $t \in \mathcal{N}_{\dim F}$.

Case 3. $\dim Z = 0$ and ϕ contracts no component of $\lfloor B_1 \rfloor$. Since E is a component of $\lfloor B_1 \rfloor$ and ϕ contracts no component of $\lfloor B_1 \rfloor$, $\lfloor B_Y \rfloor \neq 0$. Let S_Y be a component of $\lfloor B_Y \rfloor$ and S_Y^ν the normalization of S_Y. Since $\rho(Y) = 1$, $\mathbf{M}_Y$ is ample, so $\mathbf{M}_Y|_{S_Y^\nu} \not\equiv 0$. Moreover, $\dim S_Y^\nu = d-1 \geq 1$ and $(K_Y + B_Y + t\mathbf{M}_Y)|_{S_Y^\nu} \equiv 0$. By Lemma 3.9(2) with Z a point, $t \in \mathcal{N}_{d-1}$.

In all cases, $t \in \mathcal{N}_{d-1}$ by Lemma 3.5(3). □

Lemma 3.10 is the dimension-decreasing part of [HL21, Lemma 3.3] (cf. [Li21, (3.1)]).

4. Generalized lc thresholds

In this section, we prove Propositions 4.2 and 4.4 and Corollary 4.3. The constant η of Corollary 4.3 is used in the proof of Theorem 8.2.

A finite generalized lc threshold of $\mathbf{M}$ is computed by a prime divisor E exceptional over X with $e(E, X, \mathbf{M}) > 0$, and after extracting E, the trace of $\mathbf{M}$ is ample over X. Hence $\mathbf{M}$ is not numerically trivial on a general fiber of E over the center of E on X.

Lemma 4.1. *Let X be a $\mathbb{Q}$-factorial klt projective variety, $\mathbf{M}$ a nef $\boldsymbol{b}$-divisor over X, and $B \geq 0$ an $\mathbb{R}$-divisor on X such that (X, B) is lc. Let λ be the generalized lc threshold of $\mathbf{M}$ with respect to (X, B), and assume that $\lambda < +\infty$. Then $(X, B, \lambda\mathbf{M})$ is generalized lc, and there exists a prime*

divisor E over X such that E is exceptional over X, $e(E, X, \mathbf{M}) > 0$, and $a(E, X, B, \lambda\mathbf{M}) = 0$. Moreover, for any such E, there exists an extremal divisorial contraction $h : X_1 \to X$ from a $\mathbb{Q}$-factorial klt projective variety X_1 such that E is the unique h-exceptional prime divisor,

$$K_{X_1} + h_*^{-1}B + E + \lambda\mathbf{M}_{X_1} = h^*(K_X + B + \lambda\mathbf{M}_X),$$

$h^*\mathbf{M}_X = \mathbf{M}_{X_1} + e(E, X, \mathbf{M})E$, and $\mathbf{M}_{X_1}$ is ample over X.

Proof. Since X is $\mathbb{Q}$-factorial, K_X, B, and $\mathbf{M}_X$ are $\mathbb{R}$-Cartier, and $(X, B, s\mathbf{M})$ is a generalized pair for any real number $s \geq 0$. By Definition-Proposition 2.4, applied to s and to 0, for any prime divisor E over X and any real number $s \geq 0$, we have $e(E, X, \mathbf{M}) \geq 0$ and $a(E, X, B, s\mathbf{M}) = a(E, X, B) - se(E, X, \mathbf{M})$.

Let $f : X' \to X$ be a log resolution of $(X, \operatorname{Supp} B)$ such that $\mathbf{M}$ descends to X'. Thus $\operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1}B$ is a simple normal crossing divisor. For any real number $s \geq 0$, write

$$K_{X'} + B'_s + s\mathbf{M}_{X'} := f^*(K_X + B + s\mathbf{M}_X).$$

Then $f_*B'_s = B$, so $\operatorname{Supp} B'_s \subset \operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1}B$, and $\operatorname{mult}_P B'_s = 1 - a(P, X, B, s\mathbf{M})$ for any prime divisor P on X'. The coefficients of B'_s may be negative. Since $\mathbf{M}$ descends to X', for any prime divisor E over X, $a(E, X, B, s\mathbf{M})$ is the log discrepancy of E with respect to the sub-pair (X', B'_s). By [KM98, Corollary 2.31(3)] and [Kol13, Corollary 2.11 and (2.20)], a sub-pair on a smooth variety with simple normal crossing support and with all coefficients at most 1 is sub-lc, where negative and real coefficients are allowed. Therefore, $(X, B, s\mathbf{M})$ is generalized lc if and only if

$$a(P, X, B) - se(P, X, \mathbf{M}) \geq 0 \quad \text{for any prime divisor } P \text{ on } X'.$$

Since (X, B) is lc, $a(P, X, B) \geq 0$ for any prime divisor P on X'. Thus the inequality for P holds for any $s \geq 0$ when $e(P, X, \mathbf{M}) = 0$. Since $e(P, X, \mathbf{M}) = \operatorname{mult}_P(f^*\mathbf{M}_X - \mathbf{M}_{X'})$, only finitely many prime divisors P on X' satisfy $e(P, X, \mathbf{M}) > 0$. If $e(P, X, \mathbf{M}) = 0$ for any prime divisor P on X', then $\lambda = +\infty$, which is not the case. Hence

$$\lambda = \min\left\{\frac{a(P, X, B)}{e(P, X, \mathbf{M})}\middle| P \text{ is a prime divisor on } X' \text{ such that } e(P, X, \mathbf{M}) > 0\right\},$$

and $(X, B, s\mathbf{M})$ is generalized lc if and only if $0 \leq s \leq \lambda$. In particular, $(X, B, \lambda\mathbf{M})$ is generalized lc. Let E be a prime divisor on X' which attains the minimum. Then $e(E, X, \mathbf{M}) > 0$ and $a(E, X, B, \lambda\mathbf{M}) = a(E, X, B) - \lambda e(E, X, \mathbf{M}) = 0$. By Definition-Proposition 2.4, $e(D, X, \mathbf{M}) = 0$ for any prime divisor D on X, so E is exceptional over X.

Now let E be any prime divisor over X such that E is exceptional over X, $e(E, X, \mathbf{M}) > 0$, and $a(E, X, B, \lambda\mathbf{M}) = 0$. Since $a(E, X, B, \lambda\mathbf{M}) = 0 < 1$, Lemma 2.8, applied to the generalized lc pair $(X, B, \lambda\mathbf{M})$, implies that there exists an extremal divisorial contraction $h : X_1 \to X$ from a $\mathbb{Q}$-factorial klt projective variety X_1 such that E is the unique h-exceptional prime divisor,

$$K_{X_1} + h_*^{-1}B + E + \lambda\mathbf{M}_{X_1} = h^*(K_X + B + \lambda\mathbf{M}_X),$$

and $-E$ is ample over X.

Let $g : X'' \to X_1$ be a projective birational morphism from a normal variety such that $\mathbf{M}$ descends to X''. Since $g_*\mathbf{M}_{X''} = \mathbf{M}_{X_1}$, we have

$$g_*\big((h \circ g)^*\mathbf{M}_X - \mathbf{M}_{X''}\big) = h^*\mathbf{M}_X - \mathbf{M}_{X_1}.$$

Since $h_*\mathbf{M}_{X_1} = \mathbf{M}_X$, we have $h_*(h^*\mathbf{M}_X - \mathbf{M}_{X_1}) = 0$, so $h^*\mathbf{M}_X - \mathbf{M}_{X_1}$ is h-exceptional. Since E is the unique h-exceptional prime divisor, by Definition-Proposition 2.4, applied to $h \circ g$, we have

$$h^*\mathbf{M}_X - \mathbf{M}_{X_1} = \operatorname{mult}_E\big((h \circ g)^*\mathbf{M}_X - \mathbf{M}_{X''}\big)E = e(E, X, \mathbf{M})E.$$

Since $-E$ is ample over X and $e(E, X, \mathbf{M}) > 0$, $\mathbf{M}_{X_1} = h^*\mathbf{M}_X - e(E, X, \mathbf{M})E$ is ample over X. □

Proposition 4.2. *Let $d \geq 2$ be an integer, X a $\mathbb{Q}$-factorial klt projective variety of dimension d, B a reduced divisor on X, and $\mathbf{M}$ a nef $\boldsymbol{b}$-divisor over X induced by an ample Cartier divisor. Let λ be the generalized lc threshold of $\mathbf{M}$ with respect to (X, B). Assume that $(X, B, u\mathbf{M})$ is generalized lc for some real number $u > 0$ and that $\lambda < +\infty$. Then $(X, B, \lambda\mathbf{M})$ is generalized lc and $\lambda \in \mathcal{N}_{d-1}$.*

Proof. Since X is $\mathbb{Q}$-factorial, (X, B) is lc by Definition-Proposition 2.4. By Lemma 4.1, $(X, B, \lambda\mathbf{M})$ is generalized lc, and there exist a prime divisor E over X which is exceptional over X and an extremal divisorial contraction $h : X_1 \to X$ from a $\mathbb{Q}$-factorial klt projective variety X_1, such that E is the unique h-exceptional prime divisor,

$$K_{X_1} + B_1 + \lambda\mathbf{M}_{X_1} = h^*(K_X + B + \lambda\mathbf{M}_X),$$

where $B_1 := h_*^{-1}B + E$, and $\mathbf{M}_{X_1}$ is ample over X. By the definition of λ, $\lambda \geq u > 0$.

Let $V := \operatorname{center}_X E = h(E)$. Since E is exceptional over X, $\operatorname{codim}_X V \geq 2$. Let E^ν be the normalization of E and V^ν the normalization of V. Since E^ν is normal and $E^\nu \to V$ is surjective, $E^\nu \to V$ factors through V^ν. Let $E^\nu \to T \to V^\nu$ be the Stein factorization of $E^\nu \to V^\nu$, and let F be a general fiber of the contraction $E^\nu \to T$. Then

$$1 \leq \dim F = d - 1 - \dim V \leq d - 1.$$

We check the hypotheses of Lemma 3.9 for the generalized pair $(X_1, B_1, \lambda\mathbf{M})$, the real number $t = \lambda$, the component E of $\lfloor B_1 \rfloor$, and the contraction $E^\nu \to T$.

First, $\mathbf{M}$, regarded as a nef $\boldsymbol{b}$-divisor over X_1, is induced by an ample Cartier divisor by Lemma 3.2, applied to a smooth variety dominating X_1 and the higher model in Definition 3.1.

Second, $(X_1, B_1, \lambda\mathbf{M})$ is generalized lc and $B_1 \in \mathcal{S} \cup \mathcal{S}_\lambda$. Since X_1 is $\mathbb{Q}$-factorial, $K_{X_1} + B_1 + \lambda\mathbf{M}_{X_1}$ is $\mathbb{R}$-Cartier. By the crepant equality above,

$$a(P, X_1, B_1, \lambda\mathbf{M}) = a(P, X, B, \lambda\mathbf{M}) \geq 0$$

for any prime divisor P over X. Moreover, $B_1 \geq 0$ is a reduced divisor, so $B_1 \in \mathcal{S}$.

Third, $t = \lambda > 0$, and $\dim F \geq 1$.

Fourth, since h maps the image of F in X_1 to a point,

$$(K_{X_1} + B_1 + \lambda\mathbf{M}_{X_1})|_F = \big(h^*(K_X + B + \lambda\mathbf{M}_X)\big)|_F \equiv 0.$$

Fifth, the morphism $F \to X_1$ is finite onto the image, and h maps the image to a point. Since $\mathbf{M}_{X_1}$ is ample over X, $\mathbf{M}_{X_1}|_F$ is ample [Laz04, Theorem 1.7.8 and Proposition 1.2.13]. Since $\dim F \geq 1$, $\mathbf{M}_{X_1}|_F \not\equiv 0$.

By Lemma 3.9(2), $\lambda \in \mathcal{N}_{\dim F}$. By Definition 3.3, $\mathcal{N}_k \subset \mathcal{N}_{d-1}$ for any integer $1 \leq k \leq d - 1$. Thus $\lambda \in \mathcal{N}_{d-1}$. □

When the parameter decreases from u to c, the generalized log discrepancy of a prime divisor E over X increases by $(u - c)e(E, X, \mathbf{M})$, and $e(E, X, \mathbf{M})$ has no a priori upper bound. If $c > 0$ and $c \notin \mathcal{N}_{d-2}$, the following corollary proves the bound $e(E, X, \mathbf{M}) \leq \frac{1}{\eta}a(E, X, B, u\mathbf{M})$ used in the proof of Theorem 8.2, where $\eta > 0$ depends only on d and c.

Corollary 4.3. *Let $d \geq 2$ be an integer and $c > 0$ a real number such that $c \notin \mathcal{N}_{d-2}$. Then there exists a real number $\eta > 0$ depending only on d and c satisfying the following. Let X be a $\mathbb{Q}$-factorial klt projective variety of dimension d, B a reduced divisor on X, and $\mathbf{M}$ a nef $\boldsymbol{b}$-divisor over X induced by an ample Cartier divisor, such that $(X, B, u\mathbf{M})$ is generalized lc for some real number $c < u < c + \eta$. Then $(X, B, (c + 2\eta)\mathbf{M})$ is generalized lc, and for any prime divisor E over X,*

$$e(E, X, \mathbf{M}) \leq \frac{a(E, X, B, u\mathbf{M})}{\eta} \quad \text{and} \quad a(E, X, B, c\mathbf{M}) \leq \Big(1 + \frac{u - c}{\eta}\Big)a(E, X, B, u\mathbf{M}).$$

Proof. We first show that there exists a real number $\eta > 0$ such that

$$\mathcal{N}_{d-1} \cap (c, c + 2\eta) = \emptyset.$$

Suppose that $\mathcal{N}_{d-1} \cap (c, c+2\eta) \neq \emptyset$ for any real number $\eta > 0$. Let $t_1 \in \mathcal{N}_{d-1} \cap (c, c+1)$. For any integer $k \geq 1$, given $t_k \in \mathcal{N}_{d-1}$ with $t_k > c$, let

$$t_{k+1} \in \mathcal{N}_{d-1} \cap \Big(c, c + \min\Big\{t_k - c, \frac{1}{k+1}\Big\}\Big).$$

Then $(t_k)_{k\geq 1}$ is a strictly decreasing sequence in $\mathcal{N}_{d-1}$ with $\lim_{k\to+\infty} t_k = c > 0$. Since $d-1 \geq 1$, Lemma 3.7 implies that $c \in \mathcal{N}_{d-2}$, a contradiction.

We fix such an η. By construction, η depends only on d and c, and does not depend on X, B, or $\mathbf{M}$.

Now let X, B, $\mathbf{M}$, and u satisfy the hypotheses of the second part of the statement, and let λ be the generalized lc threshold of $\mathbf{M}$ with respect to (X, B). Then $\lambda \geq u > c$. By Definition-Proposition 2.4, for any prime divisor E over X and any real numbers $s, s' \geq 0$,

$$a(E, X, B, s'\mathbf{M}) = a(E, X, B, s\mathbf{M}) + (s - s')e(E, X, \mathbf{M}), \tag{4.1}$$

and $e(E, X, \mathbf{M}) \geq 0$.

If $\lambda = +\infty$, then $(X, B, s\mathbf{M})$ is generalized lc for some real number $s \geq c + 2\eta$. If $\lambda < +\infty$, then by Proposition 4.2, $(X, B, \lambda\mathbf{M})$ is generalized lc and $\lambda \in \mathcal{N}_{d-1}$. Since $\lambda > c$ and $\mathcal{N}_{d-1} \cap (c, c+2\eta) = \emptyset$, we have $\lambda \geq c + 2\eta$. In both cases, $(X, B, (c+2\eta)\mathbf{M})$ is generalized lc by Definition-Proposition 2.4.

Let E be a prime divisor over X. By (4.1),

$$0 \leq a(E, X, B, (c+2\eta)\mathbf{M}) = a(E, X, B, u\mathbf{M}) - (c + 2\eta - u)e(E, X, \mathbf{M}).$$

Since $u < c + \eta$, we have $c + 2\eta - u > \eta > 0$. Since $e(E, X, \mathbf{M}) \geq 0$, we have

$$\eta e(E, X, \mathbf{M}) \leq (c + 2\eta - u)e(E, X, \mathbf{M}) \leq a(E, X, B, u\mathbf{M}),$$

which implies the first inequality. Since $u > c$, (4.1) and the first inequality imply

$$a(E, X, B, c\mathbf{M}) = a(E, X, B, u\mathbf{M}) + (u - c)e(E, X, \mathbf{M}) \leq \Big(1 + \frac{u-c}{\eta}\Big)a(E, X, B, u\mathbf{M}).$$

□

For a reduced boundary, Proposition 4.2 gives $\lambda \in \mathcal{N}_{d-1}$. The following proposition shows that the generalized lc threshold obtained after rounding up the coefficients of a part of the boundary with coefficients in $(0, 1)$ belongs to $\mathcal{N}_{d-2}$, including the case $\lambda \leq 1$, which cannot be avoided (Remark 1.3). Paragraph (b) of the sketch of the proof in Section 1 explains why $\mathcal{N}_{d-1}$ does not suffice.

Proposition 4.4. *Let $d \geq 2$ be an integer, X a $\mathbb{Q}$-factorial klt projective variety of dimension d, and $\mathbf{M}$ a nef $\boldsymbol{b}$-divisor over X induced by an ample Cartier divisor. Let B and D be $\mathbb{R}$-divisors on X with no common components, such that $B \in \mathcal{S}$, $D \neq 0$, and $D \in (0, 1)$. Let $u > 0$ be a real number, and let λ be the generalized lc threshold of $\mathbf{M}$ with respect to $(X, B + \lceil D \rceil)$. Assume that $(X, B + D, u\mathbf{M})$ is generalized klt and $0 < \lambda < u$. Then $\lambda \in \mathcal{N}_{d-2}$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we find a prime divisor E over X with $a(E, X, B + \lceil D \rceil, \lambda\mathbf{M}) = 0$, and we show that $\operatorname{center}_X E \subset \operatorname{Supp} D$.

Since $\lambda > 0$, there exists a real number $s > 0$ such that $(X, B + \lceil D \rceil, s\mathbf{M})$ is generalized lc. Since X is $\mathbb{Q}$-factorial, $(X, B + \lceil D \rceil)$ is lc by Definition-Proposition 2.4. Since $\lambda < u$, Lemma 4.1, applied to $(X, B + \lceil D \rceil)$ and $\mathbf{M}$, implies that $(X, B + \lceil D \rceil, \lambda\mathbf{M})$ is generalized lc and that there exists a prime divisor E over X such that E is exceptional over X, $e := e(E, X, \mathbf{M}) > 0$, and $a(E, X, B + \lceil D \rceil, \lambda\mathbf{M}) = 0$. By Definition-Proposition 2.4, applied to $B + \lceil D \rceil$ with $t = \lambda$ and to $B + D$ with $t = u$,

$$0 = a(E, X, B + \lceil D \rceil, \lambda\mathbf{M}) = a(E, X, B + D, u\mathbf{M}) - \operatorname{mult}_E(\lceil D \rceil - D) + (u - \lambda)e. \tag{4.2}$$

Since $(X, B+D, u\mathbf{M})$ is generalized klt, $a(E, X, B+D, u\mathbf{M}) > 0$. Since $u > \lambda$ and $e > 0$, $(u-\lambda)e > 0$. Thus (4.2) implies that $\operatorname{mult}_E(\lceil D\rceil - D) > 0$. Since $D \in (0,1)$, $\operatorname{Supp}(\lceil D\rceil - D) = \operatorname{Supp} D$, hence $\operatorname{center}_X E \subset \operatorname{Supp} D$. In particular,

$$b := \operatorname{mult}_E \lceil D\rceil \geq \operatorname{mult}_E(\lceil D\rceil - D) > 0.$$

Step 2. In this step, we extract E and show that the strict transform of $\lceil D\rceil$ is ample over X.

By Lemma 4.1, there exists an extremal divisorial contraction $h : X_1 \to X$ from a $\mathbb{Q}$-factorial klt projective variety X_1 such that E is the unique h-exceptional prime divisor, $h^*\mathbf{M}_X = \mathbf{M}_{X_1} + eE$, $\mathbf{M}_{X_1}$ is ample over X, and

$$K_{X_1} + B_1 + \lambda\mathbf{M}_{X_1} = h^*(K_X + B + \lceil D\rceil + \lambda\mathbf{M}_X), \tag{4.3}$$

where $B_1 := h_*^{-1}B + h_*^{-1}\lceil D\rceil + E$. Since $\mathbf{M}_{X_1} = h^*\mathbf{M}_X - eE$ is ample over X and $e > 0$, $-E$ is ample over X. Since

$$h^*\lceil D\rceil = h_*^{-1}\lceil D\rceil + bE$$

and $b > 0$, $h_*^{-1}\lceil D\rceil = h^*\lceil D\rceil - bE$ is ample over X.

By (4.3),

$$a(P, X_1, B_1, \lambda\mathbf{M}) = a(P, X, B + \lceil D\rceil, \lambda\mathbf{M}) \geq 0$$

for any prime divisor P over X_1, hence $(X_1, B_1, \lambda\mathbf{M})$ is generalized lc. Since X_1 is $\mathbb{Q}$-factorial, Definition-Proposition 2.4 implies that $(X_1, B_1, s\mathbf{M})$ is generalized lc for any $s \in [0, \lambda]$ and that (X_1, B_1) is lc. Since $B \in \mathcal{S}$, $\lceil D\rceil$ is reduced, and B and D have no common components, $B_1 \in \mathcal{S}$.

Step 3. In this step, we restrict $(X_1, B_1, s\mathbf{M})$ to a general fiber F of E over $\operatorname{center}_X E$, keeping the ample divisor $H := \mathbf{M}_{X_1}|_F$ apart from the trace of the restricted nef part, whose difference R_F enters the boundary with coefficient s.

Let $V := \operatorname{center}_X E$, let V^ν be the normalization of V, and let E^ν be the normalization of E. Since E^ν is normal and the morphism $E^\nu \to V$ induced by h is surjective, $E^\nu \to V$ factors through V^ν. Let $E^\nu \to T \to V^\nu$ be the Stein factorization of $E^\nu \to V^\nu$, let F be a general fiber of $E^\nu \to T$, and let $r := \dim F$. Since E is exceptional over X, $\dim V \leq d-2$, hence

$$1 \leq r = d - 1 - \dim V \leq d - 1.$$

Moreover, h maps the image of F in X_1 to a point.

Let $f : W_1 \to X_1$ be a log resolution of $(X_1, \operatorname{Supp} B_1)$. Since $\mathbf{M}$ is induced by an ample Cartier divisor, we may choose f such that W_1 dominates the higher model X' of Definition 3.1. Then $\mathbf{M}$ descends to W_1, and $\mathbf{M}_{W_1}$ is the pullback of an ample Cartier divisor by a contraction from W_1. In particular, $\mathbf{M}_{W_1}$ is a nef Cartier divisor. Let $E' := f_*^{-1}E$. By Definition-Proposition 2.4, $f^*\mathbf{M}_{X_1} - \mathbf{M}_{W_1} \geq 0$ and $f^*\mathbf{M}_{X_1} - \mathbf{M}_{W_1}$ is f-exceptional, hence E' is not a component of $f^*\mathbf{M}_{X_1} - \mathbf{M}_{W_1}$. Write $K_{W_1} + B_{W_1} := f^*(K_{X_1} + B_1)$. Then

$$K_{W_1} + B_{W_1} + s(f^*\mathbf{M}_{X_1} - \mathbf{M}_{W_1}) + s\mathbf{M}_{W_1} = f^*(K_{X_1} + B_1 + s\mathbf{M}_{X_1})$$

for any real number s. Let $B_{E^\nu}(0)$ be the push-forward of $(B_{W_1} - E')|_{E'}$ to E^ν, which is the different of $B_1 - E$ on E^ν [Kol13, Definition 4.2 and (4.7.1)], let $R_{E^\nu} \geq 0$ be the push-forward of $(f^*\mathbf{M}_{X_1} - \mathbf{M}_{W_1})|_{E'}$ to E^ν, and let $B_{E^\nu}(s) := B_{E^\nu}(0) + sR_{E^\nu}$ for any real number s. By Definition-Proposition 2.5, applied to $(X_1, B_1, s\mathbf{M})$ and E, the generalized pair $(E^\nu, B_{E^\nu}(s), s\mathbf{M}|_{E^\nu})$ is generalized lc for any $s \in [0, \lambda]$, and

$$K_{E^\nu} + B_{E^\nu}(s) + s(\mathbf{M}|_{E^\nu})_{E^\nu} = (K_{X_1} + B_1 + s\mathbf{M}_{X_1})|_{E^\nu}.$$

Let P be a prime divisor on E^ν. If $\operatorname{mult}_P B_{E^\nu}(0) = 1$, then $\operatorname{mult}_P R_{E^\nu} = 0$, since $\operatorname{mult}_P B_{E^\nu}(\lambda) \leq 1$ and $\lambda > 0$. If $\operatorname{mult}_P B_{E^\nu}(0) < 1$, then by [HL21, Definition 2.2 and (2)], there exist a positive integer m, which is the Cartier index along P [Kol+92, (16.6.3)], non-negative integers k and n_j, and coefficients b_j of B_1, such that $\operatorname{mult}_P R_{E^\nu} = \frac{k}{m}$ and

$$\operatorname{mult}_P B_{E^\nu}(s) = \frac{m - 1 + \sum_j n_j b_j + ks}{m} \quad \text{for any real number } s. \tag{4.4}$$

Since $B_1 \in \mathcal{S}$, (4.4) and Lemma 3.4 imply that $B_{E^\nu}(0) \in \mathcal{S}$,

$$B_{E^\nu}(\lambda) \in \mathcal{S} \cup \mathcal{S}_\lambda,$$

and

$$\operatorname{mult}_P B_{E^\nu}(\lambda) \in \mathcal{S}_\lambda$$

for any prime divisor P on E^ν with $\operatorname{mult}_P R_{E^\nu} > 0$, including the case $\operatorname{mult}_P B_{E^\nu}(\lambda) = 1$, since $k \geq 1$ for such P.

Let F' be the fiber of $E' \to T$ over the image of F in T, and let $g : F' \to F$ be the induced birational morphism. Since F is general and E' is smooth, F' is smooth. Let

$$G_0 := B_{E^\nu}(0)|_F, \quad R_F := R_{E^\nu}|_F, \quad E_F := (f^*\mathbf{M}_{X_1} - \mathbf{M}_{W_1})|_{F'} \geq 0.$$

Then $R_F = g_*E_F$. Here E_F need not be g-exceptional, and R_F may be non-zero. Since F is general, the restriction to F does not change coefficients: any coefficient of $G_0 + sR_F$ is the coefficient of $B_{E^\nu}(s)$ along a prime divisor on E^ν dominating T. Thus $G_0 \in \mathcal{S}$, $G_0 + \lambda R_F \in \mathcal{S} \cup \mathcal{S}_\lambda$, any coefficient of $G_0 + \lambda R_F$ along a component of R_F belongs to $\mathcal{S}_\lambda$, and (4.4) also describes the coefficients of $G_0 + sR_F$. Let $\mathbf{M}|_F$ be the nef $\boldsymbol{b}$-divisor over F that descends to F' with $(\mathbf{M}|_F)_{F'} = \mathbf{M}_{W_1}|_{F'}$. By [BZ16, Remark 4.2(6)], $(F, G_0 + sR_F, s\mathbf{M}|_F)$ is a generalized pair for any real number $s \geq 0$, and its boundary part on F' is $(B_{W_1} - E')|_{F'} + sE_F$, the restriction to F' of the boundary part $(B_{W_1} - E')|_{E'} + s(f^*\mathbf{M}_{X_1} - \mathbf{M}_{W_1})|_{E'}$ of $(E^\nu, B_{E^\nu}(s), s\mathbf{M}|_{E^\nu})$ on E'. Since f is a log resolution of $(X_1, \operatorname{Supp} B_1)$ and F is general, the restriction to F' of the simple normal crossing divisor $\operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1}(B_1 - E)$ is a simple normal crossing divisor on the smooth variety F' [Har77, III, Corollary 10.7], and it contains $\operatorname{Supp}(B_{W_1} - E')|_{F'}$ and $\operatorname{Supp} E_F$. Since $(E^\nu, B_{E^\nu}(s), s\mathbf{M}|_{E^\nu})$ is generalized lc for any $s \in [0, \lambda]$, the coefficients of $(B_{W_1} - E')|_{F'} + sE_F$ are at most 1 for any $s \in [0, \lambda]$. Thus, computing generalized log discrepancies on F', to which $\mathbf{M}|_F$ descends, [BZ16, Definition 4.1] and [Kol13, Corollary 2.11 and (2.20)] imply that $(F, G_0 + sR_F, s\mathbf{M}|_F)$ is generalized lc for any $s \in [0, \lambda]$. Taking $s = 0$, (F, G_0) is lc. Restricting $f^*\mathbf{M}_{X_1} = \mathbf{M}_{W_1} + (f^*\mathbf{M}_{X_1} - \mathbf{M}_{W_1})$ to F', we obtain

$$g^*H = (\mathbf{M}|_F)_{F'} + E_F, \quad \text{hence} \quad H = (\mathbf{M}|_F)_F + R_F. \tag{4.5}$$

By [BZ16, Remark 4.2(6)] and (4.5),

$$K_F + G_0 + sR_F + s(\mathbf{M}|_F)_F = (K_{X_1} + B_1 + s\mathbf{M}_{X_1})|_F = K_F + G_0 + sH \tag{4.6}$$

for any real number s. Since h maps the image of F to a point, (4.3) and (4.6) imply that

$$K_F + G_0 + \lambda R_F + \lambda(\mathbf{M}|_F)_F = K_F + G_0 + \lambda H \equiv 0. \tag{4.7}$$

Since $\mathbf{M}_{X_1}$ and $h_*^{-1}\lceil D \rceil$ are ample over X, the morphism from F to X_1 is finite onto the image, and h maps the image of F to a point, H and $h_*^{-1}\lceil D \rceil|_F$ are ample [Laz04, Theorem 1.7.8 and Proposition 1.2.13]. In particular, $h_*^{-1}\lceil D \rceil|_F \geq 0$ is a non-zero divisor, since $r \geq 1$. Let P be a prime divisor on E^ν contained in $\operatorname{Supp}(h_*^{-1}\lceil D \rceil|_{E^\nu})$. Then the image of P in X_1 is contained in the intersection of E and a component of $h_*^{-1}\lceil D \rceil$, and both have coefficient 1 in B_1. Localizing at the generic point of the image of P in X_1, which has codimension 2 in X_1, we may assume that X_1 is a surface. Then [Kol13, Corollary 3.45(3)], applied to the curve germs of E and of a component D_1 of $h_*^{-1}\lceil D \rceil$ containing the image of P, shows that the coefficient of P in the different of D_1 on E^ν is 1, and by [Kol13, Proposition 2.35(2)], adding $B_1 - E - D_1 \geq 0$ does not decrease it. Thus $\operatorname{mult}_P B_{E^\nu}(0) \geq 1$. Since $(E^\nu, B_{E^\nu}(\lambda), \lambda\mathbf{M}|_{E^\nu})$ is generalized lc, $R_{E^\nu} \geq 0$, and $\lambda > 0$, we obtain $\operatorname{mult}_P B_{E^\nu}(0) = 1$ and $\operatorname{mult}_P R_{E^\nu} = 0$. Since F is general, any component of $\operatorname{Supp}(h_*^{-1}\lceil D \rceil|_F)$ has coefficient 1 in G_0 and coefficient 0 in R_F.

Step 4. In this step, we treat the case $r \geq 2$ and show that $\lambda \in \mathcal{N}_{r-1}$.

Let S be a component of $\operatorname{Supp}(h_*^{-1}\lceil D \rceil|_F)$, and let S^ν be the normalization of S. Then $\dim S^\nu = r - 1 \geq 1$. By **Step 3**, S is a component of $\lfloor G_0 + \lambda R_F \rfloor$ and $\operatorname{mult}_S R_F = 0$. Let

$g_1 : F'' \to F$ be a log resolution of $(F, \mathrm{Supp}(G_0+R_F))$ which factors through g, let $S'' := (g_1)_*^{-1}S$, and let $E_{F''} \geq 0$ be the pullback of E_F to F''. Since $\mathbf{M}|_F$ descends to F', (4.5) implies that

$$g_1^*H = (\mathbf{M}|_F)_{F''} + E_{F''}.$$

The coefficient of S'' in $E_{F''}$ is equal to $\mathrm{mult}_S(g_*E_F) = \mathrm{mult}_S R_F = 0$. Write

$$K_{F''} + \Gamma_{F''} := g_1^*(K_F + G_0).$$

By (4.6),

$$K_{F''} + \Gamma_{F''} + sE_{F''} + s(\mathbf{M}|_F)_{F''} = g_1^*\big(K_F + G_0 + sR_F + s(\mathbf{M}|_F)_F\big)$$

for any real number s. Let $\Delta_S(0)$ be the push-forward of $(\Gamma_{F''} - S'')|_{S''}$ to S^ν, let $R_S \geq 0$ be the push-forward of $E_{F''}|_{S''}$ to S^ν, and let

$$\Delta_S(s) := \Delta_S(0) + sR_S.$$

By Definition-Proposition 2.5, applied to $(F, G_0 + \lambda R_F, \lambda\mathbf{M}|_F)$ and S, the generalized pair $(S^\nu, \Delta_S(\lambda), \lambda\mathbf{M}|_{S^\nu})$ is generalized lc, where $\mathbf{M}|_{S^\nu}$ descends to S'' with $(\mathbf{M}|_{S^\nu})_{S''} = (\mathbf{M}|_F)_{F''}|_{S''}$, and by (4.7),

$$K_{S^\nu} + \Delta_S(\lambda) + \lambda(\mathbf{M}|_{S^\nu})_{S^\nu} = (K_F + G_0 + \lambda H)|_{S^\nu} \equiv 0.$$

The coefficient of a prime divisor P on S^ν in $\Delta_S(s)$ depends only on a neighborhood of the generic point ξ_P of the image of P in F, which has codimension 2 in F. Since (F, G_0) is lc, G_0 has rational coefficients, and $\xi_P \in S$, where S is a component of $\lfloor G_0 \rfloor$, [Kol13, Proposition 2.28], applied to the localization of (F, G_0) at ξ_P, shows that F has a rational singularity at ξ_P, and hence F is $\mathbb{Q}$-factorial at ξ_P [Kol13, Proposition 10.9(3)]. Thus $(\mathbf{M}|_F)_F$ is $\mathbb{Q}$-Cartier on a neighborhood of ξ_P, and we may use [HL21, Definition 2.2 and (2)] for $(F, G_0 + sR_F, s\mathbf{M}|_F)$ near ξ_P for any $s \in [0, \lambda]$. Since $G_0 + \lambda R_F \in \mathcal{S} \cup \mathcal{S}_\lambda$, [HL21, Definition 2.2 and (2)] and Lemma 3.4 imply that $\Delta_S(\lambda) \in \mathcal{S} \cup \mathcal{S}_\lambda$.

Let P be a prime divisor on S^ν with $\mathrm{mult}_P R_S > 0$. We show that $v := \mathrm{mult}_P \Delta_S(\lambda) \in \mathcal{S}_\lambda$. If $v = 1$, then

$$\mathrm{mult}_P \Delta_S(s) = 1 - (\lambda - s)\,\mathrm{mult}_P R_S < 1$$

for any $s \in [0, \lambda)$, and if $v < 1$, then $\mathrm{mult}_P \Delta_S(s) \leq v < 1$ for any $s \in [0, \lambda]$, since $R_S \geq 0$. Let $Q_1, \dots, Q_l$ be the components of $\mathrm{Supp}(G_0 + R_F)$ other than S containing ξ_P. By [HL21, Definition 2.2 and (2)], there exist a positive integer m and non-negative integers $k, n_1, \dots, n_l$ such that

$$\mathrm{mult}_P \Delta_S(s) = \frac{m - 1 + \sum_{j=1}^l n_j\,\mathrm{mult}_{Q_j}(G_0 + sR_F) + ks}{m}$$

for any $s \in [0, \lambda)$, and m, n_j, and k do not depend on s, since they are determined by the germ of F at ξ_P, by S, by the components of $\mathrm{Supp}(G_0 + R_F)$ through ξ_P, and by $\mathbf{M}|_F$, none of which depends on s. Both sides of the formula [HL21, (2)] for $\mathrm{mult}_P \Delta_S(s)$ are affine functions of s. Letting $s \to \lambda$, we obtain it for $s = \lambda$, and comparing the slopes of both sides, we obtain

$$\mathrm{mult}_P R_S = \frac{\sum_{j=1}^l n_j\,\mathrm{mult}_{Q_j} R_F + k}{m} > 0.$$

Let $b_j := \mathrm{mult}_{Q_j}(G_0 + \lambda R_F)$, so that

$$v = \frac{m - 1 + \sum_{j=1}^l n_j b_j + k\lambda}{m} \leq 1.$$

By **Step 3**, $b_j \in \mathcal{S}$ if $\mathrm{mult}_{Q_j} R_F = 0$, and $b_j \in \mathcal{S}_\lambda$ if $\mathrm{mult}_{Q_j} R_F > 0$. Since $\lambda > 0$, any element of $\mathcal{S}_\lambda$ is of the form $\frac{M-1+K\lambda}{M}$ with positive integers M and K (Definition 3.3). Moreover, $b_j = 0$ implies that $\mathrm{mult}_{Q_j} R_F = 0$.

- Assume that $b_{j_0} = 1$ and $n_{j_0} \geq 1$ for some j_0. Since $v \leq 1$, we have $n_{j_0} = 1$, $k = 0$, and $n_j b_j = 0$ for any $j \neq j_0$, hence $n_j\,\mathrm{mult}_{Q_j} R_F = 0$ for any $j \neq j_0$. Since $\mathrm{mult}_P R_S > 0$, we obtain $\mathrm{mult}_{Q_{j_0}} R_F > 0$, hence $1 = b_{j_0} \in \mathcal{S}_\lambda$, and $v = 1 \in \mathcal{S}_\lambda$.

- Assume that $b_j < 1$ for any j with $n_j \geq 1$. Then any non-zero b_j with $n_j \geq 1$ is of one of the following forms: $1 - \frac{1}{M}$ with an integer $M \geq 2$, if $\operatorname{mult}_{Q_j} R_F = 0$; $\frac{M-1+K\lambda}{M}$ with integers $M \geq 2$ and $K \geq 1$, if $\operatorname{mult}_{Q_j} R_F > 0$; $K\lambda$ with an integer $K \geq 1$, if $\operatorname{mult}_{Q_j} R_F > 0$. The b_j of the first two forms are at least $\frac{1}{2}$, and those of the second form are larger than $\frac{1}{2}$. Since $\sum_j n_j b_j \leq 1$, if the b_j of the first two forms occur at least twice, counted with the multiplicities n_j, then they occur exactly twice, both are equal to $\frac{1}{2}$, hence of the first form, $k = 0$, and all the other $n_j b_j$ are zero, which contradicts $\operatorname{mult}_P R_S > 0$. Thus they occur at most once. Let N be the sum of k and of the integers $n_j K$ over the $b_j = K\lambda$ of the third form. If no b_j of the first two forms occurs, then $N \geq 1$, since $\operatorname{mult}_P R_S > 0$, and
$$v = \frac{m - 1 + N\lambda}{m}.$$
If $b_j = 1 - \frac{1}{M}$ of the first form occurs, then $N \geq 1$, since $\operatorname{mult}_P R_S > 0$, and
$$v = \frac{m - 1 + 1 - \frac{1}{M} + N\lambda}{m} = \frac{mM - 1 + MN\lambda}{mM}.$$
If $b_j = \frac{M-1+K\lambda}{M}$ of the second form occurs, then
$$v = \frac{m - 1 + \frac{M-1+K\lambda}{M} + N\lambda}{m} = \frac{mM - 1 + (K + MN)\lambda}{mM}.$$
In each case, $v = \frac{\tilde{m}-1+\tilde{N}\lambda}{\tilde{m}} \leq 1$ with positive integers $\tilde{m}$ and $\tilde{N}$, hence $v \in \mathcal{S}_\lambda$ by Definition 3.3.

Thus $\operatorname{mult}_P \Delta_S(\lambda) \in \mathcal{S}_\lambda$ for any prime divisor P on S^ν with $\operatorname{mult}_P R_S > 0$, that is, for any P whose coefficient in $\Delta_S(s)$ depends on s with positive slope.

Let $\varphi : \hat{S} \to S^\nu$ be a $\mathbb{Q}$-factorial modification of $(S^\nu, \Delta_S(\lambda), \lambda\mathbf{M}|_{S^\nu})$ with an empty list of divisors to extract [BZ16, Lemma 4.5] (cf. [HL21, Proposition 2.4]): $\hat{S}$ is $\mathbb{Q}$-factorial, φ is crepant, and any φ-exceptional prime divisor is a component of $\lfloor \Delta_{\hat{S}} \rfloor$, hence a generalized lc place of $(S^\nu, \Delta_S(\lambda), \lambda\mathbf{M}|_{S^\nu})$. Write
$$K_{\hat{S}} + \Delta_{\hat{S}} + \lambda(\mathbf{M}|_{S^\nu})_{\hat{S}} := \varphi^*\big(K_{S^\nu} + \Delta_S(\lambda) + \lambda(\mathbf{M}|_{S^\nu})_{S^\nu}\big).$$
Then
$$\Delta_{\hat{S}} = \varphi_*^{-1}\Delta_S(\lambda) + \operatorname{Exc}(\varphi),$$
where $\operatorname{Exc}(\varphi)$ is the reduced sum of the φ-exceptional prime divisors. We check that $(\hat{S}, \Delta_{\hat{S}}, \lambda\mathbf{M}|_{S^\nu})$ satisfies condition (†).

(1) Since φ is crepant, $(\hat{S}, \Delta_{\hat{S}}, \lambda\mathbf{M}|_{S^\nu})$ is generalized lc and
$$K_{\hat{S}} + \Delta_{\hat{S}} + \lambda(\mathbf{M}|_{S^\nu})_{\hat{S}} \equiv 0.$$
(2) $\Delta_{\hat{S}} \in \mathcal{S} \cup \mathcal{S}_\lambda$, since $\Delta_S(\lambda) \in \mathcal{S} \cup \mathcal{S}_\lambda$ and $1 \in \mathcal{S}$.
(3) $\mathbf{M}|_{S^\nu}$ is induced by an ample Cartier divisor (Lemma 3.2), and $(\mathbf{M}|_{S^\nu})_{\hat{S}}$ is $\mathbb{Q}$-Cartier, since $\hat{S}$ is $\mathbb{Q}$-factorial.
(4) Assume that $(\mathbf{M}|_{S^\nu})_{\hat{S}} \equiv 0$. Let $\tilde{S}$ be a resolution of $\hat{S}$ such that $\tilde{S} \to S^\nu$ factors through S''. Pulling back $g_1^* H = (\mathbf{M}|_F)_{F''} + E_{F''}$ to $\tilde{S}$, we obtain that the pullback of $H|_{S^\nu}$ to $\tilde{S}$ is $(\mathbf{M}|_{S^\nu})_{\tilde{S}}$ plus the pullback of $E_{F''}|_{S''}$. Pushing forward to $\hat{S}$, we obtain
$$\varphi^*(H|_{S^\nu}) = (\mathbf{M}|_{S^\nu})_{\hat{S}} + R_{\hat{S}}, \tag{4.8}$$
where $R_{\hat{S}} \geq 0$ is the push-forward to $\hat{S}$ of the pullback of $E_{F''}|_{S''}$ to $\tilde{S}$. In particular, $\varphi_* R_{\hat{S}} = R_S$. Let C be a general complete intersection curve of very ample divisors on S^ν, where $C = S^\nu$ if $r = 2$. Let $Z \subset S^\nu$ be the closed subset over which φ is not an isomorphism. Since S^ν is normal, Z has codimension at least 2 in S^ν by Zariski's main theorem, and $Z = \emptyset$ if $r = 2$. Thus $C \cap Z = \emptyset$, φ is an isomorphism over a

neighborhood of C, and the φ-exceptional prime divisors, whose images are contained in Z, are disjoint from $\varphi^{-1}C$. Since H is ample and $S^\nu \to F$ is finite, $H|_{S^\nu}$ is ample [Laz04, Proposition 1.2.13]. By (4.8),

$$(R_{\hat{S}} \cdot \varphi^{-1}C) = (\varphi^*(H|_{S^\nu}) \cdot \varphi^{-1}C) - ((\mathbf{M}|_{S^\nu})_{\hat{S}} \cdot \varphi^{-1}C) = (H|_{S^\nu} \cdot C) > 0.$$

Thus there exists a component P' of $R_{\hat{S}}$ with $(P' \cdot \varphi^{-1}C) \neq 0$. Then $P' \cap \varphi^{-1}C \neq \emptyset$, hence P' is not φ-exceptional, and $P := \varphi(P')$ is a prime divisor on S^ν with $\operatorname{mult}_P R_S = \operatorname{mult}_{P'} R_{\hat{S}} > 0$. Thus

$$\operatorname{mult}_{P'} \Delta_{\hat{S}} = \operatorname{mult}_P \Delta_S(\lambda) \in \mathcal{S}_\lambda.$$

Since $\dim \hat{S} = r - 1$ and $1 \leq r - 1 \leq d - 2$, we obtain $\lambda \in \mathcal{N}_{r-1} \subset \mathcal{N}_{d-2}$ (Lemma 3.5).

Step 5. In this step, we treat the case $r = 1$ and show that $\lambda \in \mathcal{N}_0$.

Since F is a normal curve, F is smooth and g is an isomorphism. Thus $(\mathbf{M}|_F)_F = \mathbf{M}_{W_1}|_{F'}$ is a nef Cartier divisor, and

$$\delta := \deg(\mathbf{M}|_F)_F \in \mathbb{Z}_{\geq 0}.$$

By **Step 3**, there exists a point $p_0 \in \operatorname{Supp}(h_*^{-1}\lceil D \rceil|_F)$ with $\operatorname{mult}_{p_0} G_0 = 1$ and $\operatorname{mult}_{p_0} R_F = 0$. Since H is ample, $\deg H > 0$. By (4.7),

$$\deg K_F = -\deg G_0 - \lambda \deg H < -\deg G_0 \leq -1,$$

hence $F \cong \mathbb{P}^1$ and

$$\deg G_0 + \lambda \deg H = 2. \tag{4.9}$$

Let $q_1, \ldots, q_w$ be the points of $\operatorname{Supp}(G_0 + R_F)$ other than p_0. If $\operatorname{mult}_{q_l} G_0 = 1$ for some l, then $\deg G_0 \geq 2$, which contradicts (4.9) and $\lambda \deg H > 0$. Thus $\operatorname{mult}_{q_l} G_0 < 1$ for any l, and by **Step 3**, $\operatorname{mult}_{q_l}(G_0 + sR_F)$ is of the form (4.4), with m, k, n_j, and b_j depending on l. Since $R_F \geq 0$ and $(F, G_0 + \lambda R_F, \lambda \mathbf{M}|_F)$ is generalized lc,

$$\frac{m - 1 + \sum_j n_j b_j}{m} = \operatorname{mult}_{q_l} G_0 \leq \operatorname{mult}_{q_l}(G_0 + \lambda R_F) \leq 1,$$

hence $\sum_j n_j b_j \leq 1$. Since $b_j \in \mathcal{S}$ and any non-zero element of $\mathcal{S}$ is at least $\frac{1}{2}$, either $\sum_j n_j b_j \in \{0, 1\}$, or $\sum_j n_j b_j = 1 - \frac{1}{l'}$ for some integer $l' \geq 2$. If $\sum_j n_j b_j = 1$, then $\operatorname{mult}_{q_l} G_0 = 1$, which is excluded. If $\sum_j n_j b_j = 0$, then

$$\frac{m - 1 + ks}{m} = 1 - \frac{1}{m} + \frac{k}{m}s,$$

and if $\sum_j n_j b_j = 1 - \frac{1}{l'}$, then

$$\frac{m - 1 + 1 - \frac{1}{l'} + ks}{m} = 1 - \frac{1}{ml'} + \frac{kl'}{ml'}s.$$

Thus there exist positive integers m_l and non-negative integers k_l such that

$$\operatorname{mult}_{q_l}(G_0 + sR_F) = 1 - \frac{1}{m_l} + \frac{k_l}{m_l}s \quad \text{and} \quad \operatorname{mult}_{q_l} R_F = \frac{k_l}{m_l}. \tag{4.10}$$

Since $\operatorname{mult}_{p_0} R_F = 0$, (4.5) and (4.10) imply that

$$\deg H = \delta + \deg R_F = \delta + \sum_{l=1}^{w} \frac{k_l}{m_l} > 0.$$

By (4.7) and (4.10),

$$-2 + 1 + \sum_{l=1}^{w} \left(1 - \frac{1}{m_l} + \frac{k_l}{m_l}\lambda\right) + \lambda\delta = 0,$$

where $-2 = \deg K_F$ and $1 = \operatorname{mult}_{p_0} G_0$, that is,

$$\lambda\Big(\delta + \sum_{l=1}^{w} \frac{k_l}{m_l}\Big) = 1 - \sum_{l=1}^{w} \Big(1 - \frac{1}{m_l}\Big). \tag{4.11}$$

(4.11) also includes the points q_l with $\operatorname{mult}_{q_l}(G_0 + \lambda R_F) = 1$, that is, with $k_l\lambda = 1$: for example, if $\delta = 0$, $w = 1$, $m_1 = 1$, and $k_1 = k$, then $\lambda = \frac{1}{k}$. Since the left-hand side of (4.11) is $\lambda \deg H > 0$, we have $\sum_{l=1}^{w}(1 - \frac{1}{m_l}) < 1$. Since $1 - \frac{1}{m_l} \geq \frac{1}{2}$ when $m_l \geq 2$, there exists at most one l with $m_l \geq 2$.

- If $m_l = 1$ for $1 \leq l \leq w$, then (4.11) implies that $\lambda = \frac{1}{N}$, where $N := \delta + \sum_{l=1}^{w} k_l = \deg H$ is a positive integer.
- If $m := m_{l_0} \geq 2$ for exactly one l_0, then multiplying (4.11) by m, we obtain $\lambda N = 1$, where
$$N := m\delta + k_{l_0} + m\sum_{l \neq l_0} k_l = m \deg H$$
is a positive integer.

Thus $\lambda \in \mathcal{N}_0 \subset \mathcal{N}_{d-2}$ (Lemma 3.5).

Steps 4 and 5 cover the cases $r \geq 2$ and $r = 1$, which completes the proof. □

5. Good models and bounded families

In this section, we prove two results on pseudo-effective thresholds. Proposition 5.1 constructs a generalized lc model at the exact threshold, and Lemma 5.2 shows that pseudo-effective thresholds take finitely many values on a bounded family. Proposition 5.1 is the starting point of Section 7, and Lemma 5.2 is used only in Section 9.

As in paragraph (a) of the sketch of the proof in Section 1, Proposition 5.1 replaces cL with $\frac{c}{m}H$ for a general member $H \in |mL|$ with $m > c$, which keeps the threshold c and makes the boundary on the model reduced.

Proposition 5.1. *Let $d \geq 2$ be an integer, X a smooth projective variety of dimension d, Δ a reduced simple normal crossing divisor on X, and L an ample Cartier divisor on X such that $c := \operatorname{pet}(X, \Delta; L) > 0$. Then there exist a $\mathbb{Q}$-factorial klt projective variety Y, a reduced divisor B_Y on Y, and projective birational morphisms $f : W \to X$ and $g : W \to Y$ from a smooth projective variety W, satisfying the following.*

(1) $B_Y = g_* f_*^{-1}\Delta + E_Y$, where E_Y is the sum of the prime divisors on Y that are exceptional over X.

(2) $(Y, B_Y, c\overline{L})$ is generalized lc, and there exist a canonical divisor K_W on W and a g-exceptional $\mathbb{R}$-divisor $N \geq 0$ on W such that
$$f^*(K_X + \Delta + cL) = g^*(K_Y + B_Y + c\overline{L}_Y) + N,$$
where $K_X := f_* K_W$ and $K_Y := g_* K_W$.

(3) If $K_Y + B_Y + c\overline{L}_Y \not\equiv 0$, then $c \in \mathcal{N}_{d-1}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct Y, W, f, g, and N.

Let $m > c$ be an integer such that mL is very ample, and let $H \in |mL|$ be a general member. By [Har77, II, Theorem 8.18] applied to X and to any stratum of Δ, H is smooth and $\Delta + H$ is a reduced simple normal crossing divisor. Let $\Theta := \Delta + \frac{c}{m}H$. Since H is Cartier, for any prime divisor E over X, we have

$$a(E, X, \Theta) = a(E, X, \Delta + H) + \Big(1 - \frac{c}{m}\Big)\operatorname{mult}_E H \geq \Big(1 - \frac{c}{m}\Big)\operatorname{mult}_E H \geq 0, \tag{5.1}$$

where the first inequality follows from Lemma 2.7. Thus (X, Θ) is lc. Since $K_X + \Delta + tL$ is pseudo-effective for any $t > c$ and the pseudo-effective cone is closed, $K_X + \Delta + cL$ is pseudo-effective. Since $H \sim mL$, $K_X + \Theta \sim_{\mathbb{R}} K_X + \Delta + cL$, so $K_X + \Theta$ is pseudo-effective. By [Laz04, Example 1.3.14], there exists a real number $0 < \epsilon < 1$ such that $\frac{c}{m}H + \epsilon\Delta$ is ample. Write $\frac{c}{m}H + \epsilon\Delta = \sum_{l=1}^{s} r_l A_l$, where $r_l > 0$ are real numbers and A_l are ample Cartier divisors [Laz04, Definition 1.3.11], let $k > \max_l r_l$ be an integer such that kA_l is very ample for any l, and let $A' := \sum_l \frac{r_l}{k} G_l$ for general members $G_l \in |kA_l|$. Let $\Theta' := (1-\epsilon)\Delta + A'$. As for H, $\Delta + \sum_l G_l$ is a reduced simple normal crossing divisor, so (X, Θ') is klt by [KM98, Corollary 2.31(3)] and [Kol13, Corollary 2.11 and (2.20)]. Since $A' \sim_{\mathbb{R}} \frac{c}{m}H + \epsilon\Delta$, we may write $\Theta - \Theta' = \sum_j t_j \operatorname{div}_X(\psi_j)$ for real numbers t_j and rational functions ψ_j on X. Thus $K_X + \Theta'$ is pseudo-effective, and $\Theta' \geq A'$ is big. By [BCHM10, Theorem 1.2(1)], (X, Θ') has a log terminal model $\phi : X \dashrightarrow Y_0$, and by [BCHM10, Lemma 3.9.3], $K_{Y_0} + \phi_*\Theta'$ is $\mathbb{R}$-linearly equivalent to the pullback of an ample $\mathbb{R}$-divisor by a contraction. Let $\Theta_{Y_0} := \phi_*\Theta$. Since ϕ does not extract divisors, we have

$$\phi_*(\Theta - \Theta') = \sum_j t_j \operatorname{div}_{Y_0}(\psi_j),$$

so $K_{Y_0} + \Theta_{Y_0} \sim_{\mathbb{R}} K_{Y_0} + \phi_*\Theta'$.

Let $p : V \to X$ and $q : V \to Y_0$ be projective birational morphisms from a normal variety V with $q = \phi \circ p$, let K_V be a canonical divisor on V, and let $K_X := p_*K_V$ and $K_{Y_0} := q_*K_V$. Let $N_V := p^*(K_X + \Theta') - q^*(K_{Y_0} + \phi_*\Theta')$. Since ϕ does not extract divisors, N_V is q-exceptional, and since $K_{Y_0} + \phi_*\Theta'$ is nef, $-N_V$ is nef over X. For any prime divisor D on X, the coefficient of D in p_*N_V is $a(D, Y_0, \phi_*\Theta') - a(D, X, \Theta')$, which is 0 if D is not ϕ-exceptional and positive otherwise, since ϕ is a log terminal model. By the negativity lemma [BCHM10, Lemma 3.6.2], $N_V \geq 0$. Since $\Theta - \Theta'$ and $\phi_*(\Theta - \Theta')$ both pull back to $\sum_j t_j \operatorname{div}_V(\psi_j)$ on V, we obtain

$$p^*(K_X + \Theta) = q^*(K_{Y_0} + \Theta_{Y_0}) + N_V.$$

Thus $a(E, Y_0, \Theta_{Y_0}) = a(E, X, \Theta) + \operatorname{mult}_E N_V \geq 0$ for any prime divisor E on V, so (Y_0, Θ_{Y_0}) is lc. Hence (Y_0, Θ_{Y_0}) is a good log minimal model of (X, Θ), and no prime divisor on Y_0 is exceptional over X.

Since (Y_0, Θ_{Y_0}) is lc, by [Fuj11, Theorem 10.4], there exists a projective birational morphism $h_0 : Y \to Y_0$ from a $\mathbb{Q}$-factorial variety Y such that, for $\Theta_Y := (h_0)_*^{-1}\Theta_{Y_0} + \operatorname{Exc}(h_0)$, the pair (Y, Θ_Y) is dlt and $K_Y + \Theta_Y = h_0^*(K_{Y_0} + \Theta_{Y_0})$. Let $f : W \to X$ and $g : W \to Y$ be projective birational morphisms from a smooth projective variety W, let K_W be a canonical divisor on W, and let $K_X := f_*K_W$, $K_Y := g_*K_W$, and $K_{Y_0} := h_{0*}K_Y$. Let $N := N_W$ for $V := W$, $p := f$, and $q := h_0 \circ g$. Then $N \geq 0$ is an $\mathbb{R}$-divisor on W that is exceptional over Y_0, such that

$$f^*(K_X + \Theta) = (h_0 \circ g)^*(K_{Y_0} + \Theta_{Y_0}) + N = g^*(K_Y + \Theta_Y) + N. \tag{5.2}$$

Step 2. In this step, we show that N is g-exceptional and prove (1).

By (5.2), for any prime divisor P on W, we have

$$a(P, X, \Theta) = a(P, Y, \Theta_Y) - \operatorname{mult}_P N. \tag{5.3}$$

Let P be a prime divisor on W that is not g-exceptional. If $g(P)$ is not h_0-exceptional, then $\operatorname{mult}_P N = 0$, since N is exceptional over Y_0. If $g(P)$ is h_0-exceptional, then the coefficient of $g(P)$ in Θ_Y is 1, so $a(P, Y, \Theta_Y) = 0$. Since (X, Θ) is lc, (5.3) implies

$$0 \leq a(P, X, \Theta) = -\operatorname{mult}_P N \leq 0.$$

In either case, $\operatorname{mult}_P N = 0$ and $a(P, X, \Theta) = a(P, Y, \Theta_Y)$. In particular, N is g-exceptional.

Assume moreover that P is f-exceptional. Since no prime divisor on Y_0 is exceptional over X, $g(P)$ is h_0-exceptional, so the coefficient of $g(P)$ in Θ_Y is 1. Thus $a(P, X, \Theta) = a(P, Y, \Theta_Y) = 0$, and (5.1) implies $\operatorname{mult}_P H = 0$. If P is not f-exceptional, then the coefficient of $g(P)$ in Θ_Y is $1 - a(P, Y, \Theta_Y) = 1 - a(P, X, \Theta)$, which is the coefficient of $f(P)$ in Θ. Thus

$$\Theta_Y = B_Y + \frac{c}{m}H_Y, \quad \text{where} \quad B_Y := g_*f_*^{-1}\Delta + E_Y \quad \text{and} \quad H_Y := g_*f_*^{-1}H,$$

and E_Y is the sum of the prime divisors on Y that are exceptional over X. Since Δ is reduced, B_Y is reduced, and (1) holds. Moreover, since $\operatorname{mult}_P H = 0$ for any prime divisor P on W that is f-exceptional but not g-exceptional, we have $g_* f^* H = H_Y$.

Step 3. In this step, we show that Y is klt and prove (2).

Since Y is $\mathbb{Q}$-factorial, H_Y is $\mathbb{Q}$-Cartier, and for any prime divisor E over Y, we have

$$a(E, Y, B_Y) = a(E, Y, \Theta_Y) + \frac{c}{m} \operatorname{mult}_E H_Y \geq a(E, Y, \Theta_Y).$$

Thus (Y, B_Y) and $(Y, 0)$ are dlt [KM98, Corollary 2.39(1)], and Y is klt [KM98, Proposition 2.41].

Write $H = mL + \operatorname{div}_X(\varphi)$ for a rational function φ on X. Then $f^* H = m f^* L + \operatorname{div}_W(\varphi)$. Since $\overline{L}_Y = g_* f^* L$ and $g_* f^* H = H_Y$, we obtain

$$H_Y = m\overline{L}_Y + \operatorname{div}_Y(\varphi).$$

Thus $\overline{L}_Y$ is $\mathbb{Q}$-Cartier and $g^* H_Y = m g^* \overline{L}_Y + \operatorname{div}_W(\varphi)$. Substituting $\Theta = \Delta + \frac{c}{m} H$ and $\Theta_Y = B_Y + \frac{c}{m} H_Y$ into (5.2) and cancelling $\frac{c}{m} \operatorname{div}_W(\varphi)$, we obtain

$$f^*(K_X + \Delta + cL) = g^*(K_Y + B_Y + c\overline{L}_Y) + N. \tag{5.4}$$

Since Y is $\mathbb{Q}$-factorial, $K_Y + B_Y + c\overline{L}_Y$ is $\mathbb{R}$-Cartier, and $(Y, B_Y, c\overline{L})$ is a generalized pair. Let E be a prime divisor over Y, and let $u : W' \to W$ be a projective birational morphism from a smooth variety W' such that E is a divisor on W'. Write

$$K_{W'} + \Delta_{W'} := (f \circ u)^*(K_X + \Delta) \quad \text{and} \quad K_{W'} + B_{W'} + c(f \circ u)^* L := (g \circ u)^*(K_Y + B_Y + c\overline{L}_Y).$$

Since $\overline{L}$ descends to W' with trace $(f \circ u)^* L$, (5.4) implies $B_{W'} = \Delta_{W'} - u^* N$. Since $N \geq 0$ and (X, Δ) is lc (Lemma 2.7), we have

$$a(E, Y, B_Y, c\overline{L}) = 1 - \operatorname{mult}_E B_{W'} = a(E, X, \Delta) + \operatorname{mult}_E N \geq 0.$$

Thus $(Y, B_Y, c\overline{L})$ is generalized lc, and (2) follows from (5.4).

Step 4. In this step, we prove (3).

Let $D_Y := K_Y + B_Y + c\overline{L}_Y$. By **Step 3**,

$$D_Y = K_Y + \Theta_Y - \frac{c}{m} \operatorname{div}_Y(\varphi) \sim_{\mathbb{R}} h_0^*(K_{Y_0} + \Theta_{Y_0}).$$

By **Step 1**, $K_{Y_0} + \Theta_{Y_0} \sim_{\mathbb{R}} K_{Y_0} + \phi_* \Theta'$, and $K_{Y_0} + \phi_* \Theta'$ is $\mathbb{R}$-linearly equivalent to the pullback of an ample $\mathbb{R}$-divisor by a contraction [BCHM10, Lemma 3.9.3]. Thus there exist a contraction $h : Y \to Z$ and an ample $\mathbb{R}$-divisor A_Z on Z such that $D_Y \sim_{\mathbb{R}} h^* A_Z$.

We show that D_Y is not big. Suppose that D_Y is big. Since $N \geq 0$, (5.4) implies that $f^*(K_X + \Delta + cL)$ is big, so $K_X + \Delta + cL$ is big. Since the big cone is open [Laz04, Theorem 2.2.26], $K_X + \Delta + (c - \delta)L$ is big for some real number $0 < \delta < c$, which contradicts $c = \operatorname{pet}(X, \Delta; L)$. Thus D_Y is not big, and $\dim Z < d$.

Assume that $D_Y \not\equiv 0$. Then $\dim Z > 0$. Let F be a general fiber of h. Then $1 \leq \dim F \leq d - 1$ and $D_Y|_F \sim_{\mathbb{R}} 0$. Since $f^* L$ is big and

$$H^0(W, \mathcal{O}_W(k f^* L)) \subset H^0(Y, \mathcal{O}_Y(k\overline{L}_Y))$$

for any positive integer k, $\overline{L}_Y$ is big. Since Y is $\mathbb{Q}$-factorial, by Kodaira's lemma [KM98, Lemma 2.60], there exist a positive integer r, an ample Cartier divisor A on Y, and a Cartier divisor $G \geq 0$ on Y such that $r\overline{L}_Y \sim A + G$. Since F is general, $F \not\subset \operatorname{Supp} G$, so

$$(\overline{L}_Y|_F \cdot (A|_F)^{\dim F - 1}) = \frac{1}{r}\Big((A|_F)^{\dim F} + (G|_F \cdot (A|_F)^{\dim F - 1})\Big) \geq \frac{1}{r}(A|_F)^{\dim F} > 0.$$

Thus $\overline{L}_Y|_F \not\equiv 0$. Moreover, $\overline{L}$ is induced by an ample Cartier divisor (with the birational morphism $g : W \to Y$, the contraction $f : W \to X$, and the ample Cartier divisor L), and $B_Y \in \mathcal{S}$ since B_Y is reduced. By Lemma 3.9(1) applied to $(Y, B_Y, c\overline{L})$ and h, we have $c \in \mathcal{N}_{\dim F} \subset \mathcal{N}_{d-1}$. □

In the proof of Lemma 5.2, we describe pseudo-effective thresholds by volumes, since volumes are constant in log smooth families [HMX13, Theorem 1.8(3)].

Lemma 5.2. *Let d, m, and r be positive integers. Let $\mathcal{P}$ be the set of triples (X, Δ, L), where X is a smooth projective variety of dimension d, Δ is a reduced simple normal crossing divisor on X, and L is an ample Cartier divisor on X, such that mL is very ample, $(mL)^d \leq r$, and $\Delta \cdot (mL)^{d-1} \leq r$. Then the set $\{\mathrm{pet}(X, \Delta; L) \mid (X, \Delta, L) \in \mathcal{P}\}$ is finite.*

Proof. **Step 1.** In this step, we construct finitely many families that contain the members of $\mathcal{P}$ as fibers.

For any $(X, \Delta, L) \in \mathcal{P}$, let $H \in |mL|$ be a general member. By [Har77, II, Theorem 8.18] applied to X and to any stratum of Δ, $\Delta + H$ is a reduced simple normal crossing divisor, so $(X, \Delta + H)$ is a log smooth couple. Moreover, mL is very ample,

$$(mL)^d \leq r, \quad \text{and} \quad (\Delta + H) \cdot (mL)^{d-1} = \Delta \cdot (mL)^{d-1} + (mL)^d \leq 2r.$$

Thus the couples $(X, \Delta + H)$ form a bounded set (Definition 2.11). By [Bir21a, 2.21], there exist finitely many projective morphisms $V^j \to T^j$ of varieties and reduced divisors C^j on V^j such that for any $(X, \Delta, L) \in \mathcal{P}$ and any general member $H \in |mL|$, there exist j, a closed point $s \in T^j$, and an isomorphism $\phi : V^j_s \to X$ such that $\phi_* C^j_s \geq \Delta + H$.

Fix j. By generic flatness [Eis95, Theorem 14.4] and Noetherian induction on T^j_{red}, we decompose T^j into finitely many smooth irreducible locally closed subsets. After finite base changes of the subsets and relabeling the components of C^j, we may assume that over each subset Σ, the variety V^j, any component of C^j, and any intersection of components of C^j are flat over Σ, and the fibers over Σ of any component of C^j are irreducible.

For any component C of C^j over Σ that dominates Σ, the irreducible components of the geometric generic fiber of $C \to \Sigma$ are defined over a finite extension K_C of the function field of Σ. After the base change by the normalization of Σ in the compositum of the fields K_C and relabeling the components, the generic fiber of any component D over Σ is geometrically irreducible. Let $\tilde{D} \to D$ be a resolution. Since the generic fiber of $\tilde{D} \to \Sigma$ is geometrically irreducible and Σ is normal, the finite part of the Stein factorization of $\tilde{D} \to \Sigma$ [Har77, III, Corollary 11.5] is an isomorphism, so the fibers of $\tilde{D} \to \Sigma$ are connected. By generic smoothness [Har77, III, Corollary 10.7], possibly shrinking Σ, the fibers of $\tilde{D} \to \Sigma$ are smooth and connected, hence irreducible, and so are the fibers of $D \to \Sigma$, which are their images. Since D is integral and the characteristic is 0, the generic fiber of $D \to \Sigma$ is geometrically reduced, and since $D \to \Sigma$ is proper and flat, possibly shrinking Σ, all fibers of $D \to \Sigma$ are geometrically reduced [Gro66, Théorème 12.2.4(v)]. The rest of Σ is treated by Noetherian induction.

Let I be a set of components of C^j. The points of $V^j_\Sigma := V^j \times_{T^j} \Sigma$ at which the fiber of V^j_Σ over Σ, or the fiber of an intersection of components in I, is not smooth form a closed subset. Since $V^j \to T^j$ is projective, the image of the closed subset in Σ is closed. Removing the image, we obtain an open subset $\Sigma_I \subset \Sigma$ over which V^j and any intersection of components in I are flat with smooth fibers, hence smooth [Har77, III, Theorem 10.2].

Let $(X, \Delta, L) \in \mathcal{P}$, let $H \in |mL|$ be a general member, let j, a closed point $s \in T^j$, and an isomorphism $\phi : V^j_s \to X$ satisfy $\phi_* C^j_s \geq \Delta + H$, and let Σ be the subset of T^j in the decomposition that contains s. For any component P of $\Delta + H$, fix a component C_P of C^j whose fiber over s is mapped by ϕ onto P. Let I_Δ (resp. I_H) be the set of the components C_P, where P runs over the components of Δ (resp. H), and let $I := I_\Delta \cup I_H$. Since the fibers over Σ of the components of C^j are irreducible and reduced, the fiber of each C_P over s is the reduced divisor $\phi^{-1}(P)$. Since $V^j_s \cong X$ is smooth and $\Delta + H$ is a simple normal crossing divisor, $s \in \Sigma_I$. Let $\mathfrak{D}$ (resp. $\mathfrak{H}$) be the sum of the components in I_Δ (resp. I_H), restricted over Σ_I. Then $\psi := \phi^{-1}$ satisfies $\psi(\Delta) = \mathfrak{D}_s$ and $\psi(H) = \mathfrak{H}_s$, so $mL \sim H = \psi^* \mathfrak{H}_s$. Since $V^j_{\Sigma_I}$ is smooth, $\mathfrak{H}$ is Cartier. By [Laz04, Theorem 1.2.17], the set Σ'_I of points of Σ_I over which the fiber of $\mathfrak{H}$ is ample is open in Σ_I. It contains s since $\mathfrak{H}_s$ is ample, and $\mathfrak{H}$ is ample over Σ'_I [Laz04, Theorem 1.7.8]. Let $T := \Sigma'_I$, $\mathfrak{X} := V^j \times_{T^j} T$, and restrict $\mathfrak{D}$ and $\mathfrak{H}$ to $\mathfrak{X}$. The triple $(\pi : \mathfrak{X} \to T, \mathfrak{D}, \mathfrak{H})$ depends only on j, the subset Σ, and the pair (I_Δ, I_H), and there are finitely many of them. Thus we obtain finitely many triples $(\pi : \mathfrak{X} \to T, \mathfrak{D}, \mathfrak{H})$ satisfying the following: T is a smooth irreducible variety,

π is a smooth projective morphism, $\mathfrak{D}$ is a reduced divisor on $\mathfrak{X}$ with simple normal crossings over T [HMX13, Section 2.1], and $\mathfrak{H}$ is a Cartier divisor on $\mathfrak{X}$ that is ample over T. Moreover, for any $(X, \Delta, L) \in \mathcal{P}$, there exist one of the triples, a closed point $s \in T$, and an isomorphism $\psi : X \to \mathfrak{X}_s$ such that $\psi(\Delta) = \mathfrak{D}_s$ and $mL \sim \psi^*\mathfrak{H}_s$.

Step 2. In this step, we show that $\mathrm{vol}(K_{\mathfrak{X}_s} + \mathfrak{D}_s + \lambda\mathfrak{H}_s)$ does not depend on $s \in T$ for any triple $(\pi : \mathfrak{X} \to T, \mathfrak{D}, \mathfrak{H})$ in **Step 1** and any positive rational number λ.

Fix a triple $(\pi : \mathfrak{X} \to T, \mathfrak{D}, \mathfrak{H})$ and a positive rational number λ. Since $\mathfrak{H}$ is ample over T, there exists an integer $k > \lambda$ such that $k\mathfrak{H}$ is very ample over T [Laz04, Theorem 1.7.6], $R^i\pi_*\mathcal{O}_{\mathfrak{X}}(k\mathfrak{H}) = 0$ for any $i > 0$, and the natural morphism

$$\pi^*\pi_*\mathcal{O}_{\mathfrak{X}}(k\mathfrak{H}) \to \mathcal{O}_{\mathfrak{X}}(k\mathfrak{H})$$

is surjective [Har77, III, Theorem 8.8]. By cohomology and base change [Har77, III, Theorem 12.11], $\pi_*\mathcal{O}_{\mathfrak{X}}(k\mathfrak{H})$ is locally free and commutes with base change. Thus for any closed point $s \in T$, the complete linear system $|k\mathfrak{H}_s|$ defines a closed embedding of $\mathfrak{X}_s$ into a projective space, so its members are hyperplane sections of $\mathfrak{X}_s$, and $|k\mathfrak{H}_s|$ is parametrized by the fiber over s of the projective bundle $\mathbb{P}(\mathcal{V}) \to T$, where $\mathcal{V} := \pi_*\mathcal{O}_{\mathfrak{X}}(k\mathfrak{H})$. Let $U \subset \mathbb{P}(\mathcal{V})$ be the open subset of the points parametrizing divisors $J_u \in |k\mathfrak{H}_s|$ that are smooth and transverse to any stratum of $\mathfrak{D}_s$ (U is open: for the projective families over $\mathbb{P}(\mathcal{V})$ cut out by the universal divisor on $\mathfrak{X} \times_T \mathbb{P}(\mathcal{V})$ and on the pullbacks of the strata of $\mathfrak{D}$, the points of $\mathbb{P}(\mathcal{V})$ over which the fibers are smooth of the expected dimension form an open set). By Bertini's theorem [Har77, III, Corollary 10.9 and Remark 10.9.2] applied to $\mathfrak{X}_s$ and to any stratum of $\mathfrak{D}_s$, the morphism $U \to T$ is surjective. Since U is a non-empty open subset of a projective bundle over T, U is smooth and irreducible. Let $\mathfrak{X}_U := \mathfrak{X} \times_T U$, let $\mathfrak{D}_U$ be the pullback of $\mathfrak{D}$ to $\mathfrak{X}_U$, and let $J \subset \mathfrak{X}_U$ be the universal divisor.

By **Step 1** and base change, $\mathfrak{X}_U$ and any stratum of $\mathfrak{D}_U$ are smooth over U. Let S be $\mathfrak{X}_U$ or a stratum of $\mathfrak{D}_U$. Since J is Cartier and $J \geq 0$, for any $u \in U$ over $s \in T$, the fiber $J_u \cap S_u$ of $J \cap S$ over u is a hyperplane section of S_u containing no component of S_u. Thus the Hilbert polynomial of $J_u \cap S_u$ with respect to $k\mathfrak{H}_s$ is $P_{S_u}(n) - P_{S_u}(n-1)$, where P_{S_u} is the Hilbert polynomial of S_u with respect to $k\mathfrak{H}_s$. Since S is flat over U, P_{S_u} does not depend on u. Since U is integral, $k\mathfrak{H}$ is very ample over T, and flatness is local on U, $J \cap S$ is flat over U [Har77, III, Theorem 9.9]. Since J_u is transverse to S_u, $J_u \cap S_u$ is smooth, so $J \cap S$ is smooth over U [Har77, III, Theorem 10.2]. Hence $(\mathfrak{X}_U, \mathfrak{D}_U + J)$ has simple normal crossings over U [HMX13, Section 2.1]. Since $\frac{\lambda}{k} < 1$, $(\mathfrak{X}_U, \mathfrak{D}_U + \frac{\lambda}{k}J)$ is lc and has simple normal crossings over U. By [HMX13, Theorem 1.8(3)],

$$\mathrm{vol}\left(K_{\mathfrak{X}_s} + \mathfrak{D}_s + \frac{\lambda}{k}J_u\right)$$

does not depend on $u \in U$, where s is the image of u in T. Since $J_u \sim k\mathfrak{H}_s$ and the volume depends only on the numerical class [Laz04, Proposition 2.2.41 and Corollary 2.2.45], we have

$$\mathrm{vol}\left(K_{\mathfrak{X}_s} + \mathfrak{D}_s + \frac{\lambda}{k}J_u\right) = \mathrm{vol}(K_{\mathfrak{X}_s} + \mathfrak{D}_s + \lambda\mathfrak{H}_s).$$

Since $U \to T$ is surjective, $\mathrm{vol}(K_{\mathfrak{X}_s} + \mathfrak{D}_s + \lambda\mathfrak{H}_s)$ does not depend on $s \in T$. The integer k, the open subset U, and the universal divisor J depend on λ.

Step 3. In this step, we conclude the proof.

Fix a triple $(\pi : \mathfrak{X} \to T, \mathfrak{D}, \mathfrak{H})$ in **Step 1**. By **Step 2**, for any positive rational number λ, there exists a real number $v(\lambda)$ such that

$$\mathrm{vol}(K_{\mathfrak{X}_s} + \mathfrak{D}_s + \lambda\mathfrak{H}_s) = v(\lambda)$$

for any closed point $s \in T$. Let $(X, \Delta, L) \in \mathcal{P}$, $s \in T$, and $\psi : X \to \mathfrak{X}_s$ be as in **Step 1**. Since $mL \sim \psi^*\mathfrak{H}_s$, for any positive rational number λ, we have $\mathrm{vol}(K_X + \Delta + \lambda mL) = v(\lambda)$. If $\lambda > \mathrm{pet}(X, \Delta; mL)$, then $K_X + \Delta + \lambda mL$ is the sum of a pseudo-effective divisor and an ample divisor, hence big [Laz04, Theorem 2.2.26]. If $\lambda < \mathrm{pet}(X, \Delta; mL)$, then $K_X + \Delta + \lambda mL$ is not

pseudo-effective, hence not big. Since a $\mathbb{Q}$-divisor D on X is big if and only if $\mathrm{vol}(D) > 0$ [Laz04, Section 2.2.C], we have

$$\mathrm{pet}(X, \Delta; mL) = \inf\{\lambda \in \mathbb{Q}_{>0} \mid v(\lambda) > 0\}.$$

Thus $\mathrm{pet}(X, \Delta; L) = m\,\mathrm{pet}(X, \Delta; mL)$ depends only on the triple. Since there exist finitely many triples, the set $\{\mathrm{pet}(X, \Delta; L) \mid (X, \Delta, L) \in \mathcal{P}\}$ is finite. □

6. Models of polarized log smooth pairs

In this section, we study the models in Set-up 6.1 and prove Proposition 6.5, which relates the generalized log discrepancies of $(Y, S, c\overline{L})$ to the log discrepancies of (X, Δ). Set-up 6.1 is used in Section 7, and Lemma 6.4 and Proposition 6.5 are used in Section 8 to control the generalized lc places of $(Y, S, c\overline{L})$. Lemma 6.4 is also used in Section 9.

Set-up 6.1. Let d, X, Δ, L, Y, W, f, g, K_W, S, P, α, c, u, and N be as follows.

(1) X is a smooth projective variety of dimension $d \geq 2$.
(2) $\Delta \neq 0$ is a reduced simple normal crossing divisor on X.
(3) L is an ample Cartier divisor on X.
(4) Y is a $\mathbb{Q}$-factorial klt projective variety and $X \dashrightarrow Y$ is a birational map such that no prime divisor on Y is exceptional over X and $\overline{L}_Y$ is ample.
(5) $f : W \to X$ and $g : W \to Y$ are projective birational morphisms from a smooth projective variety W resolving $X \dashrightarrow Y$.
(6) K_W is a canonical divisor on W such that $K_X = f_*K_W$ and $K_Y = g_*K_W$.
(7) S is a prime divisor on Y and P is a component of Δ such that $g_*f^*\Delta = S$ and $g_*f_*^{-1}P = S$.
(8) α, c, and u are real numbers such that $0 < \alpha < 1$ and $0 < c < u$.
(9) $N \geq \alpha\Delta$ is an $\mathbb{R}$-divisor on X such that

$$f^*(K_X + \Delta + uL) = g^*(K_Y + (1-\alpha)S + u\overline{L}_Y) + f^*N.$$

(10) $K_Y + (1-\alpha)S + u\overline{L}_Y \equiv 0$ and $K_Y + S + c\overline{L}_Y \equiv 0$.
(11) $(Y, S, u\overline{L})$ is generalized lc.

The condition in Set-up 6.1(4) that no prime divisor on Y is exceptional over X, together with condition (6), implies $g_*f^*K_X = K_Y$, so the $\mathbb{Q}$-divisor E' in Definition-Proposition 6.3 is exceptional over Y.

Lemma 6.2. *Let X be a smooth projective variety of dimension d and L an ample Cartier divisor on X. Then $K_X + (d+1)L$ is nef.*

Proof. By the cone theorem [KM98, Theorem 1.24], there exist rational curves C_i on X such that $0 < -K_X \cdot C_i \leq d+1$ for any i and

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{K_X \geq 0} + \sum_i \mathbb{R}_{\geq 0}[C_i].$$

Since L is an ample Cartier divisor, $L \cdot C_i \geq 1$, and hence

$$(K_X + (d+1)L) \cdot C_i \geq -(d+1) + (d+1) = 0$$

for any i. Since L is nef, $(K_X + (d+1)L) \cdot \gamma \geq K_X \cdot \gamma \geq 0$ for any $\gamma \in \overline{\mathrm{NE}}(X)_{K_X \geq 0}$. Thus $K_X + (d+1)L$ is nef. See also [And13, Proposition 4.2 and Theorem 5.1(1)]. □

Definition-Proposition 6.3. Let d be a positive integer, X a smooth projective variety of dimension d, L an ample Cartier divisor on X, and Y a normal projective variety such that K_Y is $\mathbb{Q}$-Cartier. Let $X \dashrightarrow Y$ be a birational map such that no prime divisor on Y is exceptional over X, and let $f : W \to X$ and $g : W \to Y$ be projective birational morphisms from a smooth

projective variety W resolving $X \dashrightarrow Y$, such that $K_X = f_*K_W$ and $K_Y = g_*K_W$ for a canonical divisor K_W on W. Assume that $\overline{L}_Y$ is $\mathbb{Q}$-Cartier, and let

$$E := g^*\overline{L}_Y - f^*L \quad \text{and} \quad E' := g^*\big(K_Y + (d+1)\overline{L}_Y\big) - f^*\big(K_X + (d+1)L\big).$$

Then E and E' are g-exceptional, $E \geq 0$, and $E' \geq 0$. Moreover, for any prime divisor F over X,

$$\operatorname{mult}_F E = e(F, Y, \overline{L}) \quad \text{and} \quad \operatorname{mult}_F E' = a(F, X, 0) - a(F, Y, 0) + (d+1)e(F, Y, \overline{L}).$$

In particular,

$$a(F, Y, 0) - a(F, X, 0) \leq (d+1)e(F, Y, \overline{L})$$

for any prime divisor F over X.

Proof. By Lemma 6.2, $K_X + (d+1)L$ is nef.

We show that $g_*f^*K_X = K_Y$. Since X is smooth and $K_X = f_*K_W$, the divisor $K_W - f^*K_X$ is f-exceptional. Let Q be a component of $K_W - f^*K_X$. Suppose that Q is not g-exceptional. Then $g(Q)$ is a prime divisor on Y. Since no prime divisor on Y is exceptional over X, $\operatorname{center}_X Q$ is a prime divisor on X, and Q is not f-exceptional, a contradiction. Thus $g_*(K_W - f^*K_X) = 0$. Since $K_Y = g_*K_W$, we have

$$g_*f^*K_X = K_Y.$$

The nef $\boldsymbol{b}$-divisor $\overline{L}$ descends to W with trace f^*L, so $\overline{L}_Y = g_*f^*L$. Therefore $g_*E = 0$ and

$$g_*E' = K_Y + (d+1)\overline{L}_Y - g_*f^*\big(K_X + (d+1)L\big) = 0.$$

In particular, E and E' are g-exceptional. Since f^*L and $f^*(K_X + (d+1)L)$ are nef, and $g^*\overline{L}_Y$ and $g^*(K_Y + (d+1)\overline{L}_Y)$ are numerically trivial over Y, $-E$ and $-E'$ are g-nef. Since K_Y and $\overline{L}_Y$ are $\mathbb{Q}$-Cartier, E and E' are $\mathbb{Q}$-divisors. By the negativity lemma [KM98, Lemma 3.39], $E \geq 0$ and $E' \geq 0$.

Let F be a prime divisor over X, and let $h : W' \to W$ be a projective birational morphism from a smooth variety W' such that F is a divisor on W'. Since $\overline{L}$ descends to W' with trace h^*f^*L, Definition-Proposition 2.4 implies

$$\operatorname{mult}_F E = \operatorname{mult}_F \big((g \circ h)^*\overline{L}_Y - h^*f^*L\big) = e(F, Y, \overline{L}).$$

Let $K_{W'}$ be the canonical divisor on W' such that $h_*K_{W'} = K_W$. Then $(f \circ h)_*K_{W'} = K_X$ and $(g \circ h)_*K_{W'} = K_Y$, so by the definition of log discrepancies,

$$\operatorname{mult}_F \big((g \circ h)^*K_Y - (f \circ h)^*K_X\big) = a(F, X, 0) - a(F, Y, 0).$$

Since $E' = g^*K_Y - f^*K_X + (d+1)E$, we obtain

$$\operatorname{mult}_F E' = a(F, X, 0) - a(F, Y, 0) + (d+1)e(F, Y, \overline{L}).$$

Since $E' \geq 0$, $\operatorname{mult}_F E' \geq 0$, which is the last inequality. □

Lemma 6.4. *Notation and conditions as in Set-up 6.1. Then $e(P', Y, \overline{L}) > 0$ for any prime divisor P' on X that is exceptional over Y. Moreover, any component $P' \neq P$ of Δ is exceptional over Y, and hence $e(P', Y, \overline{L}) > 0$.*

Proof. Since Y is klt, K_Y is $\mathbb{Q}$-Cartier, and since Y is $\mathbb{Q}$-factorial, $\overline{L}_Y$ is $\mathbb{Q}$-Cartier. Thus X, L, Y, f, g, and K_W satisfy the conditions of Definition-Proposition 6.3. Let E be the $\mathbb{Q}$-divisor in Definition-Proposition 6.3.

Let P' be a prime divisor on X that is exceptional over Y, and let $P'_W := f_*^{-1}P'$. Then P'_W is g-exceptional, and by Definition-Proposition 6.3,

$$e(P', Y, \overline{L}) = \operatorname{mult}_{P'_W} E \geq 0.$$

Suppose that $\operatorname{mult}_{P'_W} E = 0$. Since X is normal, f is an isomorphism over a neighborhood of the generic point of P'. Since P'_W is not a component of E, there exists a closed point $w \in P'_W \setminus \operatorname{Supp} E$ such that f is an isomorphism near w. Since P'_W is g-exceptional, $\dim g(P'_W) \leq d-2$, so $\dim Z \geq 1$ for any irreducible component Z of $g^{-1}(g(w)) \cap P'_W$ containing w. Cutting such a component Z

by general hyperplane sections through w, we obtain an irreducible projective curve $C \subset P'_W$ such that $w \in C$ and $g(C)$ is a point. Since $w \notin \operatorname{Supp} E$, $C \not\subset \operatorname{Supp} E$. Since $E \geq 0$ is $\mathbb{Q}$-Cartier, $E \cdot C \geq 0$. Since f is an isomorphism near w, $f(C)$ is a curve and $f_*C = f(C)$. Since $g(C)$ is a point and L is ample, the projection formula [Ful98, Proposition 2.5(c)] implies

$$E \cdot C = g^*\overline{L}_Y \cdot C - f^*L \cdot C = -L \cdot f(C) < 0,$$

a contradiction. Thus $e(P', Y, \overline{L}) > 0$.

Let $P' \neq P$ be a component of Δ. Suppose that P' is not exceptional over Y. Then $S' := g_*f_*^{-1}P'$ is a prime divisor on Y. The prime divisors $f_*^{-1}P' \neq f_*^{-1}P$ are not g-exceptional ($f_*^{-1}P$ by condition (7) of Set-up 6.1), so their images S' and S under the birational morphism g are distinct. Since $f^*\Delta \geq f_*^{-1}P + f_*^{-1}P'$, we have

$$g_*f^*\Delta \geq S + S',$$

which contradicts condition (7) of Set-up 6.1. Thus P' is exceptional over Y, and $e(P', Y, \overline{L}) > 0$. □

Proposition 6.5 uses the nef Cartier divisor $K_X + (d+1)L$ together with L, and with both divisors the correction terms on both sides of the identity in Proposition 6.5 have fixed signs, so, together with condition (11) of Set-up 6.1, all of them vanish at a generalized lc place of $(Y, S, c\overline{L})$. The identity is condition (9) of Set-up 6.1 divided by $1-\alpha$, so that S has coefficient 1, and rewritten using the $\mathbb{Q}$-divisors E and E' in Definition-Proposition 6.3.

Proposition 6.5. *Notation and conditions as in Set-up 6.1. Let E' be the $\mathbb{Q}$-divisor in Definition-Proposition 6.3, and let*

$$\sigma := \frac{\alpha}{1-\alpha} \quad \text{and} \quad \kappa := \sigma\Big(d+1-c-\frac{u-c}{\alpha}\Big).$$

Then $0 \leq \kappa \leq \frac{\alpha(d+1-c)}{1-\alpha}$, and for any prime divisor F over X,

$$a(F, Y, S, c\overline{L}) + \kappa e(F, Y, \overline{L}) - \sigma \operatorname{mult}_F E' = a(F, X, \Delta) + \frac{\operatorname{mult}_F(N - \alpha\Delta)}{1-\alpha}.$$

Moreover, $a(F, Y, S, c\overline{L}) = 0$ if and only if

$$a(F, X, \Delta) = \operatorname{mult}_F(N - \alpha\Delta) = e(F, Y, \overline{L}) = \operatorname{mult}_F E' = 0.$$

In this case, $\operatorname{center}_X F$ is a stratum of Δ, and $\operatorname{mult}_{P'} N = \alpha$ for any component P' of Δ containing $\operatorname{center}_X F$.

Proof. Since Y is klt and $\mathbb{Q}$-factorial, K_Y, S, and $\overline{L}_Y$ are $\mathbb{Q}$-Cartier. Thus X, L, Y, f, g, and K_W satisfy the conditions of Definition-Proposition 6.3. Let E be the $\mathbb{Q}$-divisor in Definition-Proposition 6.3. Since $0 < \alpha < 1$, $\sigma > 0$.

We first prove the bounds on κ. Subtracting $K_Y + S + c\overline{L}_Y \equiv 0$ from $K_Y + (1-\alpha)S + u\overline{L}_Y \equiv 0$ (condition (10) of Set-up 6.1), we obtain $\alpha S \equiv (u-c)\overline{L}_Y$, and hence

$$K_Y + (d+1)\overline{L}_Y \equiv (d+1-c)\overline{L}_Y - S \equiv \frac{\kappa}{\sigma}\overline{L}_Y.$$

By the definition of E',

$$g^*\big(K_Y + (d+1)\overline{L}_Y\big) = f^*\big(K_X + (d+1)L\big) + E',$$

and E' is g-exceptional by Definition-Proposition 6.3. By the projection formula [Ful98, Proposition 2.5(c)], $E' \cdot (g^*\overline{L}_Y)^{d-1} = 0$, and hence

$$\frac{\kappa}{\sigma}\overline{L}_Y^d = \big(K_Y + (d+1)\overline{L}_Y\big) \cdot \overline{L}_Y^{d-1} = f^*\big(K_X + (d+1)L\big) \cdot (g^*\overline{L}_Y)^{d-1} \geq 0,$$

where the inequality holds because $f^*(K_X + (d+1)L)$ is nef by Lemma 6.2 and $g^*\overline{L}_Y$ is nef [Laz04, Theorem 1.4.9 and Example 1.4.16]. Since $\overline{L}_Y$ is ample, $\overline{L}_Y^d > 0$. Thus $\kappa \geq 0$. Since $u > c$ and $\alpha > 0$,

$$\kappa \leq \sigma(d+1-c) = \frac{\alpha(d+1-c)}{1-\alpha}.$$

We prove the identity. By the definitions of E and E',

$$g^*\overline{L}_Y = f^*L + E \quad \text{and} \quad g^*K_Y = f^*K_X + E' - (d+1)E.$$

Substituting the expressions for $g^*\overline{L}_Y$ and g^*K_Y into condition (9) of Set-up 6.1, we obtain

$$f^*(N - \alpha\Delta) = (1-\alpha)(f^*\Delta - g^*S) - E' + (d+1-u)E. \tag{6.1}$$

Let F be a prime divisor over X, and write $e_F := e(F, Y, \overline{L})$. By Definition-Proposition 6.3, $\operatorname{mult}_F E = e_F$, so (6.1) implies

$$\operatorname{mult}_F(N - \alpha\Delta) = (1-\alpha)(\operatorname{mult}_F \Delta - \operatorname{mult}_F S) - \operatorname{mult}_F E' + (d+1-u)e_F. \tag{6.2}$$

By Definition-Proposition 2.4,

$$a(F, X, \Delta) = a(F, X, 0) - \operatorname{mult}_F \Delta \quad \text{and} \quad a(F, Y, S, c\overline{L}) = a(F, Y, 0) - \operatorname{mult}_F S - ce_F.$$

Together with the formula for $\operatorname{mult}_F E'$ in Definition-Proposition 6.3, we have

$$\operatorname{mult}_F \Delta - \operatorname{mult}_F S = a(F, Y, S, c\overline{L}) - a(F, X, \Delta) + \operatorname{mult}_F E' - (d+1-c)e_F.$$

Substituting the last equality into (6.2), we obtain

$$\begin{aligned}\operatorname{mult}_F(N - \alpha\Delta) = (1-\alpha)\big(a(F, Y, S, c\overline{L}) - a(F, X, \Delta)\big) - \alpha \operatorname{mult}_F E' \\ + \big((d+1-u) - (1-\alpha)(d+1-c)\big)e_F.\end{aligned}$$

Since

$$(d+1-u) - (1-\alpha)(d+1-c) = \alpha(d+1-c) - (u-c) = (1-\alpha)\kappa,$$

dividing by $1-\alpha$ implies

$$a(F, Y, S, c\overline{L}) + \kappa e_F - \sigma \operatorname{mult}_F E' = a(F, X, \Delta) + \frac{\operatorname{mult}_F(N - \alpha\Delta)}{1-\alpha}.$$

We characterize the generalized lc places of $(Y, S, c\overline{L})$. By Definition-Proposition 2.4, $e_F \geq 0$, and by condition (11) of Set-up 6.1,

$$0 \leq a(F, Y, S, u\overline{L}) = a(F, Y, S, c\overline{L}) - (u-c)e_F.$$

Suppose that $a(F, Y, S, c\overline{L}) = 0$. Since $u > c$, $e_F = 0$. The identity then reads

$$-\sigma \operatorname{mult}_F E' = a(F, X, \Delta) + \frac{\operatorname{mult}_F(N - \alpha\Delta)}{1-\alpha}.$$

Since $\sigma > 0$ and $E' \geq 0$, the left-hand side is non-positive. By Lemma 2.7, $a(F, X, \Delta) \geq 0$, and since $N - \alpha\Delta \geq 0$, $\operatorname{mult}_F(N - \alpha\Delta) \geq 0$. Thus both sides are zero, and

$$a(F, X, \Delta) = \operatorname{mult}_F(N - \alpha\Delta) = e_F = \operatorname{mult}_F E' = 0.$$

Conversely, if $a(F, X, \Delta) = \operatorname{mult}_F(N - \alpha\Delta) = e_F = \operatorname{mult}_F E' = 0$, then the identity implies $a(F, Y, S, c\overline{L}) = 0$.

Finally, assume that $a(F, Y, S, c\overline{L}) = 0$. Since $a(F, X, \Delta) = 0$, $\operatorname{center}_X F$ is a stratum of Δ by Lemma 2.7. Let P' be a component of Δ containing $\operatorname{center}_X F$. Then $\operatorname{mult}_F P' \geq 1$. Since $N - \alpha\Delta - (\operatorname{mult}_{P'} N - \alpha)P' \geq 0$ and $\operatorname{mult}_{P'} N - \alpha \geq 0$,

$$0 = \operatorname{mult}_F(N - \alpha\Delta) \geq (\operatorname{mult}_{P'} N - \alpha) \operatorname{mult}_F P' \geq 0.$$

Thus $\operatorname{mult}_{P'} N = \alpha$. □

7. Thresholds approaching an accumulation point

In this section, we prove Theorem 7.1, which is step (i) of the sketch of the proof of Theorem 1.2(1) in Section 1. The proof excludes nine cases, each of which would hold for infinitely many i, and only the case in **Step 6**(b) needs the new Proposition 4.4. Of the other eight, the five in **Step 2** (two cases), **Step 3**, **Step 5**, and the anti-ample case of **Step 6**(c) are excluded by Proposition 3.6 or Lemma 3.7, and the three in **Step 4**, **Step 6**(a), and the ample case of **Step 6**(c) are excluded by known theorems.

Theorem 7.1. *Let $d \geq 2$ be an integer and $c \in \mathrm{Acc}(\mathrm{PET}_d)$ a real number such that $c > 0$ and $c \notin \mathcal{N}_{d-2}$. Then there exist a strictly decreasing sequence of real numbers u_i with $\lim_{i\to+\infty} u_i = c$ and, for each i, data X_i, Δ_i, L_i, Y_i, W_i, f_i, g_i, K_{W_i}, S_i, P_i, α_i, N_i as in Set-up 6.1 for the given d and c and with $u = u_i$, such that $\mathrm{pet}(X_i, \Delta_i; L_i) = u_i$, $\rho(Y_i) = 1$, and $\lim_{i\to+\infty} \alpha_i = 0$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we choose the sequence u_i and the triples (X_i, Δ_i, L_i).

For a smooth projective variety X and a reduced simple normal crossing divisor Δ on X, the pair (X, Δ) is lc by Lemma 2.7. Thus [HL20, Theorem 1.4], applied to lc pairs (X, Δ) of dimension d with $\Delta \in \{1\}$ and to ample Cartier divisors L on X, implies that PET_d satisfies the ACC. Since $c \in \mathrm{Acc}(\mathrm{PET}_d)$, there exist pairwise distinct real numbers $u_i \in \mathrm{PET}_d$ with $\lim_{i\to+\infty} u_i = c$. Since PET_d satisfies the ACC, no subsequence of (u_i) is strictly increasing. Thus, possibly passing to a subsequence, we may assume that u_i is strictly decreasing, and hence $u_i > c$ for any i. For each i, there exist a smooth projective variety X_i with $\dim X_i = d$, a reduced simple normal crossing divisor Δ_i on X_i (possibly $\Delta_i = 0$), and an ample Cartier divisor L_i on X_i, such that $\mathrm{pet}(X_i, \Delta_i; L_i) = u_i$.

Below we pass to subsequences several times. After each time we renumber the sequence, so the u_i are pairwise distinct and strictly decreasing with limit c.

Step 2. In this step we construct, for each i, a $\mathbb{Q}$-factorial klt projective variety $Y_{i,1}$ such that $(Y_{i,1}, 0, u_i\overline{L_i})$ is generalized klt, $K_{Y_{i,1}} + u_i\overline{L_i}_{Y_{i,1}} \equiv 0$, no prime divisor on $Y_{i,1}$ is exceptional over X_i, and any component of Δ_i is exceptional over $Y_{i,1}$.

By Proposition 5.1, applied to (X_i, Δ_i, L_i), whose pseudo-effective threshold is $\mathrm{pet}(X_i, \Delta_i; L_i) = u_i$, there exist a $\mathbb{Q}$-factorial klt projective variety $Y_{i,1}$, a reduced divisor $B_{i,1}$ on $Y_{i,1}$, projective birational morphisms $f_i : W_i \to X_i$ and $g_{i,1} : W_i \to Y_{i,1}$ from a smooth projective variety W_i, a canonical divisor K_{W_i} on W_i, and a $g_{i,1}$-exceptional $\mathbb{R}$-divisor $G_i \geq 0$ on W_i, such that

(1) $B_{i,1} = g_{i,1*}f_{i*}^{-1}\Delta_i + E_{i,1}$, where $E_{i,1}$ is the sum of the prime divisors on $Y_{i,1}$ that are exceptional over X_i,
(2) $(Y_{i,1}, B_{i,1}, u_i\overline{L_i})$ is generalized lc, $K_{X_i} = f_{i*}K_{W_i}$, $K_{Y_{i,1}} = g_{i,1*}K_{W_i}$, and

$$f_i^*(K_{X_i} + \Delta_i + u_iL_i) = g_{i,1}^*(K_{Y_{i,1}} + B_{i,1} + u_i\overline{L_i}_{Y_{i,1}}) + G_i, \tag{7.1}$$

(3) if $K_{Y_{i,1}} + B_{i,1} + u_i\overline{L_i}_{Y_{i,1}} \not\equiv 0$, then $u_i \in \mathcal{N}_{d-1}$.

Whenever we replace W_i with a higher model $h : W_i'' \to W_i$ from a smooth projective variety, we replace K_{W_i} with the canonical divisor $K_{W_i''}$ such that $h_*K_{W_i''} = K_{W_i}$, and we replace f_i, $g_{i,1}$, and G_i with $f_i \circ h$, $g_{i,1} \circ h$, and h^*G_i, so (7.1) still holds. The canonical divisor of any normal projective variety birational to X_i below is the pushforward of K_{W_i}, as in Set-up 6.1(6).

Let Y' be a $\mathbb{Q}$-factorial normal projective variety birational to X_i, and let W' be a smooth projective variety with projective birational morphisms $f' : W' \to X_i$ and $g' : W' \to Y'$. Since X_i is normal and f' is projective and birational, $f'_*\mathcal{O}_{W'}$ is a coherent $\mathcal{O}_{X_i}$-algebra contained in the function field of X_i, hence $f'_*\mathcal{O}_{W'} = \mathcal{O}_{X_i}$, and f' is a contraction. Thus g', f', and L_i show that $\overline{L_i} = \overline{f'^*L_i}$ is induced by an ample Cartier divisor as a nef $\boldsymbol{b}$-divisor over Y' (Definition 3.1). Moreover, $\overline{L_i}_{Y'} = g'_*f'^*L_i$ is the pushforward of the big divisor f'^*L_i, so $\overline{L_i}_{Y'}$ is a big $\mathbb{Q}$-Cartier divisor. In particular, $\overline{L_i}_{Y'} \not\equiv 0$.

Suppose that $K_{Y_{i,1}}+B_{i,1}+u_i\overline{L_i}_{Y_{i,1}} \not\equiv 0$ for infinitely many i. Then $u_i \in \mathcal{N}_{d-1}$ for infinitely many i, and the u_i are pairwise distinct with limit $c>0$. By Lemma 3.7, $c \in \mathcal{N}_{d-2}$, a contradiction. Thus, possibly passing to a subsequence, we may assume that $K_{Y_{i,1}} + B_{i,1} + u_i\overline{L_i}_{Y_{i,1}} \equiv 0$ for any i. Then $(Y_{i,1}, B_{i,1}, u_i\overline{L_i})$ satisfies condition (†): it is generalized lc, $B_{i,1} \in \mathcal{S}$ since $B_{i,1}$ is reduced, $\overline{L_i}$ is induced by an ample Cartier divisor and $\overline{L_i}_{Y_{i,1}}$ is $\mathbb{Q}$-Cartier, and $\overline{L_i}_{Y_{i,1}} \not\equiv 0$. Suppose that $(Y_{i,1}, B_{i,1}, u_i\overline{L_i})$ is not generalized klt for infinitely many i. Then Lemma 3.10 implies that $u_i \in \mathcal{N}_{d-1}$ for infinitely many i, and Lemma 3.7 implies that $c \in \mathcal{N}_{d-2}$, a contradiction. Thus, possibly passing to a subsequence, we may assume that $(Y_{i,1}, B_{i,1}, u_i\overline{L_i})$ is generalized klt for any i. If $B_{i,1} \neq 0$, then for a component Q of $B_{i,1}$, $a(Q, Y_{i,1}, B_{i,1}, u_i\overline{L_i}) = 1 - \operatorname{mult}_Q B_{i,1} = 0$, which contradicts the fact that $(Y_{i,1}, B_{i,1}, u_i\overline{L_i})$ is generalized klt. Thus $B_{i,1} = 0$, so $E_{i,1} = 0$ and $g_{i,1*}f_{i*}^{-1}\Delta_i = 0$. In other words, no prime divisor on $Y_{i,1}$ is exceptional over X_i, and any component of Δ_i is exceptional over $Y_{i,1}$.

Step 3. In this step we run a K-MMP and obtain a $\mathbb{Q}$-factorial klt projective variety $Y_{i,2}$ such that $\rho(Y_{i,2}) = 1$.

Since $Y_{i,1}$ is klt, the generalized pair $(Y_{i,1}, 0, \mathbf{0})$ is generalized klt. We run a $K_{Y_{i,1}}$-MMP with scaling of a general ample divisor. Since $K_{Y_{i,1}} \equiv -u_i\overline{L_i}_{Y_{i,1}}$ and $\overline{L_i}_{Y_{i,1}}$ is big, $K_{Y_{i,1}}$ is not pseudo-effective. By [BZ16, Lemma 4.4(1)], under condition (i) there, the MMP terminates with a Mori fiber space $Y_{i,2} \to T_i$. The variety $Y_{i,2}$ is $\mathbb{Q}$-factorial, klt, and projective [KM98, Propositions 3.36, 3.37 and Corollaries 3.42, 3.43]. Since any step of the MMP is a divisorial contraction or a flip, no prime divisor on $Y_{i,2}$ is exceptional over $Y_{i,1}$, so by **Step 2**, no prime divisor on $Y_{i,2}$ is exceptional over X_i, and any component of Δ_i is exceptional over $Y_{i,2}$.

Possibly replacing W_i with a higher model, we may assume that there exists a morphism $g_{i,2} : W_i \to Y_{i,2}$. Since no prime divisor on $Y_{i,2}$ is exceptional over $Y_{i,1}$, any $g_{i,1}$-exceptional prime divisor on W_i is $g_{i,2}$-exceptional. Since $K_{Y_{i,1}} = g_{i,1*}K_{W_i}$ and $\overline{L_i}_{Y_{i,1}} = g_{i,1*}f_i^*L_i$, the $\mathbb{R}$-divisor

$$g_{i,1}^*(K_{Y_{i,1}} + u_i\overline{L_i}_{Y_{i,1}}) - K_{W_i} - u_i f_i^* L_i$$

is $g_{i,1}$-exceptional. Thus

$$g_{i,2*}g_{i,1}^*(K_{Y_{i,1}} + u_i\overline{L_i}_{Y_{i,1}}) = g_{i,2*}(K_{W_i} + u_i f_i^* L_i) = K_{Y_{i,2}} + u_i\overline{L_i}_{Y_{i,2}}.$$

Let

$$D_i := g_{i,1}^*(K_{Y_{i,1}} + u_i\overline{L_i}_{Y_{i,1}}) - g_{i,2}^*(K_{Y_{i,2}} + u_i\overline{L_i}_{Y_{i,2}}).$$

Then D_i is $g_{i,2}$-exceptional, and D_i is numerically trivial over $Y_{i,2}$ since $K_{Y_{i,1}} + u_i\overline{L_i}_{Y_{i,1}} \equiv 0$. Applying the negativity lemma [BCHM10, Lemma 3.6.2] to D_i and $-D_i$, we obtain $D_i = 0$, that is,

$$g_{i,1}^*(K_{Y_{i,1}} + u_i\overline{L_i}_{Y_{i,1}}) = g_{i,2}^*(K_{Y_{i,2}} + u_i\overline{L_i}_{Y_{i,2}}). \tag{7.2}$$

Thus $K_{Y_{i,2}} + u_i\overline{L_i}_{Y_{i,2}} \equiv 0$. Moreover, for any prime divisor E over X_i,

$$a(E, Y_{i,1}, 0, u_i\overline{L_i}) = a(E, Y_{i,2}, 0, u_i\overline{L_i}),$$

since both sides are computed from (7.2) on a higher model of W_i on which E is a divisor. In particular, $(Y_{i,2}, 0, u_i\overline{L_i})$ is generalized klt.

Suppose that $\dim T_i > 0$ for infinitely many i, and let F be a general fiber of $Y_{i,2} \to T_i$ for such an i. Then $1 \leq \dim F \leq d-1$ and $(K_{Y_{i,2}} + u_i\overline{L_i}_{Y_{i,2}})|_F \equiv 0$. Since $-K_{Y_{i,2}}$ is ample over T_i, $u_i\overline{L_i}_{Y_{i,2}}|_F \equiv -K_{Y_{i,2}}|_F$ is ample, so $\overline{L_i}_{Y_{i,2}}|_F \not\equiv 0$. Since $(Y_{i,2}, 0, u_i\overline{L_i})$ is generalized lc, $Y_{i,2}$ is $\mathbb{Q}$-factorial, and $\overline{L_i}$ is induced by an ample Cartier divisor (**Step 2**), Lemma 3.9(1) implies that $u_i \in \mathcal{N}_{\dim F}$. By Lemma 3.5(3), $u_i \in \mathcal{N}_{d-1}$ for infinitely many i, and Lemma 3.7 implies that $c \in \mathcal{N}_{d-2}$, a contradiction. Thus, possibly passing to a subsequence, we may assume that T_i is a point for any i. Hence $\rho(Y_{i,2}) = 1$, $-K_{Y_{i,2}}$ is ample, and $\overline{L_i}_{Y_{i,2}} \equiv -\frac{1}{u_i}K_{Y_{i,2}}$ is ample.

Step 4. In this step we show that $G_i = f_i^*N_i$ for an $\mathbb{R}$-divisor $N_i \geq 0$ on X_i, that $\Delta_i \neq 0$, and that, possibly passing to a subsequence, the smallest coefficient α_i of N_i along the components of Δ_i satisfies $0 < \alpha_i < 1$ and $\lim_{i\to+\infty} \alpha_i = 0$.

Since $B_{i,1} = 0$, (7.1) and (7.2) imply that

$$f_i^*(K_{X_i} + \Delta_i + u_iL_i) = g_{i,2}^*(K_{Y_{i,2}} + u_i\overline{L}_{iY_{i,2}}) + G_i. \tag{7.3}$$

Let $N_i := f_{i*}G_i$. Then $N_i \geq 0$. Since $K_{Y_{i,2}} + u_i\overline{L}_{iY_{i,2}} \equiv 0$, (7.3) implies that G_i is numerically trivial over X_i. Thus $G_i - f_i^*N_i$ is f_i-exceptional and numerically trivial over X_i. Applying the negativity lemma [BCHM10, Lemma 3.6.2] to $G_i - f_i^*N_i$ and $f_i^*N_i - G_i$, we obtain $G_i = f_i^*N_i$.

Let E be a prime divisor over X_i. Possibly replacing W_i with a higher model, we may assume that E is a divisor on W_i. Write

$$K_{W_i} + \Delta_{W_i} := f_i^*(K_{X_i} + \Delta_i) \quad \text{and} \quad K_{W_i} + B_{W_i} + u_if_i^*L_i := g_{i,2}^*(K_{Y_{i,2}} + u_i\overline{L_i}_{Y_{i,2}}).$$

Since $\overline{L_i}$ descends to W_i with trace $f_i^*L_i$, (7.3) and $G_i = f_i^*N_i$ imply that $\Delta_{W_i} = B_{W_i} + f_i^*N_i$. Thus

$$a(E, Y_{i,2}, 0, u_i\overline{L_i}) = a(E, X_i, \Delta_i) + \operatorname{mult}_E N_i. \tag{7.4}$$

If $\Delta_i = 0$, then $a(E, X_i, 0) \geq 1$ [KM98, Corollary 2.31(2)], so $a(E, Y_{i,2}, 0, u_i\overline{L_i}) \geq 1$ by (7.4). Assume that $\Delta_i \neq 0$. Let α_i be the minimum of $\operatorname{mult}_D N_i$, where D runs over the components of Δ_i, and let P_i be a component of Δ_i such that $\operatorname{mult}_{P_i} N_i = \alpha_i$. Since Δ_i is reduced, $N_i \geq \alpha_i\Delta_i$. We show that

$$a(E, Y_{i,2}, 0, u_i\overline{L_i}) \geq \min\{1, \alpha_i\} \quad \text{and} \quad a(P_i, Y_{i,2}, 0, u_i\overline{L_i}) = \alpha_i. \tag{7.5}$$

The equality follows from (7.4), since $a(P_i, X_i, \Delta_i) = 0$. For the inequality, assume that $a(E, Y_{i,2}, 0, u_i\overline{L_i}) < 1$. By Lemma 2.7, $a(E, X_i, \Delta_i)$ is a non-negative integer. Since $\operatorname{mult}_E N_i \geq 0$, (7.4) implies that $a(E, X_i, \Delta_i) = 0$, and Lemma 2.7 implies that $\operatorname{center}_{X_i} E$ is a stratum of Δ_i. Thus $\operatorname{center}_{X_i} E \subset D$ for a component D of Δ_i. Since $N_i \geq (\operatorname{mult}_D N_i)D$ and $\operatorname{mult}_E D$ is a positive integer, we obtain

$$a(E, Y_{i,2}, 0, u_i\overline{L_i}) \geq \operatorname{mult}_E N_i \geq \operatorname{mult}_D N_i \geq \alpha_i.$$

Suppose that there exist a real number $0 < \epsilon \leq 1$ and infinitely many i such that $\Delta_i = 0$ or $\alpha_i \geq \epsilon$. Possibly passing to a subsequence, we may assume that $\Delta_i = 0$ or $\alpha_i \geq \epsilon$ for any i. Since any prime divisor over $Y_{i,2}$ is a prime divisor over X_i, (7.5) implies that $(Y_{i,2}, 0, u_i\overline{L_i})$ is generalized ϵ-lc. By Definition-Proposition 2.4, for any prime divisor E over $Y_{i,2}$,

$$a(E, Y_{i,2}, 0) = a(E, Y_{i,2}, 0, u_i\overline{L_i}) + u_ie(E, Y_{i,2}, \overline{L_i}) \geq a(E, Y_{i,2}, 0, u_i\overline{L_i}),$$

so $Y_{i,2}$ is ϵ-lc. Since $-K_{Y_{i,2}}$ is ample (**Step 3**), Theorem 2.12 and Definition 2.11 imply that there exists a positive integer r depending only on d and ϵ such that, for any i, there exists a very ample Cartier divisor A_i on $Y_{i,2}$ with $A_i^d \leq r$.

Let Γ_i be the intersection of $d-1$ general members of $|A_i|$. Since $Y_{i,2}$ is normal, the singular locus of $Y_{i,2}$ has codimension at least 2. Thus Γ_i is a smooth projective curve contained in the smooth locus of $Y_{i,2}$ [Har77, III, Corollary 10.9 and Remark 10.9.2]. Let $\Gamma_{i1}, \dots, \Gamma_{ik_i}$ be the connected components of Γ_i. Since $(A_i \cdot \Gamma_{ij})$ is a positive integer for any j and $\sum_{j=1}^{k_i}(A_i \cdot \Gamma_{ij}) = A_i^d$, we have $k_i \leq A_i^d$. By adjunction [Har77, II, Proposition 8.20], $K_{\Gamma_{ij}} = (K_{Y_{i,2}} + (d-1)A_i)|_{\Gamma_{ij}}$, and $\deg K_{\Gamma_{ij}} \geq -2$, since Γ_{ij} is a smooth connected projective curve [Har77, IV, Example 1.3.3]. Thus

$$\sum_{j=1}^{k_i} \deg K_{\Gamma_{ij}} = (K_{Y_{i,2}} + (d-1)A_i) \cdot A_i^{d-1} \geq -2k_i \geq -2A_i^d,$$

and hence

$$0 < -K_{Y_{i,2}} \cdot A_i^{d-1} \leq (d+1)A_i^d \leq (d+1)r.$$

Any Weil divisor on $Y_{i,2}$ is Cartier along Γ_i. Thus

$$-K_{Y_{i,2}} \cdot A_i^{d-1} = \deg(-K_{Y_{i,2}}|_{\Gamma_i}) \quad \text{and} \quad \overline{L_i}_{Y_{i,2}} \cdot A_i^{d-1} = \deg(\overline{L_i}_{Y_{i,2}}|_{\Gamma_i})$$

are integers, where we use that $\overline{L_i}_{Y_{i,2}} = g_{i,2*}f_i^*L_i$ has integer coefficients. Since $\overline{L_i}_{Y_{i,2}}$ is ample, $\overline{L_i}_{Y_{i,2}} \cdot A_i^{d-1}$ is a positive integer. Since $K_{Y_{i,2}} + u_i\overline{L_i}_{Y_{i,2}} \equiv 0$,

$$u_i = \frac{-K_{Y_{i,2}} \cdot A_i^{d-1}}{\overline{L_i}_{Y_{i,2}} \cdot A_i^{d-1}}.$$

Since $u_i > c$, we have $\overline{L_i}_{Y_{i,2}} \cdot A_i^{d-1} < \frac{(d+1)r}{c}$. Thus u_i takes only finitely many values, which contradicts the fact that the u_i are pairwise distinct.

Therefore, $\Delta_i \neq 0$ for $i \gg 0$ and $\lim_{i\to+\infty} \alpha_i = 0$. Possibly passing to a subsequence, we may assume that $\Delta_i \neq 0$ and $\alpha_i < 1$ for any i. Since $(Y_{i,2}, 0, u_i\overline{L_i})$ is generalized klt, (7.5) implies that $\alpha_i > 0$. By **Step 3**, P_i is exceptional over $Y_{i,2}$.

Step 5. In this step we extract P_i, run an MMP, and obtain Y_i with $\rho(Y_i) = 1$ together with the prime divisor S_i on Y_i.

The generalized pair $(Y_{i,2}, 0, u_i\overline{L_i})$ is generalized klt, $Y_{i,2}$ is $\mathbb{Q}$-factorial and klt, P_i is exceptional over $Y_{i,2}$, and $a(P_i, Y_{i,2}, 0, u_i\overline{L_i}) = \alpha_i < 1$ by (7.5). By Lemma 2.8, there exists an extremal divisorial contraction $h_i : Y_{i,3} \to Y_{i,2}$ from a $\mathbb{Q}$-factorial klt projective variety $Y_{i,3}$ such that P_i is the unique h_i-exceptional prime divisor and

$$K_{Y_{i,3}} + (1-\alpha_i)S_{i,3} + u_i\overline{L_i}_{Y_{i,3}} = h_i^*(K_{Y_{i,2}} + u_i\overline{L_i}_{Y_{i,2}}), \tag{7.6}$$

where $S_{i,3}$ is the prime divisor on $Y_{i,3}$ corresponding to P_i. Thus

$$K_{Y_{i,3}} + (1-\alpha_i)S_{i,3} + u_i\overline{L_i}_{Y_{i,3}} \equiv 0,$$

and, by (7.6), for any prime divisor E over $Y_{i,3}$,

$$a(E, Y_{i,3}, (1-\alpha_i)S_{i,3}, u_i\overline{L_i}) = a(E, Y_{i,2}, 0, u_i\overline{L_i}).$$

In particular, $(Y_{i,3}, (1-\alpha_i)S_{i,3}, u_i\overline{L_i})$ is generalized klt, and so is $(Y_{i,3}, 0, u_i\overline{L_i})$ by Definition-Proposition 2.4. The only prime divisor on $Y_{i,3}$ that is not a divisor on $Y_{i,2}$ is $S_{i,3}$, which corresponds to the prime divisor P_i on X_i. Thus no prime divisor on $Y_{i,3}$ is exceptional over X_i, and any component of Δ_i other than P_i is exceptional over $Y_{i,3}$.

Since $0 < \alpha_i < 1$, $S_{i,3}$ is a prime divisor, and

$$K_{Y_{i,3}} + u_i\overline{L_i}_{Y_{i,3}} \equiv -(1-\alpha_i)S_{i,3},$$

the $\mathbb{R}$-divisor $K_{Y_{i,3}} + u_i\overline{L_i}_{Y_{i,3}}$ is not pseudo-effective. We run a $(K_{Y_{i,3}} + u_i\overline{L_i}_{Y_{i,3}})$-MMP for the generalized klt pair $(Y_{i,3}, 0, u_i\overline{L_i})$ with scaling of a general ample divisor. By [BZ16, Lemma 4.4(1)], under condition (i) there, the MMP terminates with a Mori fiber space $Y_i' \to T_i'$ (by the definition of the MMP for generalized pairs in [BZ16, Section 4], each step of the MMP is a divisorial contraction or a flip of a $(K_{Y''} + B'')$-negative extremal ray for a klt pair (Y'', B''), where Y'' is the variety before the step, so the varieties appearing in the MMP are $\mathbb{Q}$-factorial [KM98, Propositions 3.36 and 3.37]). Let Y'' be a variety appearing in the MMP and S'' the birational transform of $S_{i,3}$ on Y''. No prime divisor on Y'' is exceptional over $Y_{i,3}$. The argument of **Step 3**, with $Y_{i,3}$, Y'', and $K_{Y_{i,3}} + (1-\alpha_i)S_{i,3} + u_i\overline{L_i}_{Y_{i,3}}$ in place of $Y_{i,1}$, $Y_{i,2}$, and $K_{Y_{i,1}} + u_i\overline{L_i}_{Y_{i,1}}$, shows that

$$K_{Y''} + (1-\alpha_i)S'' + u_i\overline{L_i}_{Y''} \equiv 0$$

and that the pullbacks of $K_{Y_{i,3}} + (1-\alpha_i)S_{i,3} + u_i\overline{L_i}_{Y_{i,3}}$ and $K_{Y''} + (1-\alpha_i)S'' + u_i\overline{L_i}_{Y''}$ to a common resolution coincide.

We show that $S_{i,3}$ is not contracted by the MMP. Suppose that a step of the MMP is a divisorial contraction $Y'' \to Y'''$ of an extremal ray R whose exceptional divisor is S''. Then

$$(K_{Y''} + u_i\overline{L_i}_{Y''}) \cdot R < 0 \quad \text{and} \quad K_{Y''} + (1-\alpha_i)S'' + u_i\overline{L_i}_{Y''} \equiv 0,$$

so $S'' \cdot R > 0$. Since $\rho(Y''/Y''') = 1$, S'' is nef over Y'''. Since the pushforward of $-S''$ to Y''' is zero, the negativity lemma [BCHM10, Lemma 3.6.2] implies that $-S'' \geq 0$, a contradiction. Since flips are isomorphisms in codimension one, the birational transform S'_i of $S_{i,3}$ on Y'_i is a prime divisor. Thus

$$K_{Y'_i} + (1-\alpha_i)S'_i + u_i\overline{L}_{iY'_i} \equiv 0,$$

$(Y'_i, (1-\alpha_i)S'_i, u_i\overline{L_i})$ is generalized klt, no prime divisor on Y'_i is exceptional over X_i, and any component of Δ_i other than P_i is exceptional over Y'_i.

Suppose that $\dim T'_i > 0$ for infinitely many i. Possibly passing to a subsequence, we may assume that $\dim T'_i > 0$ for any i. Possibly replacing W_i with a higher model, we may assume that there exists a morphism $g'_i : W_i \to Y'_i$ and that the union of $\mathrm{Exc}(g'_i)$ and the birational transform of S'_i on W_i is a simple normal crossing divisor. Let $t_i \in T'_i$ be a general point, let F_i be the fiber of $Y'_i \to T'_i$ over t_i, and let $\tilde{F}_i$ be the fiber of $W_i \to T'_i$ over t_i. Then $1 \leq \dim F_i \leq d-1$. The morphism $W_i \to T'_i$ is the composition of g'_i and the Mori fiber space $Y'_i \to T'_i$, and both have connected fibers (for g'_i by [Har77, III, Corollary 11.4]), so $\tilde{F}_i$ is connected. By [Har77, III, Corollary 10.7], applied to $W_i \to T'_i$, $\tilde{F}_i$ is smooth. Since $\tilde{F}_i$ is smooth and connected, $\tilde{F}_i$ is irreducible, so $\tilde{F}_i$ is a smooth projective variety. Since Y'_i is normal, $g'_i(\mathrm{Exc}(g'_i))$ has codimension at least 2 in Y'_i, and g'_i is an isomorphism over $Y'_i \backslash g'_i(\mathrm{Exc}(g'_i))$. Since F_i is general, $F_i \not\subset g'_i(\mathrm{Exc}(g'_i))$, so the induced morphism $p_i : \tilde{F}_i \to F_i$ is birational.

We show that $\overline{L}_{iY'_i}|_{F_i} \not\equiv 0$. Since Y'_i is $\mathbb{Q}$-factorial and no prime divisor on Y'_i is exceptional over X_i, Definition-Proposition 6.3, applied to X_i, L_i, Y'_i, f_i, g'_i, and K_{W_i}, implies that $g'^*_i\overline{L}_{iY'_i} - f_i^*L_i \geq 0$. Let $V_i \subset Y'_i$ be the union of $g'_i(\mathrm{Exc}(g'_i))$, $g'_i(\mathrm{Exc}(f_i))$, and $g'_i(\mathrm{Supp}(g'^*_i\overline{L}_{iY'_i} - f_i^*L_i))$. Then V_i is a proper closed subset of Y'_i, so $F_i \not\subset V_i$ since F_i is general. Let $C_i \subset F_i$ be an irreducible curve such that $C_i \not\subset V_i$, and let $\tilde{C}_i \subset W_i$ be the birational transform of C_i. Then $\tilde{C}_i$ is not contained in $\mathrm{Exc}(f_i)$ or in $\mathrm{Supp}(g'^*_i\overline{L}_{iY'_i} - f_i^*L_i)$, and $f_i(\tilde{C}_i)$ is a curve. By the projection formula [Ful98, Proposition 2.5(c)],

$$(\overline{L}_{iY'_i} \cdot C_i) = (g'^*_i\overline{L}_{iY'_i} \cdot \tilde{C}_i) \geq (f_i^*L_i \cdot \tilde{C}_i) = (L_i \cdot f_{i*}\tilde{C}_i) > 0.$$

Thus $\overline{L}_{iY'_i}|_{F_i} \not\equiv 0$.

By [BZ16, Remark 4.2(6)], the restriction of $(Y'_i, (1-\alpha_i)S'_i, u_i\overline{L_i})$ to F_i is the generalized pair $(F_i, (1-\alpha_i)S'_i|_{F_i}, u_i\overline{L_i}|_{F_i})$, where $\overline{L_i}|_{F_i}$ is the nef $\boldsymbol{b}$-divisor over F_i determined by $\tilde{F}_i \to F_i$ and $f_i^*L_i|_{\tilde{F}_i}$. The trace of $\overline{L_i}|_{F_i}$ on F_i is the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $\overline{L}_{iY'_i}|_{F_i}$, and

$$K_{F_i} + (1-\alpha_i)S'_i|_{F_i} + u_i\overline{L}_{iY'_i}|_{F_i} = (K_{Y'_i} + (1-\alpha_i)S'_i + u_i\overline{L}_{iY'_i})|_{F_i} \equiv 0.$$

Write

$$K_{W_i} + B'_{W_i} + u_if_i^*L_i := g'^*_i(K_{Y'_i} + (1-\alpha_i)S'_i + u_i\overline{L}_{iY'_i}).$$

Since $(Y'_i, (1-\alpha_i)S'_i, u_i\overline{L_i})$ is generalized klt, the coefficients of B'_{W_i} are less than 1, and $\mathrm{Supp}\, B'_{W_i}$ is a simple normal crossing divisor. Since $\tilde{F}_i$ is a general fiber of $W_i \to T'_i$, the coefficients of $B'_{W_i}|_{\tilde{F}_i}$ are less than 1, $\mathrm{Supp}\, B'_{W_i}|_{\tilde{F}_i}$ is a simple normal crossing divisor on $\tilde{F}_i$, and

$$K_{\tilde{F}_i} + B'_{W_i}|_{\tilde{F}_i} + u_if_i^*L_i|_{\tilde{F}_i} = p_i^*(K_{F_i} + (1-\alpha_i)S'_i|_{F_i} + u_i\overline{L}_{iY'_i}|_{F_i}).$$

Thus $(F_i, (1-\alpha_i)S'_i|_{F_i}, u_i\overline{L_i}|_{F_i})$ is generalized lc [BZ16, Definition 4.1], [Kol13, Corollary 2.11 and (2.20)]. Let $\tilde{S}_i \subset W_i$ be the birational transform of S'_i. Since $\tilde{S}_i$ is a component of a simple normal crossing divisor, $\tilde{S}_i$ is smooth. If $\tilde{S}_i \to T'_i$ is not surjective, then $S'_i|_{F_i} = 0$. Otherwise, by [Har77, III, Corollary 10.7], applied to $\tilde{S}_i \to T'_i$, $\tilde{S}_i|_{\tilde{F}_i}$ is smooth, hence reduced. Since $g'_i(\mathrm{Exc}(g'_i)) \cap F_i$ has codimension at least 2 in F_i, $S'_i|_{F_i} = p_{i*}(\tilde{S}_i|_{\tilde{F}_i})$ is a reduced divisor. By Lemma 3.2, applied to $f_i|_{\tilde{F}_i} : \tilde{F}_i \to X_i$, the identity contraction of X_i, and L_i, there exist a contraction $\tilde{F}_i \to Z_i$ to a normal projective variety and a finite morphism $Z_i \to X_i$ whose composition is $f_i|_{\tilde{F}_i}$, such

that the pullback H_i of L_i to Z_i is an ample Cartier divisor. Thus $\overline{L_i}|_{F_i}$ is the nef $\boldsymbol{b}$-divisor determined by the pullback of H_i to $\tilde{F}_i$, and $\overline{L_i}|_{F_i}$ is induced by an ample Cartier divisor.

Possibly passing to a subsequence, we may assume that $\dim F_i$ does not depend on i. We check the conditions of Proposition 3.6 for the generalized pairs $(F_i, (1-\alpha_i)S_i'|_{F_i}, u_i\overline{L_i}|_{F_i})$ of dimension $\dim F_i$, with u_i, c, $(1-\alpha_i)S_i'|_{F_i}$, and 0 in place of t_i, t, D_i, and B_i there. Condition (1) of Proposition 3.6 holds by the previous paragraph. Condition (2) holds since $(1-\alpha_i)S_i'|_{F_i} \in \{1-\alpha_i\}$ and $\lim_{i\to+\infty}\alpha_i = 0$. Condition (3) holds by the previous paragraph, and condition (4) does not apply since $\overline{L_{iY_i'}}|_{F_i} \not\equiv 0$. Thus $c \in \mathcal{N}_{\dim F_i - 1}$ by Proposition 3.6. Since $\dim F_i - 1 \le d-2$, Lemma 3.5(3) implies that $\mathcal{N}_{\dim F_i-1} \subset \mathcal{N}_{d-2}$, where $\mathcal{N}_{\dim F_i-1} = \mathcal{N}_0$ if $\dim F_i = 1$. Thus $c \in \mathcal{N}_{d-2}$, a contradiction.

Therefore, possibly passing to a subsequence, we may assume that T_i' is a point for any i. Let $Y_i := Y_i'$ and $S_i := S_i'$. Possibly replacing W_i with a higher model, we may assume that there exists a morphism $g_i : W_i \to Y_i$. Then Y_i is a $\mathbb{Q}$-factorial klt projective variety with $\rho(Y_i) = 1$ (Y_i is klt by Definition-Proposition 2.4 since $(Y_i, (1-\alpha_i)S_i, u_i\overline{L_i})$ is generalized klt), $(Y_i, (1-\alpha_i)S_i, u_i\overline{L_i})$ is generalized klt, $K_{Y_i} + (1-\alpha_i)S_i + u_i\overline{L_{iY_i}} \equiv 0$, and no prime divisor on Y_i is exceptional over X_i. By (7.3), $G_i = f_i^*N_i$, (7.6), and the equality of pullbacks along the MMP shown above,

$$f_i^*(K_{X_i} + \Delta_i + u_iL_i) = g_i^*(K_{Y_i} + (1-\alpha_i)S_i + u_i\overline{L_{iY_i}}) + f_i^*N_i. \tag{7.7}$$

Since no prime divisor on Y_i is exceptional over X_i, any f_i-exceptional prime divisor on W_i is g_i-exceptional, so $g_{i*}f_i^*\Delta_i = g_{i*}f_{i*}^{-1}\Delta_i$. Since any component of Δ_i other than P_i is exceptional over Y_i and S_i is the birational transform of P_i, we obtain $g_{i*}f_{i*}^{-1}P_i = S_i$ and $g_{i*}f_i^*\Delta_i = S_i$. Finally, $\overline{L_{iY_i}} = g_{i*}f_i^*L_i$ is big and $\mathbb{Q}$-Cartier (**Step 2**). Since $\rho(Y_i) = 1$, $\overline{L_{iY_i}}$ is ample.

Step 6. In this step we show that, possibly passing to a subsequence, $(Y_i, S_i, u_i\overline{L_i})$ is generalized lc and $K_{Y_i} + S_i + c\overline{L_{iY_i}} \equiv 0$, and we finish the proof.

(a) We show that, possibly passing to a subsequence, $(Y_i, S_i, c\overline{L_i})$ is generalized lc for any i. By Definition-Proposition 2.4, for any prime divisor E over Y_i,

$$a(E, Y_i, (1-\alpha_i)S_i, c\overline{L_i}) = a(E, Y_i, (1-\alpha_i)S_i, u_i\overline{L_i}) + (u_i - c)e(E, Y_i, \overline{L_i}) > 0,$$

so $(Y_i, (1-\alpha_i)S_i, c\overline{L_i})$ is generalized klt. Let s_i be the generalized lc threshold of S_i with respect to $(Y_i, 0, c\overline{L_i})$ (Definition 2.3). Then $s_i \ge 1-\alpha_i$. Suppose that $(Y_i, S_i, c\overline{L_i})$ is not generalized lc for infinitely many i. Since the generalized log discrepancies of $(Y_i, sS_i, c\overline{L_i})$ are affine in s, $(Y_i, s_iS_i, c\overline{L_i})$ is generalized lc, so $1-\alpha_i \le s_i < 1$ for those i. Since $\lim_{i\to+\infty}\alpha_i = 0$, possibly passing to a subsequence, we may assume that s_i is strictly increasing. However, $(Y_i, 0, c\overline{L_i})$ is a generalized lc pair of dimension d, $\overline{L_i}$ descends to the nef Cartier divisor $f_i^*L_i$ on W_i, and the coefficients of S_i and the coefficient c of the nef part belong to the finite set $\{1, c\}$. Thus the ACC for generalized lc thresholds [BZ16, Theorem 1.5] implies that $\{s_i\}$ satisfies the ACC, a contradiction.

(b) We show that, possibly passing to a subsequence, $(Y_i, S_i, u_i\overline{L_i})$ is generalized lc for any i. Let λ_i be the generalized lc threshold of $\overline{L_i}$ with respect to (Y_i, S_i) (Definition 2.3). By (a), $\lambda_i \ge c$. Suppose that $(Y_i, S_i, u_i\overline{L_i})$ is not generalized lc for infinitely many i. For those i, by Definition-Proposition 2.4, the set of $s \ge 0$ with $(Y_i, S_i, s\overline{L_i})$ generalized lc is the closed interval $[0, \lambda_i]$, so $c \le \lambda_i < u_i$. The conditions of Proposition 4.4 hold for Y_i, $\overline{L_i}$, $B = 0$, $D = (1-\alpha_i)S_i$, and $u = u_i$: Y_i is a $\mathbb{Q}$-factorial klt projective variety of dimension d, $\overline{L_i}$ is induced by an ample Cartier divisor (**Step 2**), $D \ne 0$, $D \in (0,1)$, $\lceil D \rceil = S_i$, $(Y_i, (1-\alpha_i)S_i, u_i\overline{L_i})$ is generalized klt (**Step 5**), and $0 < \lambda_i < u_i$. Thus $\lambda_i \in \mathcal{N}_{d-2}$ by Proposition 4.4. Since $c \le \lambda_i < u_i$ and $\lim_{i\to+\infty} u_i = c$, the λ_i converge to c. By the second part of Lemma 3.7, with $d-2$ in place of d, $c \in \mathcal{N}_{d-2}$, a contradiction.

(c) We show that, possibly passing to a subsequence, $K_{Y_i} + S_i + c\overline{L_{iY_i}} \equiv 0$ for any i. Since $\rho(Y_i) = 1$, $K_{Y_i} + S_i + c\overline{L_{iY_i}}$ is numerically trivial, anti-ample, or ample. Since S_i is a prime

divisor, Y_i is $\mathbb{Q}$-factorial, and $\rho(Y_i) = 1$, S_i is ample [KM98, Theorem 1.18], and $\overline{L_{iY_i}}$ is ample by **Step 5**.

Suppose that $K_{Y_i} + S_i + c\overline{L_{iY_i}}$ is anti-ample for infinitely many i, and consider such an i. Since $K_{Y_i} + S_i + u_i\overline{L_{iY_i}} \equiv \alpha_i S_i$ is ample and $\overline{L_{iY_i}}$ is ample, there exists a real number v_i such that

$$c < v_i < u_i \quad \text{and} \quad K_{Y_i} + S_i + v_i\overline{L_{iY_i}} \equiv 0.$$

By Definition-Proposition 2.4, the generalized log discrepancies of $(Y_i, S_i, v\overline{L_i})$ are affine in v. Since $(Y_i, S_i, c\overline{L_i})$ and $(Y_i, S_i, u_i\overline{L_i})$ are generalized lc by (a) and (b), $(Y_i, S_i, v_i\overline{L_i})$ is generalized lc. Let S_i^ν be the normalization of S_i. Then $\dim S_i^\nu = d - 1 \geq 1$, $(K_{Y_i} + S_i + v_i\overline{L_{iY_i}})|_{S_i^\nu} \equiv 0$, and $\overline{L_{iY_i}}|_{S_i^\nu}$ is ample, so $\overline{L_{iY_i}}|_{S_i^\nu} \not\equiv 0$. Since $S_i \in \mathcal{S}$, Y_i is $\mathbb{Q}$-factorial, and $\overline{L_i}$ is induced by an ample Cartier divisor, Lemma 3.9(2), applied with Z a point, implies that $v_i \in \mathcal{N}_{d-1}$. Since $c < v_i < u_i$ and $\lim_{i\to+\infty} u_i = c$, the v_i have infinitely many distinct values and converge to c. Thus $c \in \mathrm{Acc}(\mathcal{N}_{d-1}) \cap (0, +\infty)$, and Lemma 3.7 implies that $c \in \mathcal{N}_{d-2}$, a contradiction.

Suppose that $K_{Y_i} + S_i + c\overline{L_{iY_i}}$ is ample for infinitely many i, and consider such an i. Since

$$K_{Y_i} + (1 - \alpha_i)S_i + c\overline{L_{iY_i}} \equiv -(u_i - c)\overline{L_{iY_i}}$$

is anti-ample and S_i is ample, there exists a real number b_i such that

$$1 - \alpha_i < b_i < 1 \quad \text{and} \quad K_{Y_i} + b_iS_i + c\overline{L_{iY_i}} \equiv 0.$$

Since $b_i < 1$ and $(Y_i, S_i, c\overline{L_i})$ is generalized lc by (a), $(Y_i, b_iS_i, c\overline{L_i})$ is generalized lc. Since $\lim_{i\to+\infty} \alpha_i = 0$, possibly passing to a subsequence, we may assume that b_i is strictly increasing. Let $\Lambda := \{c\} \cup \{b_i \mid i \in \mathbb{N}^+\}$. Then Λ is a DCC set of non-negative real numbers. For any i, $(Y_i, b_iS_i, c\overline{L_i})$ is a generalized lc pair of dimension d, $c \in \Lambda$, $\overline{L_i}$ descends to the nef Cartier divisor $f_i^*L_i$ on W_i with $f_i^*L_i \not\equiv 0$, $b_iS_i \in \Lambda$, and $K_{Y_i} + b_iS_i + c\overline{L_{iY_i}} \equiv 0$. By Theorem 2.13, b_i belongs to a finite set for any i, which contradicts the fact that b_i is strictly increasing.

Thus, possibly passing to a subsequence, $K_{Y_i} + S_i + c\overline{L_{iY_i}} \equiv 0$ for any i.

Finally, we check that X_i, Δ_i, L_i, Y_i, W_i, f_i, g_i, K_{W_i}, S_i, P_i, α_i, c, u_i, and N_i satisfy the conditions of Set-up 6.1. Conditions (1) and (3) hold by **Step 1**. Condition (2) holds by **Steps 1** and **4**. Conditions (4) and (7) hold by **Step 5**. Conditions (5) and (6) hold by **Step 2** and the choice of g_i in **Step 5**. Condition (8) holds since $0 < \alpha_i < 1$ by **Step 4** and $0 < c < u_i$ by **Step 1**. Condition (9) holds by $N_i \geq \alpha_i\Delta_i$ (**Step 4**) and (7.7). Condition (10) holds by **Step 5** and (c), and condition (11) holds by (b). Moreover, $\mathrm{pet}(X_i, \Delta_i; L_i) = u_i$, u_i is strictly decreasing with $\lim_{i\to+\infty} u_i = c$, $\rho(Y_i) = 1$, and $\lim_{i\to+\infty} \alpha_i = 0$. □

8. A discrepancy gap at a fixed accumulation point

In this section, we prove Theorem 8.2, which is step (ii) of the sketch of the proof in Section 1 and is used in Section 9.

Lemma 8.1 bounds the positive generalized log discrepancies from below, and Theorem 8.2 shows, using Proposition 6.5 and Corollary 4.3, that S is the only generalized lc place of $(Y, S, c\overline{L})$ in Set-up 6.1 when $c \notin \mathcal{N}_{d-2}$ and both $u - c$ and α are smaller than positive real numbers depending only on d and c.

Lemma 8.1. *Let d be a positive integer and $c > 0$ a real number. Then there exists a real number $\epsilon_0 > 0$ depending only on d and c satisfying the following. Let X be a $\mathbb{Q}$-factorial klt projective variety of dimension d, B a reduced divisor on X, and $\mathbf{M}$ a nef $\boldsymbol{b}$-divisor over X that descends to a nef Cartier divisor $\mathbf{M}_{X'} \not\equiv 0$ on a higher model X' of X. Assume that $(X, B, c\mathbf{M})$ is generalized lc and $K_X + B + c\mathbf{M}_X \equiv 0$. Then $a(E, X, B, c\mathbf{M}) \geq \epsilon_0$ for any prime divisor E over X such that $a(E, X, B, c\mathbf{M}) > 0$.*

Proof. Suppose that the lemma does not hold. Then for any real number $t > 0$, there exist X, B, $\mathbf{M}$, and X' satisfying the conditions of the lemma and a prime divisor E over X such that $0 < a(E, X, B, c\mathbf{M}) < t$. Taking $t = 1$, we obtain X_1, B_1, $\mathbf{M}_1$, X_1', and E_1, and we let $\delta_1 := a(E_1, X_1, B_1, c\mathbf{M}_1)$. For any $i \in \mathbb{N}^+$, taking $t = \min\{\delta_i, \frac{1}{i+1}\}$, we obtain X_{i+1}, B_{i+1}, $\mathbf{M}_{i+1}$,

X'_{i+1}, and E_{i+1}, and we let $\delta_{i+1} := a(E_{i+1}, X_{i+1}, B_{i+1}, c\mathbf{M}_{i+1})$. Thus we obtain $\mathbb{Q}$-factorial klt projective varieties X_i of dimension d, reduced divisors B_i on X_i, nef $\boldsymbol{b}$-divisors $\mathbf{M}_i$ over X_i, higher models X'_i of X_i, and prime divisors E_i over X_i for $i \in \mathbb{N}^+$, satisfying the following.

(1) $\mathbf{M}_i$ descends to a nef Cartier divisor $\mathbf{M}_{i,X'_i} \not\equiv 0$ on X'_i.
(2) $(X_i, B_i, c\mathbf{M}_i)$ is generalized lc and $K_{X_i} + B_i + c\mathbf{M}_{i,X_i} \equiv 0$.
(3) $0 < \delta_{i+1} < \delta_i < 1$ for any i, and $\lim_{i\to+\infty} \delta_i = 0$.

Fix $i \in \mathbb{N}^+$. Since X_i is $\mathbb{Q}$-factorial and B_i is reduced, Definition-Proposition 2.4 implies that

$$a(D, X_i, B_i, c\mathbf{M}_i) = 1 - \operatorname{mult}_D B_i \in \{0, 1\}$$

for any prime divisor D on X_i. Since $0 < \delta_i < 1$, E_i is exceptional over X_i. By Lemma 2.8, there exists an extremal divisorial contraction $h_i : Z_i \to X_i$ from a $\mathbb{Q}$-factorial klt projective variety Z_i such that E_i is the unique h_i-exceptional prime divisor and

$$K_{Z_i} + h_{i*}^{-1}B_i + (1-\delta_i)E_i + c\mathbf{M}_{i,Z_i} = h_i^*(K_{X_i} + B_i + c\mathbf{M}_{i,X_i}). \tag{8.1}$$

By (8.1), the generalized pairs $(Z_i, h_{i*}^{-1}B_i + (1-\delta_i)E_i, c\mathbf{M}_i)$ and $(X_i, B_i, c\mathbf{M}_i)$ satisfy

$$a(D, Z_i, h_{i*}^{-1}B_i + (1-\delta_i)E_i, c\mathbf{M}_i) = a(D, X_i, B_i, c\mathbf{M}_i)$$

for any prime divisor D over X_i. Thus $(Z_i, h_{i*}^{-1}B_i + (1-\delta_i)E_i, c\mathbf{M}_i)$ is a generalized lc pair of dimension d, and

$$K_{Z_i} + h_{i*}^{-1}B_i + (1-\delta_i)E_i + c\mathbf{M}_{i,Z_i} \equiv 0.$$

Let Z'_i be a common resolution of X'_i and Z_i, and let $\phi_i : Z'_i \to X'_i$ be the induced morphism. Then $\mathbf{M}_i$ descends to Z'_i, and the trace $\mathbf{M}_{i,Z'_i} = \phi_i^*\mathbf{M}_{i,X'_i}$ is a nef Cartier divisor. Since $\mathbf{M}_{i,X'_i} \not\equiv 0$, there exists a curve C on X'_i such that $(\mathbf{M}_{i,X'_i} \cdot C) \neq 0$. Since ϕ_i is surjective, there exists a curve C' on Z'_i such that $\phi_i(C') = C$ (cut an irreducible component of $\phi_i^{-1}(C)$ dominating C by general hyperplane sections). By the projection formula,

$$(\phi_i^*\mathbf{M}_{i,X'_i} \cdot C') = \deg(C' \to C)(\mathbf{M}_{i,X'_i} \cdot C) \neq 0.$$

Thus $\mathbf{M}_{i,Z'_i} \not\equiv 0$.

Let

$$\Lambda := \{0, 1, c\} \cup \{1 - \delta_i \mid i \in \mathbb{N}^+\}.$$

Since $0 < 1 - \delta_i < 1$ for any i and the sequence $1 - \delta_i$ is strictly increasing, Λ is a DCC set of non-negative real numbers. For any i, $h_{i*}^{-1}B_i + (1-\delta_i)E_i \in \Lambda$ since B_i is reduced and $1 - \delta_i \in \Lambda$. Moreover, $c \in \Lambda$, and $\mathbf{M}_i$ descends to the nef Cartier divisor $\mathbf{M}_{i,Z'_i} \not\equiv 0$ on the higher model Z'_i of Z_i, so the condition that $\mu = 0$ if $\mathbf{M}_{i,Z'_i} \equiv 0$ is satisfied with $\mu := c$. Thus the hypotheses of Theorem 2.13 hold for $(Z_i, h_{i*}^{-1}B_i + (1-\delta_i)E_i, c\mathbf{M}_i)$. By Theorem 2.13 with $\mu := c$, there exists a finite subset $\Lambda_0 \subset \Lambda$ such that $1 - \delta_i \in \Lambda_0$ for any i. This contradicts the fact that the real numbers $1 - \delta_i$ are pairwise distinct. □

Theorem 8.2. *Let $d \geq 2$ be an integer and $c > 0$ a real number such that $c \notin \mathcal{N}_{d-2}$. Then there exist real numbers $\eta > 0$ and $\epsilon > 0$ depending only on d and c satisfying the following. Notation and conditions as in Set-up 6.1, with the given d and c. Assume that $u < c + \eta$ and $\alpha < \frac{\epsilon}{2}$. Then $(Y, S, c\overline{L})$ is generalized lc, S is the unique generalized lc place of $(Y, S, c\overline{L})$, and $a(F, Y, S, c\overline{L}) \geq \epsilon$ for any prime divisor $F \neq S$ over Y.*

Proof. By Corollary 4.3, there exists a real number $\eta > 0$ depending only on d and c satisfying the conclusion of Corollary 4.3. By Lemma 8.1, there exists a real number $\epsilon_0 > 0$ depending only on d and c satisfying the conclusion of Lemma 8.1. Let $\epsilon := \epsilon_0$. We show that η and ϵ satisfy the theorem.

Step 1. In this step, we show that $(Y, S, c\overline{L})$ is generalized lc and $a(F, Y, S, c\overline{L}) \geq \epsilon$ for any prime divisor F over Y such that $a(F, Y, S, c\overline{L}) > 0$.

Since Y is $\mathbb{Q}$-factorial, Definition-Proposition 2.4 implies that

$$a(F,Y,S,t\overline{L}) = a(F,Y,0) - \operatorname{mult}_F S - te(F,Y,\overline{L}) \tag{8.2}$$

for any prime divisor F over Y and any real number $t \geq 0$. Since $e(F,Y,\overline{L}) \geq 0$ and $c < u$,

$$a(F,Y,S,c\overline{L}) = a(F,Y,S,u\overline{L}) + (u-c)e(F,Y,\overline{L}) \geq a(F,Y,S,u\overline{L}) \geq 0$$

for any prime divisor F over Y, where the last inequality holds because $(Y,S,u\overline{L})$ is generalized lc. Thus $(Y,S,c\overline{L})$ is generalized lc. Moreover, $\dim Y = \dim X = d$, $K_Y + S + c\overline{L}_Y \equiv 0$, and $\overline{L}$ descends to the nef Cartier divisor f^*L on the higher model W of Y. Since L is ample,

$$((f^*L)^d) = (L^d) > 0,$$

so $f^*L \not\equiv 0$. By Lemma 8.1 with $X := Y$, $B := S$, $\mathbf{M} := \overline{L}$, and $X' := W$, $a(F,Y,S,c\overline{L}) \geq \epsilon$ for any prime divisor F over Y such that $a(F,Y,S,c\overline{L}) > 0$.

Step 2. In this step, we use the ampleness of L (Lemma 6.4) and Corollary 4.3 to show that $\operatorname{mult}_{P'} N > \frac{\epsilon}{2}$ for any component $P' \neq P$ of Δ.

Let $P' \neq P$ be a component of Δ. By Set-up 6.1(9),

$$K_W + uf^*L - g^*(K_Y + (1-\alpha)S + u\overline{L}_Y) = K_W - f^*(K_X + \Delta) + f^*N.$$

Since $\overline{L}$ descends to W with trace f^*L, the multiplicity of the left-hand side along $f_*^{-1}P'$ is $a(P',Y,(1-\alpha)S,u\overline{L}) - 1$ (Definition 2.3). Since $f_*K_W = K_X$ and $\operatorname{mult}_{P'}\Delta = 1$, the multiplicity of the right-hand side along $f_*^{-1}P'$ is $-1 + \operatorname{mult}_{P'} N$. Thus

$$a(P',Y,(1-\alpha)S,u\overline{L}) = \operatorname{mult}_{P'} N.$$

Since $S \geq 0$, Definition-Proposition 2.4 implies that

$$a(P',Y,S,u\overline{L}) = a(P',Y,(1-\alpha)S,u\overline{L}) - \alpha \operatorname{mult}_{P'} S \leq \operatorname{mult}_{P'} N.$$

By Lemma 6.4, $e(P',Y,\overline{L}) > 0$. Since $a(P',Y,S,u\overline{L}) \geq 0$ and $u > c$, (8.2) implies that

$$a(P',Y,S,c\overline{L}) = a(P',Y,S,u\overline{L}) + (u-c)e(P',Y,\overline{L}) > 0.$$

By **Step 1**, $a(P',Y,S,c\overline{L}) \geq \epsilon$.

The nef $\boldsymbol{b}$-divisor $\overline{L}$ is induced by an ample Cartier divisor: in Definition 3.1, take the projective birational morphism $g : W \to Y$ from the smooth variety W, the contraction $f : W \to X$ to the normal projective variety X, and the ample Cartier divisor L on X. Then $\overline{L} = \overline{f^*L}$. Since $(Y,S,u\overline{L})$ is generalized lc and $c < u < c + \eta$, Corollary 4.3 with $X := Y$, $B := S$, and $\mathbf{M} := \overline{L}$ implies that

$$\epsilon \leq a(P',Y,S,c\overline{L}) \leq \left(1 + \frac{u-c}{\eta}\right) a(P',Y,S,u\overline{L}) \leq \left(1 + \frac{u-c}{\eta}\right) \operatorname{mult}_{P'} N < 2 \operatorname{mult}_{P'} N,$$

where the last inequality holds because $u - c < \eta$ and $\operatorname{mult}_{P'} N \geq \alpha > 0$ by $N \geq \alpha\Delta$. Thus $\operatorname{mult}_{P'} N > \frac{\epsilon}{2} > \alpha$.

Step 3. In this step, we show that S is the unique generalized lc place of $(Y,S,c\overline{L})$ and conclude the proof.

Since S is a prime divisor on Y, $a(S,Y,0) = 1$, $\operatorname{mult}_S S = 1$, and $e(S,Y,\overline{L}) = 0$ (Definition-Proposition 2.4). By (8.2), $a(S,Y,S,c\overline{L}) = 0$. Thus S is a generalized lc place of $(Y,S,c\overline{L})$ (Definition 2.3).

Let F be a prime divisor over Y such that $a(F,Y,S,c\overline{L}) = 0$. Since X and Y are birational, F is a prime divisor over X. By Proposition 6.5, $\operatorname{center}_X F$ is a stratum of Δ, and $\operatorname{mult}_{P'} N = \alpha$ for any component P' of Δ containing $\operatorname{center}_X F$. By Definition 2.6, $\operatorname{center}_X F$ is an irreducible component of $P_1 \cap \cdots \cap P_r$ for some integer $r \geq 1$ and pairwise distinct components $P_1, \ldots, P_r$ of Δ, each of which contains $\operatorname{center}_X F$. Thus $\operatorname{mult}_{P_j} N = \alpha$ for any j, and **Step 2** implies that $P_j = P$ for any j. Hence $r = 1$ and $\operatorname{center}_X F$ is an irreducible component of P, that is, $\operatorname{center}_X F = P$. Since $\operatorname{center}_X F = P$ is a prime divisor on X, $F = P$ as prime divisors over X.

Hence $F = S$ as prime divisors over Y, since $g_* f_*^{-1} P = S$. Therefore, S is the unique generalized lc place of $(Y, S, c\overline{L})$.

Finally, let $F \neq S$ be a prime divisor over Y. By **Step 1**, $a(F, Y, S, c\overline{L}) \geq 0$, and $a(F, Y, S, c\overline{L}) \neq 0$ since S is the unique generalized lc place of $(Y, S, c\overline{L})$. By **Step 1** again, $a(F, Y, S, c\overline{L}) \geq \epsilon$. □

9. Proof of the main theorem

In this section, we prove Theorem 1.2. Part (1) follows from Theorems 7.1 and 8.2 together with boundedness, and part (2) follows from part (1) and the properties of the sets $\mathcal{N}_d$ in Section 3.

Proof of Theorem 1.2. The proof is long, so we divide it into several steps.

Step 1. In this step, we assume that (1) fails and fix a sequence of data as in Set-up 6.1.

Assume that there exist an integer $d \geq 2$ and a real number $c \in \mathrm{Acc}(\mathrm{PET}_d)$ such that $c > 0$ and $c \notin \mathcal{N}_{d-2}$. By Theorem 7.1, there exist a strictly decreasing sequence of real numbers u_i with $\lim_{i\to+\infty} u_i = c$ and, for each i, data X_i, Δ_i, L_i, Y_i, W_i, f_i, g_i, K_{W_i}, S_i, P_i, α_i, N_i as in Set-up 6.1 for d and c and with $u = u_i$, such that

$$\mathrm{pet}(X_i, \Delta_i; L_i) = u_i, \quad \rho(Y_i) = 1, \quad \text{and} \quad \lim_{i\to+\infty} \alpha_i = 0.$$

Let $\eta > 0$ and $\epsilon > 0$ be the real numbers in Theorem 8.2, which depend only on d and c. When ϵ decreases, the assumption $\alpha < \frac{\epsilon}{2}$ of Theorem 8.2 becomes stronger and the conclusion $a(F, Y, S, c\overline{L}) \geq \epsilon$ becomes weaker. Thus, possibly replacing ϵ with $\min\{\epsilon, 1\}$, we may assume that $\epsilon \leq 1$. Since $\lim_{i\to+\infty} u_i = c$ and $\lim_{i\to+\infty} \alpha_i = 0$, possibly removing finitely many terms of the sequence, we may assume that $u_i < c + \eta$ and $\alpha_i < \frac{\epsilon}{2}$ for any i. By Theorem 8.2, for any i, $(Y_i, S_i, c\overline{L_i})$ is generalized lc and

$$a(F, Y_i, S_i, c\overline{L_i}) \geq \epsilon \quad \text{for any prime divisor } F \neq S_i \text{ over } Y_i. \tag{9.1}$$

Step 2. In this step, we show that $(Y_i, 0)$ is ϵ-lc and $-K_{Y_i}$ is ample, and we find very ample divisors of bounded degree on the varieties Y_i.

Since Y_i is $\mathbb{Q}$-factorial, K_{Y_i}, S_i, and $\overline{L_i}_{Y_i}$ are $\mathbb{Q}$-Cartier. By Definition-Proposition 2.4, for any prime divisor F over Y_i,

$$a(F, Y_i, 0) = a(F, Y_i, S_i, c\overline{L_i}) + \mathrm{mult}_F S_i + ce(F, Y_i, \overline{L_i}), \tag{9.2}$$

and $e(F, Y_i, \overline{L_i}) \geq 0$. Since $S_i \geq 0$, $\mathrm{mult}_F S_i \geq 0$. If $F \neq S_i$, then (9.1) and (9.2) imply that $a(F, Y_i, 0) \geq \epsilon$. If $F = S_i$, then $a(F, Y_i, 0) = 1 \geq \epsilon$. Thus $(Y_i, 0)$ is ϵ-lc.

Since $\rho(Y_i) = 1$ and S_i is a $\mathbb{Q}$-Cartier prime divisor, S_i is ample [KM98, Theorem 1.18]. By Set-up 6.1(4), $\overline{L_i}_{Y_i}$ is ample. By Set-up 6.1(10),

$$-K_{Y_i} \equiv S_i + c\overline{L_i}_{Y_i},$$

and $c > 0$, so $-K_{Y_i}$ is ample [KM98, Theorem 1.18]. Since Y_i is birational to X_i, $\dim Y_i = d$. By Theorem 2.12, applied to d, ϵ, and the pairs $(Y_i, 0)$, and by Definition 2.11, there exists a positive integer v depending only on d and ϵ, hence only on d and c, such that for any i, there exists a very ample Cartier divisor A_i on Y_i with

$$A_i^d \leq v. \tag{9.3}$$

Step 3. In this step, we find log smooth couples (X_i', Σ_i) in a bounded set of couples, birational to Y_i, such that $\overline{L_i}$ and $\overline{A_i}$ descend to X_i' and Σ_i contains the birational transform of Δ_i.

Let F be a prime divisor over Y_i, and let $h : Y' \to Y_i$ be a projective birational morphism from a normal variety such that F is a divisor on Y' and $\overline{L_i}$ descends to Y'. Since $\overline{A_i}$ descends to Y_i with trace A_i, $\overline{A_i}_{Y'} = h^* A_i$. Thus

$$e(F, Y_i, c\overline{L_i} + \overline{A_i}) = \mathrm{mult}_F \left(h^*(c\overline{L_i}_{Y_i} + A_i) - c\overline{L_i}_{Y'} - h^* A_i\right) = ce(F, Y_i, \overline{L_i}).$$

By Definition-Proposition 2.4, applied with $t = 1$ and with $t = c$,

$$a(F, Y_i, S_i, c\overline{L_i} + \overline{A_i}) = a(F, Y_i, 0) - \operatorname{mult}_F S_i - ce(F, Y_i, \overline{L_i}) = a(F, Y_i, S_i, c\overline{L_i}).$$

Since $(Y_i, S_i, c\overline{L_i})$ is generalized lc, $(Y_i, S_i, c\overline{L_i} + \overline{A_i})$ is generalized lc.

By Set-up 6.1(10),

$$K_{Y_i} + S_i + c\overline{L_i}_{Y_i} + A_i \equiv A_i.$$

Since A_i is ample, $K_{Y_i} + S_i + c\overline{L_i}_{Y_i} + A_i$ is ample [Laz04, Proposition 1.3.13], hence big. Since $\operatorname{vol}(A_i) = A_i^d$ [Laz04, Corollary 1.4.41] and the volume of an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D depends only on the numerical class of D [Laz04, Proposition 2.2.41 and Corollary 2.2.45], (9.3) implies that

$$\operatorname{vol}(K_{Y_i} + S_i + c\overline{L_i}_{Y_i} + A_i) = A_i^d \le v < v + 1.$$

Let $\Phi := \{1, c\}$. Since Φ is a finite subset of $\mathbb{R}_{\ge 0}$, Φ is a DCC set. By Set-up 6.1(5), W_i is a common resolution of X_i and Y_i, so the nef $\boldsymbol{b}$-divisor $c\overline{L_i} + \overline{A_i}$ is determined by $g_i : W_i \to Y_i$ and the $\mathbb{R}$-divisor $cf_i^*L_i + g_i^*A_i$ on W_i. Here $f_i^*L_i$ and $g_i^*A_i$ are nef Cartier divisors on W_i, and their coefficients c and 1 are positive and belong to Φ. Moreover, Y_i is projective, $\dim Y_i = d$, and $S_i \in \Phi$. Thus the generalized pair $(Y_i, S_i, c\overline{L_i} + \overline{A_i})$, with the data g_i, $cf_i^*L_i + g_i^*A_i$, satisfies the hypotheses of [Bir21a, Theorem 1.2] for d, Φ, and $v + 1$. By [Bir21a, Theorem 1.2], there exists a bounded set $\mathcal{P}$ of couples, depending only on d, Φ, and $v + 1$, hence only on d and c, such that for any i, there exist a log smooth couple $(X_i', \Sigma_i) \in \mathcal{P}$ and a birational map $X_i' \dashrightarrow Y_i$ with the following properties. The divisor Σ_i contains the birational transform of S_i and any prime divisor on X_i' which is exceptional over Y_i. Both $\overline{L_i}$ and $\overline{A_i}$ descend to X_i'. There exists a very ample divisor $H_i \le \Sigma_i$ on X_i' such that $H_i - c\overline{L_i}_{X_i'} - \overline{A_i}_{X_i'}$ is pseudo-effective. Here descent is in the sense of [Bir21a, 3.1(1)] with Z a point, which is descent in the sense of Definition 2.2.

Let $h_i' : W_i' \to X_i'$ and $h_i : W_i' \to W_i$ be projective birational morphisms from a smooth projective variety W_i' resolving the birational map $X_i' \dashrightarrow W_i$ induced by $X_i' \dashrightarrow Y_i$ and g_i. Since $\overline{L_i}$ and $\overline{A_i}$ descend to X_i', and $f_i^*L_i$ and $g_i^*A_i$ are the traces of $\overline{L_i}$ and $\overline{A_i}$ on W_i, Definition 2.2 implies that

$$h_i'^*\overline{L_i}_{X_i'} = h_i^*f_i^*L_i \quad \text{and} \quad h_i'^*\overline{A_i}_{X_i'} = h_i^*g_i^*A_i. \tag{9.4}$$

Since h_i' is birational, the traces

$$\overline{L_i}_{X_i'} = h_{i*}'h_i^*f_i^*L_i \quad \text{and} \quad \overline{A_i}_{X_i'} = h_{i*}'h_i^*g_i^*A_i$$

have integer coefficients. Since X_i' is smooth, $\overline{L_i}_{X_i'}$ and $\overline{A_i}_{X_i'}$ are Cartier divisors. By Lemma 2.9, applied to h_i', $f_i \circ h_i$, L_i, and $\overline{L_i}_{X_i'}$, there exists a morphism $f_i' : X_i' \to X_i$ such that $f_i \circ h_i = f_i' \circ h_i'$. By Lemma 2.9, applied to h_i', $g_i \circ h_i$, A_i, and $\overline{A_i}_{X_i'}$, there exists a morphism $g_i' : X_i' \to Y_i$ such that $g_i \circ h_i = g_i' \circ h_i'$. Since $f_i \circ h_i$, $g_i \circ h_i$, and h_i' are birational, f_i' and g_i' are projective birational morphisms, and g_i' induces the birational map $X_i' \dashrightarrow Y_i$. Since h_i' is birational, (9.4) implies that

$$\overline{L_i}_{X_i'} = h_{i*}'h_i'^*\overline{L_i}_{X_i'} = h_{i*}'h_i^*f_i^*L_i = h_{i*}'h_i'^*f_i'^*L_i = f_i'^*L_i$$

and

$$\overline{A_i}_{X_i'} = h_{i*}'h_i'^*\overline{A_i}_{X_i'} = h_{i*}'h_i^*g_i^*A_i = h_{i*}'h_i'^*g_i'^*A_i = g_i'^*A_i.$$

In particular,

$$H_i - cf_i'^*L_i - g_i'^*A_i \text{ is pseudo-effective.} \tag{9.5}$$

We show that

$$(f_i')_*^{-1}\Delta_i \le \Sigma_i. \tag{9.6}$$

Since $f_i \circ h_i = f_i' \circ h_i'$ and $g_i \circ h_i = g_i' \circ h_i'$, for any prime divisor Q on X_i,

$$g_i'\big((f_i')_*^{-1}Q\big) = g_i\big(f_{i*}^{-1}Q\big).$$

By Set-up 6.1(7), $g_i(f_{i*}^{-1}P_i) = S_i$, so $(f_i')_*^{-1}P_i$ is the birational transform of S_i on X_i'. Let $Q \ne P_i$ be a component of Δ_i. By Lemma 6.4, Q is exceptional over Y_i, that is, $g_i(f_{i*}^{-1}Q)$ is not a divisor

on Y_i. Hence $(f_i')_*^{-1}Q$ is a prime divisor on X_i' which is exceptional over Y_i. Therefore any component of $(f_i')_*^{-1}\Delta_i$ is a component of Σ_i. Since Δ_i and Σ_i are reduced, (9.6) holds.

Step 4. In this step, we bound L_i^d, $K_{X_i}\cdot L_i^{d-1}$, and $\Delta_i\cdot L_i^{d-1}$, and we find a positive integer m such that mL_i is very ample for any i.

Let $\mathcal{P}_0$ be the set of log smooth couples in $\mathcal{P}$. Then $\mathcal{P}_0$ is a bounded set of log smooth couples, and $(X_i',\Sigma_i)\in\mathcal{P}_0$ for any i. By Lemma 2.14, there exists a positive integer b depending only on $\mathcal{P}_0$, hence only on d and c, such that for any i, there exist a very ample Cartier divisor A_i' on X_i' and an $\mathbb{R}$-divisor $C_i\geq\Sigma_i$ on X_i' such that $A_i'^d\leq b$, and $bA_i'-C_i$, $bA_i'-K_{X_i'}$, and $bA_i'+K_{X_i'}$ are nef. Here $\dim X_i'=d$ since X_i' is birational to Y_i.

Fix an index i. For any integer $0\leq k\leq d$, let

$$s_k:=A_i'^{d-k}\cdot(f_i'^*L_i)^k.$$

Since A_i' is very ample, A_i' is nef and big. Since L_i is ample and f_i' is birational, $f_i'^*L_i$ is nef and big. Since A_i' is an ample Cartier divisor, $s_0=A_i'^d$ is a positive integer. By the projection formula [Ful98, Proposition 2.5(c)], $s_d=L_i^d>0$. By [Laz04, Theorem 1.6.1], $s_k^d\geq s_0^{d-k}s_d^k>0$ for any $0\leq k\leq d$. By [Laz04, Example 1.6.4],

$$s_k^2\geq s_{k-1}s_{k+1}\quad\text{for any integer } 1\leq k\leq d-1.$$

Thus $\frac{s_{k+1}}{s_k}\leq\frac{s_k}{s_{k-1}}$ for any $1\leq k\leq d-1$, so $\frac{s_k}{s_{k-1}}\leq\frac{s_1}{s_0}$ for any $1\leq k\leq d$. Since $s_0\geq 1$, for any $1\leq k\leq d$,

$$s_k=s_0\prod_{j=1}^{k}\frac{s_j}{s_{j-1}}\leq\frac{s_1^k}{s_0^{k-1}}\leq s_1^k. \tag{9.7}$$

Recall from **Step 3** that $H_i\leq\Sigma_i$ is very ample. Since A_i' is nef, (9.5) implies that $(H_i-cf_i'^*L_i-g_i'^*A_i)\cdot A_i'^{d-1}\geq 0$. Since $g_i'^*A_i$ is nef, $g_i'^*A_i\cdot A_i'^{d-1}\geq 0$. Since $\Sigma_i-H_i\geq 0$, $C_i-\Sigma_i\geq 0$, and $bA_i'-C_i$ and A_i' are nef, we obtain

$$cs_1\leq H_i\cdot A_i'^{d-1}\leq\Sigma_i\cdot A_i'^{d-1}\leq C_i\cdot A_i'^{d-1}\leq bA_i'^d\leq b^2.$$

Thus $s_1\leq\frac{b^2}{c}$, and (9.7) implies that $s_k\leq\left(\frac{b^2}{c}\right)^k$ for any $1\leq k\leq d$. Since L_i is an ample Cartier divisor on the smooth variety X_i, L_i^d is a positive integer. Taking $k=d$ and $k=d-1$, we obtain

$$1\leq L_i^d=s_d\leq\left(\frac{b^2}{c}\right)^d\quad\text{and}\quad A_i'\cdot(f_i'^*L_i)^{d-1}=s_{d-1}\leq\left(\frac{b^2}{c}\right)^{d-1}. \tag{9.8}$$

Since X_i and X_i' are smooth and f_i' is birational, $K_{X_i'}-f_i'^*K_{X_i}$ is f_i'-exceptional. By the projection formula [Ful98, Proposition 2.5(c)], $(K_{X_i'}-f_i'^*K_{X_i})\cdot(f_i'^*L_i)^{d-1}=0$, so $K_{X_i}\cdot L_i^{d-1}=K_{X_i'}\cdot(f_i'^*L_i)^{d-1}$. Since $bA_i'-K_{X_i'}$, $bA_i'+K_{X_i'}$, and $f_i'^*L_i$ are nef, (9.8) implies that

$$|K_{X_i}\cdot L_i^{d-1}|=|K_{X_i'}\cdot(f_i'^*L_i)^{d-1}|\leq bA_i'\cdot(f_i'^*L_i)^{d-1}\leq b\Big(\frac{b^2}{c}\Big)^{d-1}. \tag{9.9}$$

Since $f_{i*}'(f_i')_*^{-1}\Delta_i=\Delta_i$, the projection formula [Ful98, Proposition 2.5(c)] implies that

$$\Delta_i\cdot L_i^{d-1}=(f_i')_*^{-1}\Delta_i\cdot(f_i'^*L_i)^{d-1}.$$

Since $\Delta_i\geq 0$ and L_i is ample, $\Delta_i\cdot L_i^{d-1}\geq 0$. By (9.6), $C_i\geq\Sigma_i\geq(f_i')_*^{-1}\Delta_i$. Since $bA_i'-C_i$ and $f_i'^*L_i$ are nef, (9.8) implies that

$$\begin{aligned}0\leq\Delta_i\cdot L_i^{d-1}&=(f_i')_*^{-1}\Delta_i\cdot(f_i'^*L_i)^{d-1}\\&\leq C_i\cdot(f_i'^*L_i)^{d-1}\leq bA_i'\cdot(f_i'^*L_i)^{d-1}\leq b\Big(\frac{b^2}{c}\Big)^{d-1}.\end{aligned} \tag{9.10}$$

Since X_i is smooth and K_{X_i} and L_i are Cartier, L_i^d and $K_{X_i}\cdot L_i^{d-1}$ are integers. By (9.8) and (9.9), the pairs $(L_i^d,K_{X_i}\cdot L_i^{d-1})$ belong to a finite set depending only on d, b, and c. By [Siu93,

Theorem (0.1)], for a smooth projective variety X of dimension d and an ample Cartier divisor L on X, $m'L$ is very ample for any integer m' which is at least a bound depending only on d, L^d, and $K_X \cdot L^{d-1}$. Let m be the smallest positive integer that is at least the bounds for the elements of the finite set above. Then m depends only on d and c, and mL_i is very ample for any i, since m is at least the bound for $(L_i^d, K_{X_i} \cdot L_i^{d-1})$. Let r be a positive integer such that $m^d\big(\frac{b^2}{c}\big)^d \leq r$ and $m^{d-1}b\big(\frac{b^2}{c}\big)^{d-1} \leq r$. Then r depends only on d and c, and by (9.8) and (9.10), for any i,

$$(mL_i)^d = m^d L_i^d \leq r \quad \text{and} \quad \Delta_i \cdot (mL_i)^{d-1} = m^{d-1}\Delta_i \cdot L_i^{d-1} \leq r. \tag{9.11}$$

Step 5. In this step, we derive a contradiction, which proves (1).

By Set-up 6.1(1)–(3), for any i, X_i is a smooth projective variety of dimension d, Δ_i is a reduced simple normal crossing divisor on X_i, and L_i is an ample Cartier divisor on X_i. By **Step 4**, mL_i is very ample and (9.11) holds for any i. By Lemma 5.2, applied to d, m, and r, the set $\{\mathrm{pet}(X_i, \Delta_i; L_i) \mid i\}$ is finite. Recall from **Step 1** that $\mathrm{pet}(X_i, \Delta_i; L_i) = u_i$ and the sequence u_i is strictly decreasing, so the set $\{\mathrm{pet}(X_i, \Delta_i; L_i) \mid i\}$ is infinite. This is a contradiction. Thus (1) holds.

Step 6. In this step, we prove (2), that is, Conjecture 1.1.

The case $d = 1$ is trivial. Assume that $d \geq 2$.

Let Γ and Γ' be subsets of $\mathbb{R}$ such that $\Gamma \subset \{0\} \cup \Gamma'$. If $x > 0$ is the limit of a sequence of pairwise distinct elements of Γ, then at most one term of the sequence is 0, and after removing the term equal to 0, the sequence consists of pairwise distinct elements of Γ'. Thus

$$\mathrm{Acc}(\Gamma) \cap (0, +\infty) \subset \mathrm{Acc}(\Gamma') \cap (0, +\infty). \tag{9.12}$$

By definition, $\mathrm{PET}_d \subset [0, +\infty)$. Since $[0, +\infty)$ is closed,

$$\mathrm{Acc}^k(\mathrm{PET}_d) \subset [0, +\infty) \quad \text{for any integer } k \geq 0.$$

We show by induction on k that

$$\mathrm{Acc}^k(\mathrm{PET}_d) \cap (0, +\infty) \subset \mathcal{N}_{d-k-1} \quad \text{for any integer } 1 \leq k \leq d-1. \tag{9.13}$$

The case $k = 1$ is (1). Assume that (9.13) holds for some integer $1 \leq k \leq d-2$. Then

$$\mathrm{Acc}^k(\mathrm{PET}_d) \subset \{0\} \cup \mathcal{N}_{d-k-1}.$$

Since $d - k - 1 \geq 1$, (9.12) and Lemma 3.7 imply that

$$\mathrm{Acc}^{k+1}(\mathrm{PET}_d) \cap (0, +\infty) \subset \mathrm{Acc}(\mathcal{N}_{d-k-1}) \cap (0, +\infty) \subset \mathcal{N}_{d-k-2}.$$

Thus (9.13) holds for $k + 1$.

By Lemma 3.8, $\mathcal{N}_{d-k-1} \subset (0, d-k]$ for any integer $1 \leq k \leq d-1$. Thus (9.13) implies that $\mathrm{Acc}^k(\mathrm{PET}_d) \subset [0, d-k]$ for any integer $1 \leq k \leq d-1$. By (9.13) for $k = d-1$,

$$\mathrm{Acc}^{d-1}(\mathrm{PET}_d) \subset \{0\} \cup \mathcal{N}_0.$$

Since $\mathcal{N}_0$ has no positive accumulation point (Definition 3.3), (9.12) implies that

$$\mathrm{Acc}^d(\mathrm{PET}_d) \cap (0, +\infty) \subset \mathrm{Acc}(\mathcal{N}_0) \cap (0, +\infty) = \emptyset.$$

Hence

$$\mathrm{Acc}^d(\mathrm{PET}_d) \subset \{0\} = [0, d-d].$$

Therefore $\mathrm{Acc}^k(\mathrm{PET}_d) \subset [0, d-k]$ for any integer $1 \leq k \leq d$, which is Conjecture 1.1. □

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